



RMC.BestFit 2.0 Verification Report

Analytical checks, independent benchmarks, estimation recovery,
and uncertainty diagnostics

U.S. ARMY CORPS OF ENGINEERS, RISK MANAGEMENT CENTER

Draft for external peer review - not for public release
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Contents

Each analysis chapter explains the data, comparison procedure, results, and limitations. Appendix A provides a complete index of the verification tests.

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CHAPTER 01

Report Documentation

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Authors

- C. Haden Smith, U.S. Army Corps of Engineers, Risk Management Center, Lakewood, Colorado, USA.
- Woodrow L. Fields, U.S. Army Corps of Engineers, Risk Management Center, Lakewood, Colorado, USA.
- Brian Skahill, Fariborz Maseeh Department of Mathematics and Statistics, Portland State University, Portland, Oregon, USA.

Abstract

RMC.BestFit is a Bayesian-first framework for flood-frequency and water-resources analysis. This report explains how its calculations are verified and what the results mean for an engineering reader. Each analysis is presented through its data, test setup, comparison procedure, results, and limitations. Evidence includes direct mathematical answers, independent numerical calculations, comparisons with other scientific software, published flood-frequency examples, and experiments in which the generating model is known. The scope includes distribution fitting and frequency analysis, multivariate models, rating curves, time series, and spatial extremes. Appendix A maps all 329 current verification methods to the report. Numerical verification establishes the behavior of the tested calculations; model suitability for a particular engineering application remains a separate judgment.

Scope and review status

This edition describes the repository identified above. Numerical results retain the execution configurations recorded with their evidence; the report date identifies the source and documentation review. The draft is supplied for independent scientific and engineering review. Report number, reviewers, signatures, and final distribution remain subject to publication approval.

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CHAPTER 02

Executive Summary

Purpose

This report explains how RMC.BestFit's statistical analyses are tested, what the comparisons show, and where the evidence stops. It is intended for engineers and reviewers who need to assess the software without reading its code. Each analysis chapter describes the observations or generated data, the calculation performed by BestFit, the independent reference, and the meaning of the result.

The central question is whether the software performs its specified calculations correctly. Choosing a model that adequately represents a watershed, measurement process, or risk decision remains a separate engineering judgment.

Scope and principal findings

The tests combine direct mathematical answers, comparisons with independent R and Python software, published flood-frequency examples, and experiments with known generating models. For example, seven Bulletin 17C examples reproduce 21 published parameter values within 0.001; diagnostic calculations reproduce independent R results to tight numerical tolerances; and generated-data experiments check whether known parameters or responses lie within their stated uncertainty intervals.

Capability	Scientific evidence	Supported result
Univariate families and distribution fitting	Fifteen-family recovery, SciPy and R comparisons, analytical identities, and common-data model ranking	Distribution parameterizations, likelihoods, fitted results, and criteria satisfy the specified checks.
Estimation and diagnostics	Analytical posteriors and covariance; R <code>loo</code> , <code>posterior</code> , <code>gmm</code> , and <code>bbmle</code>	Estimator objectives, profiling, information criteria, and convergence diagnostics reproduce independent references.
Bulletin 17C	Published examples, PeakFQ plotting positions, moment covariance, regional penalties, and synthetic recovery	Seven example parameter vectors and the specified moment and penalty calculations satisfy their acceptance rules.
Point process, competing risks, mixtures, and composites	Independent likelihoods, dependence identities, external packages, and generating-model recovery	Event rates, combinations of processes, and propagation of uncertainty satisfy the checks for the specified designs.
Bivariate and coincident frequency	Independent dependence-model fits, simulated recovery, and response calculations	The tested fits and probabilities of responses driven by two variables match their references.
Rating curves	SciPy likelihoods and optima, continuity identities, and parameter and response recovery	One-, two-, and three-control designs satisfy the likelihood and recovery criteria.
Time-series models	Independent transforms, recurrences, likelihoods, forecasts, and recovery	AR, MA, ARIMA, and ARIMAX calculations satisfy the specified checks.
Spatial extremes	Independent Gaussian-copula and process calculations, uncertainty propagation, and ten-site recovery	The tested likelihood, prediction, distance, and uncertainty calculations match their references.

The main chapters cover all fifteen analysis types: distribution fitting, univariate, Bulletin 17C, point process, competing risk, mixture, composite, bivariate, coincident frequency, rating curve, autoregressive, moving average, ARIMA, ARIMAX, and spatial extremes. The estimation chapter explains the shared fitting and diagnostic calculations. The data chapters explain checks that preserve measurements, dates, missing values, and observation types before analysis.

Appendix A maps all **329 current test methods in 71 classes** to these explanations. The catalog records 327 verified methods and two tests of accepted limitations. These are inventory counts, not probabilities that a model is correct or percentages of statistical interval coverage.

How to interpret the results

Analytical and external-package comparisons provide the most direct checks of a numerical calculation. Recovery experiments ask whether a fitted model can recover its generating parameters or responses under a declared sample size, seed, and estimator configuration. A single successful recovery experiment does not demonstrate repeated-sample confidence-interval coverage.

The two accepted limitations concern joint-prior sampling and one-step GMM deletion influence. The generic sampler draws parameter priors independently and therefore does not represent additional coupled prior factors. GMM influence correctly ranks the tested outlier but its magnitude is not calibrated to exact deletion scale. The relevant checks document these boundaries; they do not remove them.

Bulletin 17C bootstrap diagnostics also distinguish delivered fits from converged fits. In the inspected 1,000-refit experiment, all outputs were accepted by the delivery workflow, while 947 met outer convergence and 53 reached the iteration cap. Interval coverage and universal bootstrap convergence are not established by that experiment.

The analysis chapters explain these and other model-specific boundaries. The report's conclusions are restricted to the stated calculations, data designs, and acceptance rules.

CHAPTER 03

System Under Test and Verification Methodology

How to read this report

The report is written for engineers and reviewers who need to understand what was tested without reading the software. Each analysis chapter begins with the question the analysis answers, describes the data and the comparison procedure, and explains the results. The equations identify the quantity being checked; the accompanying text explains its meaning. Appendix A provides the complete test index for readers who need to trace a result to its implementation.

Two kinds of data appear throughout the report. **Observed data** come from a published example or a retained measurement record. **Synthetic data** are generated from a model whose parameters are known in advance. Observed data connect the tests to engineering applications. Synthetic data make it possible to ask whether an analysis can recover a known answer. Synthetic magnitudes have no physical unit unless the chapter assigns one.

The following terms describe different parts of an analysis and should not be read as interchangeable results.

Term	Meaning in this report
Parameter	A quantity that defines a model, such as a mean, a standard deviation, or a trend slope.
Generating value, or parent value	The parameter chosen before generating synthetic observations; it is the known target in a recovery experiment.
Likelihood	A measure of how well a specified set of parameters explains the observed data. A log likelihood is the logarithm of this measure.
Prior and posterior	The prior represents information about parameters before using the study data. The posterior combines that information with the data.
Posterior draw	One sampled set of parameter values from the posterior. Many draws describe uncertainty; they are not additional observations.
Quantile	The value associated with a stated probability. For annual maxima, the 0.99 quantile is exceeded with annual probability 0.01, often called the 100-year event.
Confidence interval	An interval constructed by a procedure intended to include the fixed true value at a stated rate over repeated samples.
Credible interval	An interval containing a stated share of the posterior probability for a parameter or response.
Predictive interval or band	An interval or curve band for new observations, including their variability as well as the uncertainty included by the specified method.
Independent reference, also called an oracle	An analytical answer, published result, or calculation made separately from the BestFit calculation being checked.

An exceedance probability is a probability for each year under the model; a 100-year event is not a prediction that one such event occurs at regular 100-year intervals. A confidence or credible interval for a flood quantile is also different from a predictive interval for a future flood.

System under test

RMC.BestFit combines data management, statistical models, estimation, and presentation of results. Its numerical calculations use the RMC.Numerics library. This report examines both individual calculations and complete analyses: a correct distribution formula is necessary, but the data preparation, fitting, and uncertainty calculations must also work together.

Component	Source-review configuration
RMC.BestFit	Version 2.0.0; commit <code>8808f19f3bfa712241e55a0dfa47407ff5a60970</code>
RMC.Numerics source	Commit <code>7e8e8d1c5f26e045a35ec9fc09367de95ed05b02</code>
Declared package baseline	RMC.Numerics 2.2.0
Runtime	.NET on Windows, 64-bit process
Principal Bayesian sampler	DEMCzs, with the configuration declared for each test design
Reproducibility	Explicit generators, retained sample sizes, seeds, and acceptance rules

Local builds can resolve the sibling Numerics source project instead of the package. These are distinct dependency configurations. A run's recorded configuration identifies which was used; package compatibility and numerical results are not inferred solely from a source build.

The report describes the current tests and their supporting evidence. A result marked **passed** means that the recorded check met its stated numerical or statistical acceptance rule. A table headed **reference value** reports the comparison target, not an unrecorded BestFit estimate. Where an execution record preserves the pass outcome but not the fitted values, the chapter reports the target and the accepted comparison without inventing an estimate. Source-review dates do not establish that every method has been rerun against one common binary.

Verification and validation

Verification asks whether software implements its specified equations and algorithms correctly. **Validation** asks whether a model adequately represents a physical process for its intended use. This report primarily provides verification. Synthetic recovery experiments assess behavior under a known data-generating law; they do not establish that the law is suitable for a particular watershed.

Evidence types

Evidence	Question answered
Analytical calculation	Does the implementation reproduce a value that can be calculated directly from a formula?
Independent calculation	Does it agree with a separately written numerical calculation?
External scientific package	Does it reproduce an equivalent calculation in a documented R or Python package?
Published source	Does it reproduce a worked example or benchmark with a defined parameterization?
Recovery experiment	Does estimation recover known generating parameters or responses under the specified design?

The comparison procedure has four parts: specify the data and model, establish an independent reference, perform the BestFit calculation, and compare the results using a rule fixed for that test. Finishing an analysis or producing a plausible graph does not establish accuracy. Conversely, a test deliberately designed to expose an unsupported behavior can pass by confirming that limitation; the report identifies those cases explicitly.

Deterministic comparisons

For a reference value r and software value s , absolute and relative errors are

$$e_{\text{abs}} = |s - r|, \quad e_{\text{rel}} = \frac{|s - r|}{\max(|r|, e_0)}. \quad (\text{M.1})$$

Here r is the reference, s is the BestFit result, and e_0 prevents division by zero when the reference is close to zero. For example, an absolute tolerance of 10^{-8} permits a difference no greater than 0.00000001. This is a check of arithmetic agreement, not an allowance for measurement error. For a matrix, the rule ordinarily applies to every entry.

Comparing two calculations at the **same parameter values** isolates formula implementation. Comparing two **fitted sets of parameters** also tests optimization and can use an uncertainty-based criterion. A fit accepted within a 95% region is not thereby shown to reproduce every printed reference digit.

Recovery experiments

In a recovery experiment, a known model generates the observations and BestFit estimates that model from those observations. Random sampling means that an estimate ordinarily differs from its generating value. The tests therefore assess the difference relative to its uncertainty. For directly estimated maximum-likelihood parameters, a common rule is

$$\left| \frac{\hat{\theta}_j - \theta_{0j}}{\text{SE}(\hat{\theta}_j)} \right| \leq 1.96. \quad (\text{M.2})$$

The symbol $\hat{\theta}_j$ denotes the fitted parameter, θ_{0j} its generating value, and SE its estimated standard error. Thus equation (M.2) requires the difference to be no more than 1.96 standard errors. Other tests assess the entire parameter set using a joint likelihood-ratio region, or vary one parameter while refitting the others to obtain a profile interval. The chapter states which rule applies.

Bayesian recovery checks ordinarily require the generating value inside the central 95% posterior interval, together with the sampling diagnostics described below. Sometimes several combinations of parameters describe nearly the same response. In those cases the test can assess the engineering quantity directly, such as the 0.99 quantile or the discharge curve, rather than requiring every coefficient to be determined separately.

Sample size counts observations, not MCMC draws. For spatial recovery, 100 ten-site vectors contain 1,000 scalar observations but only 100 multivariate row/year likelihood contributions. For point processes, the observation period and event count are reported separately.

Interval coverage

Recovery and interval coverage answer different questions. Across B independent repetitions, let I_b indicate whether an interval contains its generating truth. Empirical coverage and its Monte Carlo standard error at a nominal target C_0 are

$$\hat{C} = \frac{1}{B} \sum_{b=1}^B I_b, \quad \text{MCSE}(\hat{C}) = \sqrt{\frac{C_0(1-C_0)}{B}}. \quad (\text{M.3})$$

Here $\hat{C}$ is the observed fraction of successful inclusions, and MCSE describes its sampling uncertainty. For example, testing one generated dataset and finding its true value inside a 95% interval is a recovery result. Showing that such intervals include the truth about 95% of the time requires many independently generated datasets. This distinction applies to the Bulletin 17C, rating-curve, and spatial results in this report.

Bayesian diagnostics

Bayesian estimation uses Markov chain Monte Carlo (MCMC) to sample parameter uncertainty. Several chains explore the same posterior. The convergence statistic $\hat{R}$ compares variation within and between chains; a value close to one is desirable. Effective sample size (ESS) estimates how much information remains after accounting for dependence between successive draws. A long chain can therefore have a small ESS.

The usual recovery requirements are $\hat{R} < 1.10$ and ESS of at least 100 for every monitored parameter. These diagnostics support interpretation of the posterior intervals but do not prove model suitability. The diagnostic calculations themselves are compared with the independent R package `posterior`. Predictive checks based on leave-one-out calculations are compared with R `loo`; the estimation chapter explains both their numerical results and their reliability warnings.

Reproducibility and report coverage

Independent data and numerical references are stored beneath `verification/data/` with generators, environments, parameter conversions, and hashes. The verification catalog (source baseline; current snapshot attached) records method identities, scientific references, acceptance rules, and evidence locations. Appendix A maps every current test method to a report chapter. The PDF also embeds the current catalog and document metadata as attachments, retaining the complete acceptance and provenance record with the report.

The test index is a traceability aid. The analysis chapters carry the scientific explanation; a reader need not understand a test name or open a source file to understand the experiment. Separate software regression tests check data preservation, saved projects, and error handling. Documentation checks protect the report's links, metadata, and code descriptions. These controls support the analysis workflow and are distinguished from the numerical comparisons reported here.

CHAPTER 04

Time-Series Data

Test objective

Time-series data are measurements with dates or times, such as daily discharge or an irregular sequence of stage measurements. Before fitting a model, the software must preserve the values, their timing, missing observations, and the units assigned to them. A missing daily value must not become zero, and an empty file block must not hide observations stored later in the record.

Test and result

The tests construct short records with known values and dates, import records from HEC-DSS, and save and reopen project data. Each operation is followed by a direct comparison with the supplied values and the expected dates. Additional cases use long gaps and records spread over several storage blocks to check that the import preserves the complete series.

Test setup	Procedure and expected result	Result
Three daily values: 1, a DSS missing-value marker, and 3	Import the series. Preserve 1 and 3 and represent the middle day as missing. Four supported missing-value markers are checked.	Passed; valid values agree within 10^{-12} and the missing day remains identifiable.
Three daily values: 0, 1, and 0	Import using the same path as the missing-value examples.	Passed; both zeros remain valid measurements.
Daily period-average values of 1,950	Convert the DSS end-of-period timestamp at midnight on 2 October 1905 to the represented day, 1 October 1905.	Passed; the value is unchanged and assigned to the correct day.
Regular and irregular records with omitted times	Fill absent expected steps in a regular record with missing values; preserve only the supplied timestamps in an irregular record.	Passed; the two types of record retain their different time meanings.
Sparse records, empty blocks, and adjacent storage boundaries	Read all relevant blocks and compare the complete sequence of timestamps and values.	Passed; later data remain present and boundary observations occur once.
Imported data saved and reopened	Compare values, gaps, interval information, and period conversion after reopening.	Passed; project persistence preserves the tested record.

Malformed input also receives deliberate checks. Duplicate timestamps, inconsistent array lengths, and conflicting blocks are rejected instead of being returned as a successful partial import. User-assigned labels and plot settings are checked separately when copying or reopening a series.

These are deterministic software checks: the expected answer is the original supplied record or a prescribed timestamp conversion. They passed in the software regression suites and are not included in the 329-method numerical-verification inventory.

Preserving these inputs supports the statistical comparisons in the time-series analysis chapter. Network availability, provider revisions, station quality, and the analyst's treatment of missing values remain outside these checks.

CHAPTER 05

Input Data

Test objective

Input-data processing prepares the observations used by frequency and distribution analyses. These may be entered directly, obtained from an agency record, extracted as annual or seasonal block values, or selected as peaks above a threshold. The software must preserve both the values and what is known about them. An exact measurement, a flood known only to lie between two bounds, and a year known to remain below a perception threshold contribute different information.

Test and result

The tests construct controlled input records, apply the selected processing settings, save and reopen them, and compare the resulting observations and settings with the expected values. Exact, interval, threshold, and uncertain observations are checked as separate data types. Changes to an input must also identify the project as modified so that an analyst is not unknowingly relying on an outdated saved state.

An important example concerns incomplete years. Selecting an annual maximum from a partial daily record can understate the largest event that actually occurred. The following tests assess whether BestFit warns the analyst about that condition; they do not fill in the unobserved flows.

Controlled source record	Processing setup	Expected and observed result
All 365 daily values from 1 January through 31 December 2021	Calendar-year blocks	No incomplete-record warning. Passed.
334 daily values beginning 1 February 2021	Calendar-year blocks	Warning for the incomplete first year. Passed.
426 daily values beginning 1 October 2020	Water-year blocks starting in October	Warning for the incomplete final water year. Passed.
Full 2021 calendar with one explicitly missing value	Calendar-year blocks	Warning for missing coverage within the year. Passed.
Full 2021 calendar with one omitted daily timestamp	Calendar-year blocks	Warning for the gap, even though no missing-value marker was supplied. Passed.

The warning cases remain valid input objects; the result is a warning to support the analyst's decision, not automatic removal of the year. Switching to manual observations clears a warning that applies only to block extraction.

Other tests distinguish a calculated low-outlier threshold from one chosen by the analyst. When derived data are cleared, the calculated threshold is reset; a user-defined threshold of 25 remains 25. Saving and reopening preserves the tested observations, plotting positions, and custom axis titles. These checks and the deterministic block and peak calculations passed in the software regression suites.

These checks establish preservation of supplied information. Later chapters test its statistical use through threshold and interval floods, mixed observation types, paired stage-discharge data, and missing-site observations. Input processing does not establish stationarity, independence, record completeness, correct perception thresholds, or application suitability; those remain the analyst's responsibility.

CHAPTER 06

Estimation, Model Comparison, and Diagnostics

Verification objectives

An estimate is useful only if the software combines the data and prior information correctly and reports uncertainty on the intended scale. This chapter asks three practical questions: does estimation reproduce a known answer, do the model-comparison statistics use the right information, and do the diagnostics identify unreliable sampling?

Maximum likelihood estimation (MLE) selects the parameters that best explain the observations. Maximum a posteriori estimation (MAP) also incorporates prior information and selects the highest point of the posterior. Bayesian sampling describes the whole posterior, including its uncertainty. The generalized method of moments (GMM) chooses parameters that satisfy specified relationships between model and sample quantities, with weighting when those relationships cannot all be met exactly.

MLE, MAP, and GMM worked comparison

Test setup and procedure

Seven positive observations are chosen so that their base-10 logarithms are 1.1, 1.4, 1.7, 2.0, 2.3, 2.6, and 2.9. Their log-space mean is exactly 2 and the sum of squared deviations from that mean is 2.52. This small dataset allows the reference estimates to be calculated directly, without fitting software.

The test fits the same observations by MLE, flat-prior MAP, and Bulletin 17C GMM. The flat-prior comparison explicitly disables the optional Jeffreys scale prior: it asks whether MAP reproduces MLE when the prior contributes no preference within the allowed range. Exact likelihood inference divides by $n = 7$, while the Bulletin 17C sample-moment calculation divides by $n - 1 = 6$:

$$\hat{\sigma}_{\text{MLE}}^2 = 0.36, \quad \hat{\sigma}_{\text{GMM}}^2 = 0.42. \quad (\text{E.1})$$

Results and interpretation

Calculation	Independent target in log space	Recorded result and meaning
MLE and flat-prior MAP mean	2.000000	Both fitted points satisfy the analytical 95% joint likelihood-ratio and standard-error checks. Their data log likelihoods agree within 10^{-8} .
MLE and flat-prior MAP standard deviation	0.600000	The likelihood scale uses the seven-observation divisor.
GMM mean	2.000000	Within 10^{-5} of the sample-moment answer.
GMM standard deviation	0.648074	Within 10^{-5} of the sample-moment answer; the moment discrepancy is below 10^{-10} .

The different standard deviations are the intended consequence of the two finite-sample formulas. They are not evidence of an inconsistent logarithm or a failed fit. All comparisons passed under their respective rules.

Adding prior information

The next comparison adds Normal information about the mean. Its center is m and its standard deviation is τ . Four cases distinguish a change in the best estimate from a change in uncertainty. The widths below are expressed relative to the unpenalized standard error of the mean, which is about 0.2268 for MLE and 0.2449 for GMM.

Prior or equivalent GMM penalty	Center and width	Behavior tested
Flat	No preference over the fitted region	Reproduce the unpenalized estimate and uncertainty.
Wide and centered	Center 2; standard deviation 100 times the unpenalized standard error	Add very little information.
Narrow and centered	Center 2; standard deviation 0.1 times the unpenalized standard error	Reduce uncertainty without moving the preferred location.
Narrow and shifted	Center two unpenalized standard errors above 2; same narrow width	Move the estimate toward the prior and change its uncertainty.

For each case, BestFit refits both mean and scale. Independently derived equations and the curvature of the objective supply the reference fit and covariance. The tests also compare the fitted interval with the familiar fixed-variance information-combination reference,

$$V_* = (V_L^{-1} + \tau^{-2})^{-1}, \quad \mu_* = V_*(\hat{\mu}_L/V_L + m/\tau^2). \quad (\text{E.2})$$

where $\hat{\mu}_L$ and V_L are the unpenalized location and its variance. More precise information receives more weight. Because the scale is reestimated, equation (E.2) is an interval reference for this experiment, not an exact identity for the final fitted location.

The four-regime comparisons passed. MAP fits satisfy the analytical joint 95% criterion; GMM location errors satisfy the 1.96-standard-error criterion. Covariance entries agree with the independent calculations within the larger of 2×10^{-5} and 0.2% of the reference magnitude. Equation (E.2)'s location lies within every tested central 95% interval. These results check both the direction of prior influence and its effect on reported uncertainty.

Recovery and informative priors

Recovery from generated data

Three experiments isolate the general estimation routines. Each contains 1,000 observations or paired rows. The distribution chapter gives the additional family-specific recovery designs.

Estimation experiment	Test setup and procedure	Result
Bayesian Normal analysis	Generate a Normal sample with mean 100 and standard deviation 15, using seed 12345; estimate both parameters with the production Bayesian configuration.	Both generating values are inside the central 95% posterior intervals; each monitored parameter has $\hat{R} < 1.10$ and ESS at least 100.
Normal MAP	Generate mean 100 and standard deviation 10; fit the posterior mode and calculate uncertainty from its curvature.	Both generating values are within 1.96 estimated standard errors of the fitted values.
Two-step GMM	Generate responses with mean 2 and standard deviation 1, paired with independent Uniform(-1,1) auxiliary values, using seed 24680. Fit the two conditions that the mean residual and the mean auxiliary-value-weighted residual are zero.	The generating mean 2 is within 1.96 standard errors, using the two-step sandwich covariance.

The GMM covariance calculation combines the sensitivity of the moment equations with their observed variability; this is why it is called a sandwich covariance. The known mean is used only to generate and assess the data, not supplied as the fitted answer.

Thirteen-family MLE recovery checks complement these general experiments. Most compare generating values with standard-error intervals. Generalized Normal and Generalized Logistic use 95% profile intervals; Generalized Pareto additionally checks its 0.99 quantile. A separate Ln-Normal calculation compares the fitted result with the exact natural-log mean and standard deviation of the same sample. All meet their specified criteria. Parameter tables and the distinction between original and logarithmic scales appear in the distribution chapter.

An exact Bayesian posterior

A separate experiment uses 1,000 observations generated with mean 100 and standard deviation 15, but treats the observation standard deviation as known. The prior for the unknown mean is Normal with center 120 and standard deviation 5. The intentionally displaced prior tests whether the data and prior are combined with the correct relative weights.

For this known-scale case, equation (E.2) gives an exact posterior: use the sample mean for $\hat{\mu}_L$, $V_L = 225/1000$, $m = 120$ and $\tau^2 = 25$. The posterior standard deviation is 0.472221, and the prior receives about 0.892% of the combined location weight. The data dominate because the 1,000-observation mean is much more precisely determined than the prior.

The test samples the posterior and compares its mean, standard deviation, and two central 95% endpoints with this exact Normal answer. The accepted absolute differences are 0.05, 0.03, and 0.10, respectively; all passed. The result verifies the shape and width of a posterior with informative prior input, not merely whether an interval happens to contain the generating mean.

Profile likelihood and covariance

A profile calculation varies one parameter and refits the other parameters at each step. This matters when parameters are related: holding the other parameters fixed can make an interval too narrow.

The reference example uses a smooth two-parameter quadratic log likelihood with correlation 0.8 and an optimum at zero. At a fixed first parameter t , the independently calculated optimum for the second is $0.8t$, and the profile log likelihood is $-t^2/2$. R `bbmle` supplies an independent numerical calculation at eight equally spaced midpoints from -3.5 to 3.5.

Quantity	Independent reference	BestFit comparison
Profile at $t = 0.5$	-0.125	Agrees within 10^{-10} .
Profile at $t = 1.5$	-1.125	Agrees within 10^{-10} .
Profile at $t = 3.5$	-6.125	Agrees within 10^{-10} .
90% profile interval	-1.644854 to 1.644854	Both endpoints agree within 10^{-8} .
Interval if the second parameter were incorrectly held fixed	-0.986912 to 0.986912	Comparison illustrates the narrower conditional slice; it is not the reported profile interval.

All eight profile values, the optimum, and the interval passed for MLE and for MAP with a flat prior. An informative-prior variant checks that MAP refits using the complete posterior, and a restricted-support variant checks ten valid points on a 16-point grid. Invalid points are not included in the numerical agreement claim. The 90% level is part of this particular external comparison; it is not the 95% recovery rule used elsewhere.

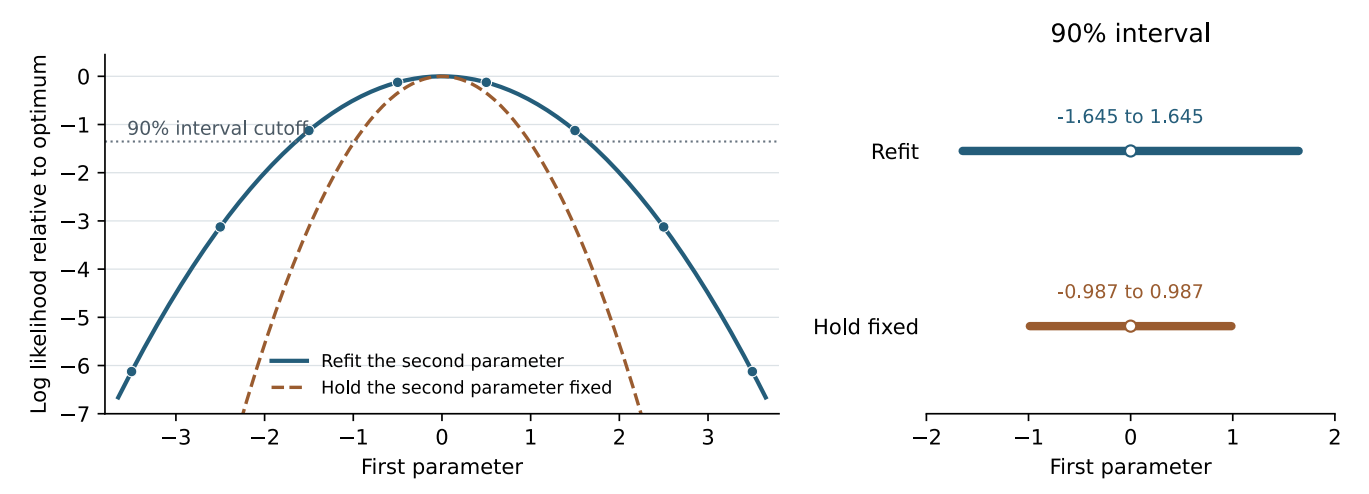


Figure E.1. Allowing the second parameter to adjust produces the wider profile interval. The curves and points show the independent analytical and R references; BestFit's tested profile values agree within 10^{-10} . They are not plotted as separately retained BestFit estimates.

Covariance is checked independently on ten Normal observations: 1.2, 0.7, 1.9, 1.1, 0.4, 1.6, 0.9, 1.3, 1.0, and 1.5, with known standard deviation 0.5. The reference mean is 1.16, its variance is $0.5^2/10 = 0.025$, and its standard error is 0.158114. The fitted mean satisfies the standard-error criterion, and both uncertainty quantities agree within 10^{-8} relative error. A separate two-parameter quadratic confirms that adding a flat prior leaves the covariance unchanged within 10^{-4} . These checks passed.

DIC and WAIC

The deviance information criterion (DIC) and widely applicable information criterion (WAIC) balance fit against model complexity. They help compare candidate models fitted to the same data; a small value by itself does not establish that a model is physically appropriate.

The setup contains five Normal observations, 2.4, 2.9, 3.5, 4.1, and 4.8, and forty fixed candidate pairs of mean and standard deviation. Using the same pairs in both calculations removes random sampling differences. BestFit and R evaluate how well each pair explains each observation, producing 200 individual log-likelihood values. The tests compare those values first and then compare the summary statistics. The independent packages are R `BayesianTools` 0.1.9 and `loo` 2.10.0.

Quantity	External result	BestFit tolerance	Result
Mean deviance	13.0457484595105	$1e-10$ absolute	Passed
Deviance at posterior mean	12.6377539947693	$1e-10$ absolute	Passed
Effective DIC parameters	0.407994464741220	$1e-10$ absolute	Passed
DIC	13.4537429242518	$1e-10$ absolute	Passed
lppd	-6.35851654848624	$1e-10$ absolute	Passed
Effective WAIC parameters	0.351001866165060	$1e-10$ absolute	Passed
WAIC	13.4190368293026	$1e-10$ absolute	Passed

All individual and summary comparisons passed. Here deviance is minus twice the log likelihood; lppd is the sum of log predictive densities for the observed data. The effective-parameter terms measure the complexity penalty, which need not be an integer. Agreement in both components establishes more than agreement in the final DIC or WAIC total.

Two additional checks fit 100 Normal observations generated with mean 100 and standard deviation 10. They independently recompute the Akaike information criterion (AIC) and Bayesian information criterion (BIC) at the MAP estimate. Both use the **data likelihood**, excluding the prior density, and agree within 10^{-10} . When a nonconstant prior moves the MAP away from the MLE, these MAP-evaluated numbers are not conventional MLE-based AIC and BIC.

PSIS-LOO

Leave-one-out cross-validation asks how well a model predicts an observation when that observation is left out of the fit. Pareto-smoothed importance sampling (PSIS) approximates this comparison by reweighting the available posterior draws. It also reports when those weights are too uneven for a reliable approximation.

The same five observations and forty fixed parameter pairs were passed through R `loo`. BestFit compares the prediction score for each omitted observation, its smoothed weights, the Pareto shape diagnostic k , and its effective sample size. ELPD means expected log predictive density; LOOIC is the same aggregate prediction score expressed as minus twice ELPD.

Quantity	R <code>loo</code> result	BestFit tolerance	Result
ELPD-LOO	-6.70094841652149	$1e-10$ absolute	Passed
Effective LOO parameters	0.342431868035246	$1e-10$ absolute	Passed
LOOIC	13.4018968330430	$1e-10$ absolute	Passed
SE(LOOIC)	1.95756902272540	$1e-10$ absolute	Passed

All aggregate comparisons passed. The five reference Pareto values are 0.0682, -0.0302, 0.4116, 0.3062, and 0.3209. With only forty draws, the applicable reliability threshold is 0.375804. The third observation therefore requires attention even though its k is below the commonly displayed 0.5 category boundary. BestFit identifies that observation correctly.

Six additional sets of 256 importance ratios test progressively heavier tails and a degenerate case. These represent increasingly uneven contributions from the posterior draws. Smoothed weights, k , and effective sample sizes agree with R within 10^{-8} ; the degenerate case correctly returns an infinite diagnostic instead of implying a reliable approximation.

Exact leave-one-out refits, moment matching, and chain-relative-efficiency adjustment are not implemented. The influence diagnostic compares the Pareto shape and the sample-size-dependent reliability threshold with R `loo`. This check is separate from the deterministic formatting and serialization of diagnostic results.

GMM specification and covariance

The GMM specification test asks whether two proposed moment relationships can both be satisfied by one parameter. Ten response values are 0.2, 0.5, 0.7, 1.1, 1.4, 1.8, 2.1, 2.6, 3.2, and 4.0. They are paired, in that order, with auxiliary values 0.5, 1.0, and so on through 5.0. The two sample moments are the average residual $x - \theta$ and the average weighted residual $z(x - \theta)$.

BestFit and R `gmm` 1.9.1 fit these same data twice. The first fit weights the two moment discrepancies equally. The second updates the weights using their covariance. For the first fit, the independent optimum and weighted discrepancy Q are

$$\hat{\theta} = 2.28992700729929, \quad Q = 0.317956204379562. \quad (\text{E.3})$$

For efficient two-step GMM, the reference values are $\hat{\theta} = 1.93548454750500$ and $Q = 1.00755078518454$. BestFit agrees within 10^{-5} for the parameter and 10^{-8} for the discrepancy. The Hansen statistic tests the remaining mismatch between the two moment conditions:

$$J = nQ = 10.0755078518454, \quad p = 0.00150253202968641. \quad (\text{E.4})$$

Comparison	R or analytical value	Tolerance	Result
Fixed-weight variance	0.145652392138090	1e-8	Passed
Efficient two-step variance	0.132600447299456	1e-8	Passed
Hansen J	10.0755078518454	1e-7	Passed
Hansen p-value	0.00150253202968641	1e-8	Passed

The variance and specification comparisons all passed. The small p-value indicates that these particular data are inconsistent with both moment conditions holding at once; a correct test should detect that disagreement. It does not indicate a software failure. The covariance reference is also reconstructed directly from the independent moment derivatives and moment variability.

BestFit does not label an arbitrary fixed-weight discrepancy as a Hansen test: the first fit's Hansen statistic and p-value are unavailable. The chi-square interpretation applies to the efficient second-step weighting used in the second fit.

Gradients and influence

An optimizer needs the direction in which its objective improves, called the gradient. A controlled GMM example uses the moment $g(\theta) = \theta^2 - 2$. The test changes the parameter by small positive and negative amounts and independently calculates the resulting slope. It verifies the full derivative when a penalty is present and the documented half-derivative convention for the unpenalized estimating equation. Both comparisons agree within 10^{-8} .

Influence diagnostics ask which observations most affect an estimate or its uncertainty. On the seven log-space observations used above, each observation's fit influence, variance influence, and combined leverage are compared with an independent R `gmm` calculation. Every value agrees within 10^{-5} . Adding an extremely weak centered penalty preserves these values within 10^{-4} , as expected when almost no information is added.

One influence check records an accepted limitation. Removing one value from a sample changes the sample mean by a known amount, so exact deletion provides a direct comparison. The test checks the seven-value sample and then adds an unusually high eighth value. BestFit and exact deletion both identify the added observation as most influential, but the diagnostic's magnitude differs from exact deletion by more than 25% on the tested scale. It can identify the leading observation in this example; it must not be interpreted as the exact amount the estimate would change if that observation were removed.

Joint-prior sampling boundary

Prior information can describe each parameter separately or describe how two parameters relate. To distinguish those cases, the test gives two parameters separate Uniform(-1,1) priors and also specifies a narrow conditional relationship requiring them to be close together.

The generic prior sampler produces 20,000 pairs. The check compares their behavior with independent Uniform draws: absolute correlation is below 0.03, mean squared separation exceeds 0.60, and fewer than 12% of pairs lie within 0.1 of one another. These findings agree with separate sampling; the independent mean squared separation is $2/3$.

This is the second accepted limitation. The additional joint relationship is not represented by these prior draws. A calculation that needs the full coupled prior, such as a corresponding prior-predictive analysis, cannot treat them as draws from that joint distribution.

MCMC diagnostics

Setup and procedure

Nine fixed collections of four chains test the diagnostic calculations themselves. Some represent well-mixed draws; others deliberately include a shifted chain, unequal spread, strong serial dependence, or a chain stuck in the tail. BestFit and R `posterior` 1.7.0 receive the identical draws. This isolates the diagnostic formula from the sampler that originally produced the values.

The tests compare rank-normalized split $\hat{R}$ directly. For ESS, R supplies bulk ESS for the main body of the distribution and tail ESS for its 5% and 95% quantiles. BestFit's single conservative ESS is compared with the smallest of those three reference values; the test does not independently verify three separate BestFit ESS outputs.

Chain design	Draws per chain used	R reference $\hat{R}$	R reference conservative ESS	Interpretation
Independent Normal draws	200	0.99873	774.55	Little loss of information to chain dependence.
One shifted chain	200	1.05158	95.16	The small ESS flags limited useful information.
Different chain spreads	100	1.28101	447.63	Chain disagreement is detected by $\hat{R}$.
Serial dependence of 0.85	240	1.04057	53.52	Many stored draws provide much less independent information.
One chain stuck in the upper tail	240	1.07719	11.70	Tail information is especially poor.
Rounded values with ties	160	1.00049	578.87	Repeated values receive consistent ranks.
Initial 80 warmup values discarded	160	0.99901	638.26	Diagnostics use the retained portion of each chain.
Repeating integer pattern	64	2.86020	5.25	Severe disagreement is detected.
Constant chains	40	Unavailable	Unavailable	No spurious convergence or ESS value is reported.

Every finite $\hat{R}$ agrees within 10^{-10} and every conservative ESS within 10^{-8} ; unavailable cases are also reproduced. A passing comparison for a deliberately poor chain means the diagnostic identified the problem correctly, not that the chain is acceptable for inference.

Sampler calculations

Two further tests examine specific sampler calculations. For the No-U-Turn Sampler (NUTS), a two-parameter posterior has an independently derived gradient. At parameter values 1.25 and -0.75, the expected gradient components are -1.50 and 0.50; both agree within 10^{-8} .

For adaptive random-walk Metropolis-Hastings (ARWMH), a controlled model rejects every proposal. A rejected proposal leaves the chain at its previous state, and that repeated state must still contribute to the adaptive covariance. The configured four-chain run requires 125 recorded transitions per chain; every acceptance count is zero and every covariance history contains exactly 125 states. Both sampler checks passed. Together with the recovery and diagnostic comparisons, they check behavior that would be missed by merely confirming that a sampling run finishes.

CHAPTER 07

Distribution Fitting, Univariate Analysis, and Bulletin 17C

Verification objectives

A flood-frequency analysis must do more than draw a smooth curve through observed floods. It must use the intended probability distribution, estimate its parameters consistently, and carry information about missing years, historical floods, and regional studies into the result. This chapter examines those steps for a single response variable, such as annual peak discharge.

Three kinds of comparison are used. Synthetic records test whether estimation recovers a known generating distribution. Independent analytical calculations and statistical packages test distribution formulas and fitted solutions. Published flood studies test complete combinations of systematic, historical, and prior information. Maximum likelihood estimation (MLE) selects the parameters that give the greatest likelihood to the observed data. Bayesian estimation also incorporates prior information and describes the remaining uncertainty with a posterior distribution.

A quantile $Q(p)$ is the value with probability p of not being exceeded. For an annual-maximum series, $Q(0.99)$ is the 1% annual exceedance probability (AEP), or 100-year, discharge. A 95% posterior interval contains the central 95% of the posterior distribution; a single synthetic experiment does not demonstrate 95% long-run coverage. Tables headed **reference** or **generating** contain comparison targets. Measured BestFit outputs are identified separately where available.

Distribution-family oracles

Stationary Bayesian family recovery

Setup. A stationary distribution has parameters that do not change through the record. Each of the fifteen families below supplies 1,000 simulated scalar observations, with random seed 12345. The sample is fitted using the matching Bayesian model and the production DEMCzs sampler settings. The tests request posterior-mode point estimates and central 95% intervals, retain the model's prior construction and parameter bounds, and explicitly enable the Jeffreys scale rule in the LP3 case. A Jeffreys scale rule assigns prior weight inversely proportional to scale.

Synthetic response magnitudes have no assigned physical unit. Location, mean, scale, and standard deviation are expressed in those response units, except where the table identifies logarithms. Shape and skewness are dimensionless. **Ln-Normal is specified here by the mean and standard deviation of the original positive response**, despite its use of natural logarithms internally. Log-Normal and Log-Pearson Type III (LP3) instead use the mean and standard deviation of the base-10 logarithm of the response. Scale is a distribution parameter and need not equal standard deviation.

Family	Generating parameters: Bayesian recovery	Generating parameters: MLE and distribution-fitting recovery
Normal	Mean 100; standard deviation 15	Mean 100; standard deviation 15
Ln-Normal	Response mean 4.5; response standard deviation 0.4	Response mean 3.5; response standard deviation 0.4
Log-Normal, base 10	Log mean 3; log standard deviation 0.5	Log mean 3; log standard deviation 0.5
Exponential	Location 0; scale 50	Location 10; scale 25
Gamma	Scale 5; shape 2	Scale 5; shape 3
Generalized Extreme Value (GEV)	Location 100; scale 20; shape -0.1	Location 50; scale 15; shape 0.1
Generalized Logistic	Location 100; scale 15; shape 0.1	Location 75; scale 10; shape 0.15
Generalized Normal	Location 100; scale 15; shape 0.1	Location 50; scale 10; shape -0.3
Generalized Pareto	Location 0; scale 50; shape 0.1	Location 0; scale 20; shape 0.15
Gumbel	Location 100; scale 20	Location 50; scale 15
Kappa Four	Location 100; scale 20; first shape -0.1; second shape 0.1	Location 100; scale 20; first shape -0.1; second shape 0.1
Logistic	Location 100; scale 10	Location 75; scale 10
Pearson Type III	Mean 100; standard deviation 20; skewness 0.5	Mean 100; standard deviation 20; skewness 0.8
Log-Pearson Type III	Log mean 3; log standard deviation 0.5; log skewness 0.2	Log mean 2; log standard deviation 0.3; log skewness 0.5
Weibull	Scale 100; shape 2.5	Scale 100; shape 2.5

Procedure. The known generating parameter must lie in its central 95% posterior interval. Each sampled parameter must also have $\hat{R} < 1.10$, which checks agreement among sampling chains, and effective sample size (ESS) at least 100, which accounts for dependence among draws. The zero-location Exponential and Generalized Pareto cases assess that boundary through the central 95% interval for $Q(0.99)$; their remaining parameters are checked directly. A secondary point-estimate rule applies only when a nonzero parameter's entire 95% interval is narrower than 5% of its generating magnitude; in that situation, point-estimate error must also be at most 5%.

Results. All fifteen recovery methods are verified. This supports recovery of the declared distributions, including positive data, skewed tails, and four-parameter flexibility, under these fixed samples and settings. The evidence supplies acceptance outcomes rather than a complete table of fitted posterior estimates and interval endpoints. These experiments do not establish recovery for every parameter combination or record length.

Distribution-fitting and MLE recovery

Setup and procedure. The right-hand column above describes fifteen additional samples, each with 1,000 observations and seed 12345. Each sample is submitted to the default Distribution Fitting analysis. The generating family must be present in the candidate list and fit successfully; the experiment then checks its estimated parameters against the known values. It does not require that the generating family rank first among all candidates.

Where an analytical parameter covariance is available, a parameter must be within 1.96 estimated standard errors of its generating value. A standard error measures uncertainty in an estimated parameter, rather than the spread of the observations. Generalized Logistic, Generalized Normal, and Kappa Four use 95% profile intervals from an additional likelihood fit: each interval must contain both the known parameter and the Distribution Fitting estimate. A profile interval allows the other parameters to adjust while the parameter under examination is varied. Ln-Normal comparisons are made in natural-log mean and variance; Pearson III and LP3 use the equivalent mean, inverse-scale, and shape coordinates appropriate to their covariance calculation. Generalized Pareto checks scale and shape directly and checks its zero-location direction through a 95% band for $Q(0.99)$. The conditional 5% point rule described above also applies to these fitting checks.

Results. All fifteen Distribution Fitting recovery methods are verified. A separate direct-MLE series uses the same right-hand parents for thirteen families: all except base-10 Log-Normal and Kappa Four. Those thirteen recovery checks, plus a fourteenth check of Ln-Normal against the closed-form fit to the same log-transformed sample, are also verified. Together they test the fitting workflow and the estimator directly. They support recovery within the stated uncertainty rules, not exact equality to generating parameters or a guarantee that model-selection criteria identify the true family.

Nonstationary univariate recovery

Setup. A nonstationary model allows a distribution parameter to vary with a covariate. These seven recovery designs use the observation index $t = 0, 1, \dots, 999$ as that covariate, 1,000 simulated responses, and seed 12345. The index is a dimensionless position in the record, not a calendar year. The Bayesian procedure uses the same production sampling settings and 95% interval convention as the stationary tests. In the table, μ denotes mean, σ standard deviation, ξ location, α scale, and κ shape.

Design	Known distribution and trend	What the comparison tests
Constant Normal	$\mu = 100, \sigma = 15$	A constant trend reproduces the stationary parameter-recovery problem.
Normal with reciprocal mean	$\mu(t) = 1/(0.01 + 0.00001t); \sigma = 15$	The mean declines from 100 to about 50 while observation variability stays fixed.
Normal with reciprocal standard deviation	$\mu = 100; \sigma(t) = 1/[1/15 + (1/10 - 1/15)t/999]$	Standard deviation declines from 15 to 10 without a change in mean.
Normal with sinusoidal standard deviation	$\mu = 100; \sigma(t) = 15 + 3\sin(2\pi 0.01t + 0.3)$	Variability cycles between 12 and 18 with a period of 100 index units.
GEV with linear shape	$\xi = 50, \alpha = 15; \kappa(t) = 0.05 + 0.0001t$	Tail shape changes while location and scale remain fixed.
Generalized Pareto with changing location and scale	$\xi(t) = 10 + 0.02t; \alpha(t) = 20 \exp(0.0005t); \kappa = 0.1$	The threshold location and spread change together.
LP3 with changing log mean and log standard deviation	$\mu(t) = 3 + 0.0002t; \sigma(t) = 0.2 \exp(0.0002t); \log \text{ skewness } 0.2$	Both the level and variability of base-10 log discharge change.

Procedure and results. Constant parameters are checked for central 95% posterior inclusion. Changing parameters are checked as predicted responses at $t = 0, 499, 999$; the sinusoidal design instead uses $t = 0, 25, 50, 75, 999$ to sample the cycle. This checks the physically interpretable curve when different combinations of coefficients can describe similar responses. All sampled coefficients must meet $\hat{R} < 1.10$ and ESS at least 100, and the conditional 5% rule applies to resolved parameters or responses. All seven recovery designs are verified. They establish recovery at the declared response points, not simultaneous coverage of every point on a continuous trend.

An eighth, analytical check examines the reciprocal formula itself. With starting index 10, coefficients 0.02 and 0.001, and evaluation index 510, the response is $1/0.52 = 1.923076923 \dots$. Its derivatives with respect to the intercept and slope are $-1/0.52^2$ and $-500/0.52^2$. The response agrees within 10^{-12} ; centered numerical derivatives using perturbation 10^{-7} agree within 10^{-6} and 10^{-4} , respectively. This verified calculation checks the trend formula independently of sampling.

Independent distribution functions and fitted solutions

Setup. Each family also has a deterministic sample comprising its quantiles at 39 evenly spaced nonexceedance probabilities, from 0.025 to 0.975. The generating distributions are specified below. These are fixed calculation examples, not random samples. SciPy 1.18.1 supplies thirteen independent fitted solutions; R 4.4.3 with Imomco 2.5.7 supplies Generalized Logistic and Generalized Normal. All synthetic magnitudes are in arbitrary response units.

Family	Distribution used to construct the 39 observations	External fitted $Q(0.9)$ reference, rounded
Normal	Mean 100; standard deviation 15	117.669110
Ln-Normal	Response mean 100; response standard deviation 30	135.351898
Log-Normal, base 10	Log mean 2; log standard deviation 0.35	258.392097
Exponential	Location 5; scale 20	48.307450
Gamma	Scale 12; shape 3.5	68.821461
GEV	Location 100; scale 20; shape -0.15	148.337810
Generalized Logistic	Location 75; scale 10; shape 0.15	92.244444
Generalized Normal	Location 50; scale 10; shape -0.3	64.255965
Generalized Pareto	Location 10; scale 5; BestFit shape -0.2	23.211643
Gumbel	Location 100; scale 20	141.590460
Kappa Four	Location 100; scale 20; first shape -0.1; second shape 0.2	148.525342
Logistic	Location 100; scale 15	130.338836
Pearson Type III	Mean 100; standard deviation 20; skewness 1	124.719447
Log-Pearson Type III	Log mean 2; log standard deviation 0.3; log skewness 0.8	233.819015
Weibull	Scale 100; shape 2.2	140.744883

Procedure. BestFit and the external implementation fit the same observations. Their maximum log likelihoods are compared using a joint 95% likelihood-ratio region, which assesses all fitted parameters together: twice the absolute log-likelihood difference must not exceed 5.991465, 7.814728, or 9.487729 for two, three, or four parameters. At the same external parameter values, probability density, cumulative distribution function (CDF), quantile, and likelihood calculations must agree within 10^{-8} absolute plus 10^{-7} relative tolerance. The CDF is the probability of a value at or below the specified magnitude. These formula checks are much tighter than the uncertainty-based comparison between separately fitted optima.

Parameter conventions are reconciled before comparison. Gamma and Weibull reverse the external shape/scale ordering; Ln-Normal converts log parameters into original-response moments; LP3 includes the base-10 transformation in its density. BestFit's GEV shape matches SciPy's GEV shape and has the opposite sign to the common Coles convention. BestFit's Generalized Pareto shape has the opposite sign to SciPy's Pareto shape. The free-location Pareto optimum is on a support boundary, so its likelihood-ratio rule is a declared comparison screen rather than proof of an exact 95% confidence region.

Results. All fifteen external comparisons are verified. The displayed quantiles are independent targets, with acceptance evaluated at full precision. Additional Kappa Four checks use location 0, scale 1, first shape 0, and second shape 0.2: the density at response 1 is independently 0.2709847912611985, and analytical CDF inversion returns response 1, both at absolute tolerance 10^{-10} . Supporting finite-shape checks use first shapes ± 0.25 , second shapes ± 0.5 , and probabilities 10^{-6} , 0.1, 0.5, 0.9, 0.999999 to verify increasing quantiles, valid support, and CDF inversion. These checks distinguish distribution validity from the separate question of whether particular moments exist.

Published real-data likelihood comparisons

Setup. Fifteen fits use observed river flows or wind speeds. The Indiana discharge records come from Rao and Hamed (2000): Tippecanoe River near Delphi, Wabash River at Lafayette, White River near Nora, Sugar Creek at Crawfordsville, White River at Mount Carmel, and East Fork White River at Seymour. Their discharge unit is cubic feet per second (ft^3/s). Harricana River at Amos, Quebec, comes from Bobée and Ashkar (1991) and uses cubic metres per second (m^3/s). The New York wind record consists of 153 daily values from May–September 1973 in the R airquality data, in miles per hour (mph).

Fitted family	Record and observation count	Reference parameters, with labels
Normal	Tippecanoe; 48 annual peaks	Mean 12,665.21 ft^3/s ; population standard deviation 4,660.42 ft^3/s , calculated from the observations
Ln-Normal	Wabash; 85 annual peaks	Natural-log mean 10.716952; natural-log standard deviation 0.447419, calculated from the observations and converted to response moments before comparison
Log-Normal, base 10	Wabash; 85 annual peaks	Log mean 4.654313; log standard deviation 0.194311, calculated from the observations
Exponential	Wabash; 85 annual peaks	Location 13,100 ft^3/s ; scale 36,122.35 ft^3/s
Gamma	Harricana; 69 annual peaks	Scale $1/0.08833 \text{ m}^3/\text{s}$; shape 16.89937
Pearson Type III	Harricana; 69 annual peaks	Mean 191.31739 m^3/s ; standard deviation 47.01925 m^3/s ; skewness 0.61897
Log-Pearson Type III	Harricana; 69 annual peaks	Base-10 log mean 2.26878; log standard deviation 0.10621; log skewness -0.02925
Gumbel	Sugar Creek; 53 annual peaks	Location 8,049.6 ft^3/s ; scale 4,478.6 ft^3/s
Weibull	Tippecanoe; 48 annual peaks	Scale 14,196.9496 ft^3/s ; shape 2.9829, from the retained R-Stan weak-prior comparison
GEV	White River near Nora; 62 annual peaks	Location 10,849 ft^3/s ; scale 5,745.6 ft^3/s ; shape 0.005
Generalized Pareto	Mount Carmel; 287 peaks above 50,000 ft^3/s	Location 50,400 ft^3/s ; scale 55,142.29 ft^3/s ; shape 0.0945

Fitted family	Record and observation count	Reference parameters, with labels
Logistic	Tippecanoe; 48 annual peaks	Location 12,628.59 ft ³ /s; scale 2,708.64 ft ³ /s
Generalized Logistic	East Fork White River; 68 annual peaks	Location 30,911.83 ft ³ /s; scale 9,305.0205 ft ³ /s; shape -0.144152
Generalized Normal	New York wind; 153 daily values	Location 9.7285364 mph; scale 3.4885029 mph; shape -0.1307169
Kappa Four	New York wind; 153 daily values	Location 8.68360234 mph; scale 3.10384972 mph; first shape 0.14470737; second shape -0.07348014

Procedure and results. Each BestFit MLE must complete, and the reference point must satisfy the joint 95% likelihood-ratio limit for its parameter count. All fifteen checks are verified. The Normal and log-Normal reference standard deviations use divisor n , as required by the closed-form MLE, rather than the sample-statistic divisor $n - 1$. The wind references originate in an R L-moment calculation, and the Weibull reference uses a weak-prior R-Stan calculation; their acceptance supports consistency with those reference points, not identity between different estimation methods. The East Fork example also retains the documented difference between textbook summary statistics and the tabulated observations. The independent-package tests above provide the separate fitted-optimum evidence.

Log10-Normal closed-form test

Setup. Seven positive observations are defined by base-10 logarithms 1.1, 1.4, 1.7, 2.0, 2.3, 2.6, and 2.9. This small example makes both the transformation and the likelihood normalization independently calculable. The mean and population standard deviation of the transformed sample are

$$\hat{\mu} = 2, \quad \hat{\sigma} = 0.6. \quad (\text{D.1})$$

Procedure and results. The original-response log likelihood must include the transformation term $-\sum_i \log(y_i \ln 10)$. Omitting it could leave a plausible-looking fit while producing the wrong likelihood for model comparison. The analytical comparison is verified under the following rules.

Quantity	Analytical reference	Required agreement
Log mean	2	Fitted error at most 1.96 standard errors, using $0.6/\sqrt{7}$
Log standard deviation	0.6	Fitted error at most 1.96 standard errors, using $0.6/\sqrt{14}$
Joint fitted point	The mean and scale above	Twice the likelihood difference at most 5.991464547107979
Original-response log likelihood at the analytical point	-44.431208784723104	Absolute error at most 10^{-10}
CDF at the median response, 100	0.5	Absolute error at most 10^{-12}
$Q(0.9)$ at the analytical point	587.3959385303014	Absolute error at most 10^{-9}

The fitted $Q(0.9)$ is also checked using its 95% band obtained by propagating parameter covariance into quantile uncertainty. This checks the practical response as well as the log-space parameters.

Common-data fitting test

Setup and procedure. Gumbel, Normal, and Logistic are fitted to the same 39-point Normal quantile sample, generated with mean 100 and standard deviation 15 at probabilities 0.025-0.975. This asks whether BestFit calculates and ranks competing fits consistently. The Akaike information criterion (AIC) and Bayesian information criterion (BIC) combine likelihood with a penalty for parameter count; smaller values are preferred. Root mean squared error (RMSE) measures the discrepancy between the ordered observations and fitted quantiles; its parameter-adjusted form is illustrated below.

Candidate	SciPy reference AIC	SciPy reference BIC	SciPy reference RMSE	Reference inverse-squared-RMSE weight
Normal	319.329424	322.656548	1.144404	0.459581
Logistic	321.099336	324.426459	1.122088	0.478043
Gumbel	323.836960	327.164084	3.106353	0.062376

Results. The verified checks require AIC ranking Normal, Logistic, Gumbel, and RMSE ranking Logistic, Normal, Gumbel, while retaining the configured display order Gumbel, Normal, Logistic. The differing rankings are expected because likelihood and quantile residuals measure different aspects of fit. At common parameter values, likelihood, AIC, and BIC agree within 10^{-8} absolute plus 10^{-7} relative tolerance. The independently recomputed criteria/RMSE and normalized weights proportional to $1/\text{RMSE}^2$ agree with BestFit within 10^{-10} and 10^{-12} , respectively.

One numerical comparison documents the small difference between independently optimized Gumbel fits. SciPy returned location 93.1123479994 and scale 13.1576285680; BestFit returned location 93.1129321295 and scale 13.1578232496. Their RMSE values were 3.1063533062 and 3.1065229843. The points satisfy the joint 95% likelihood comparison. This illustrates why agreement is assessed through the objective and uncertainty rule instead of demanding identical optimizer coordinates.

Parameter-adjusted RMSE

With four residuals of 1, 2, 3, and 4 and one fitted parameter, the independent reference is

$$\text{RMSE} = \sqrt{\frac{1^2 + 2^2 + 3^2 + 4^2}{4 - 1}} = \sqrt{10} = 3.1622776601683795. \quad (\text{D.2})$$

BestFit reproduced this value within 10^{-12} . Reordering the observed/fitted pairs together leaves the answer unchanged. A separate verified plotting-position check uses the 39 observations to recover the analytical probabilities $i/40$ for ranks $i = 1, \dots, 39$, within 10^{-12} . These checks establish the arithmetic and data ordering needed to interpret the common-data fitting comparison.

Bulletin 17C published examples

Setup. The seven official examples test the specialized generalized method of moments (GMM) route for flood-frequency analysis. GMM estimates parameters by matching selected data moments, with the weighting appropriate to this method. Each example fits LP3 to discharges expressed in ft^3/s , with parameters defined on the base-10 logarithm of those numeric discharge values. A perception threshold records the magnitude above which a flood would have been noticed during a stated period. An interval observation gives bounds on a flood whose exact magnitude is unknown. Low-outlier censoring retains information that a small observation falls below a threshold. Examples 1, 2, 4, 6, and 7 use the Multiple Grubbs-Beck Test (MGBT) for low-outlier screening.

Example and site	Record and analysis setup
1. Moose River at Victory, Vermont	68 annual peaks, 1947-2014; systematic-record example with the prescribed low-outlier processing.
2. Orestimba Creek near Newman, California	82 annual peaks, 1932-2013, including zero-flow years; low flows are handled by the prescribed censoring procedure.
3. Back Creek near Jones Springs, West Virginia	56 observed annual peaks, including 22,000 ft ³ /s in 1936; threshold periods 1932-1938, 1976-1992, and 1999-2003 use 21,000 ft ³ /s; manual low-outlier threshold 2,000 ft ³ /s.
4. Arkansas River at Pueblo, Colorado	81 systematic peaks in 1895-1976 plus intervals for 1864, 1893, 1894, and 1921; historical thresholds extend to 1165.
5. Bear Creek at Ottumwa, Iowa	50 annual peaks, 1965-2014; nine crest-stage-gage threshold periods, ranging from 560 to 1,180 ft ³ /s; manual low-outlier threshold 1,200 ft ³ /s.
6. Santa Cruz River at Lochiel, Arizona	65 annual peaks, 1949-2013; 12,000 ft ³ /s perception threshold for 1927-1948; MGBT low-outlier processing.
7. American River at Fair Oaks, California	77 systematic peaks within 1905-1997; five paleoflood/historical intervals and ten threshold periods, extending to year 1.

For Example 4 the historical bounds are 41,000-60,000 ft³/s in 1864, 20,000-25,000 in 1893, 35,000-40,000 in 1894, and 80,000-103,000 in 1921. Perception thresholds are 150,000 for 1165-1858, 40,000 for 1859-1892, 19,900 for 1893-1894, and 20,000 for 1977-2004. Example 7 uses bounds 600,000-850,000 in year 650, 400,000-550,000 in each of 1437, 1574, and 1711, and 262,000-300,000 in 1862. Its early thresholds are 599,000 for years 1-1301, 399,000 for 1302-1847, and 261,000 for 1848-1904; missing portions of the later record use 150,000 ft³/s. These long periods supply information about large floods without treating every unobserved year as a measured discharge.

Procedure and results. The fitted log mean, log standard deviation, and log skewness are compared with the published targets below, with absolute tolerance 0.001 for each. All 21 comparisons across the seven examples are verified. The table reports the published targets, not fitted BestFit coordinates.

Example	Published log mean	Published log standard deviation	Published log skewness
1	3.328623159	0.140287994	0.396626124
2	3.022663041	0.682087092	-0.929108050
3	3.759834285	0.243406211	0.144442997
4	3.885777246	0.245920859	0.817849937
5	3.278686106	0.233135027	-0.925407257
6	3.069106533	0.489820622	-0.462278724
7	4.653457000	0.376721000	-0.101163000

Three additional plotting-position comparisons reproduce PeakFQ probabilities for Examples 4, 5, and 7 within 10^{-12} . For example, the four historical intervals in Example 4 have reference plotting probabilities 0.0091324201, 0.0507219548, 0.0211032951, and 0.0045662100. The oldest paleoflood in Example 7 has reference probability 0.00025. These checks verify how observations are positioned on the frequency plot; they do not verify confidence-band coverage.

Bulletin 17C covariance, regional information, and recovery

Uncertainty in fitted moments

Setup and procedure. Thirteen checks examine covariance, the matrix describing parameter variances and their co-movement. Six families are tested at sample sizes 25 and 100, using seed 12345: Exponential with location 10 and scale 50; Gamma with scale 5 and shape 2; Normal with mean 100 and standard deviation 15; Pearson III with mean 100, standard deviation 20, and skewness 0.5; base-10 Log-Normal with log mean 3 and log standard deviation 0.5; and LP3 with log mean 3, log standard deviation 0.5, and log skewness 0.2. The thirteenth case uses Moose River Example 1 with 68 observations.

Independent calculations use central moments through order six and the finite-sample corrections for the fitted second and third moments. Frozen Python calculations are cross-checked by a separately implemented calculation within 10^{-12} scaled tolerance. BestFit's fitted covariance is then compared at the actual fitted parameters, with 10^{-5} scaled plus 10^{-10} absolute tolerance. For a concrete reference, the Exponential location/scale variances are 100 and 200 at sample size 25, with covariance -100; at sample size 100 they are 25 and 50, with covariance -25.

Results. All thirteen methods are verified. They check both uncertainty magnitude and parameter dependence, including the transformations needed for log-discharge families. Agreement at these cases does not establish that matrix regularization is harmless in every ill-conditioned fit.

Regional information

Setup. Twelve comparisons ask whether an at-site estimate and regional information receive the intended relative weight. The at-site data are deterministic quantile grids at probabilities $i/(n+1)$: Log-Normal with base-10 log mean 3 and log standard deviation 0.5, or LP3 with those same values and log skewness 0.2. Regional targets are log mean 3.25 with mean square error (MSE) 0.004, log standard deviation 0.5 with MSE 0.0015, log skewness 0.3 with MSE 0.05, and log-transformed $Q(0.99)$ of 4.5 with MSE 0.01. MSE expresses uncertainty in the regional estimate and controls its weight.

For one parameter, the independent inverse-variance reference is

$$\theta_w = \frac{\theta_a/V_a + \theta_r/V_r}{1/V_a + 1/V_r}, \quad V_w = \frac{1}{1/V_a + 1/V_r}, \quad (\text{D.3})$$

where subscripts a and r denote at-site and regional estimates and V is their variance or MSE. With several parameters, the reference retains their covariance. Quantile information is propagated through the local sensitivity of the quantile to the parameters. These nonlinear comparisons are approximate and have the declared limits below; the first percentage concerns the weighted value and the second its MSE.

Family and information applied	Sample sizes	Value/MSE comparison limits
Log-Normal: log mean only	25 and 100	Mean 1% / 1%
Log-Normal: log standard deviation only	25 and 100	Standard deviation 5% / 5%
Log-Normal: both parameters	25 and 100	Mean 5% / 5%; standard deviation 10% / 10%
Log-Normal: $Q(0.99)$ only	25 and 100	Quantile 10% / 20%

Family and information applied	Sample sizes	Value/MSE comparison limits
Log-Normal: both parameters and $Q(0.99)$	25	Mean 5% / 10%; standard deviation 10% / 15%; quantile 10% / 20%
LP3: log mean and log skewness	25	Mean 5% / 25%; skewness value 15% plus 0.01 absolute, MSE 20%
LP3: all three parameters	25	Mean 5% / 10%; standard deviation 10% / 15%; skewness value 15% plus 0.01 absolute, MSE 20%
LP3: all parameters and $Q(0.99)$	25	Mean 5%, standard deviation 10%, quantile 10%, skewness 15% plus 0.01 absolute; every compared MSE 25%

Results. All twelve comparisons are verified. The simple mean-only case closely reproduces inverse-variance weighting. The wider limits for simultaneous nonlinear information describe the approximation being tested; they are not the acceptance limits for synthetic parent recovery. These deterministic grids do not measure repeated-sample interval coverage.

Recovery of parameters and the 100-year quantile

Six additional GMM checks each use 1,000 random observations and seed 12345. Their parents are Exponential location 0/scale 50; Gamma scale 5/shape 2; Normal mean 100/standard deviation 15; Pearson III mean 100/standard deviation 20/skewness 0.5; base-10 Log-Normal log mean 3/log standard deviation 0.5; and LP3 log mean 3/log standard deviation 0.5/log skewness 0.2. Parameter error must be at most 1.96 standard errors using the fitted moment-based covariance. Pearson III and LP3 check mean and standard deviation directly and assess the weak skewness direction through inclusion of the known $Q(0.99)$ in its 95% quantile band. All six are verified.

A separate Profile-Q example uses 1,000 LP3 observations with log mean 3, log standard deviation 0.5, log skewness -0.5, and seed 12345. After fitting, the three true-profile parameter intervals must be finite and ordered. The generating $Q(0.99)$ must also lie within the fitted quantile estimate plus or minus 1.96 standard errors, calculated by propagating moment-based parameter uncertainty. The verified result therefore supports the declared 100-year response check. It does not establish generating-parameter inclusion in the profile intervals, or repeated-sample coverage of those intervals.

Published Bayesian comparisons

Australian Rainfall and Runoff Flike examples

Setup. Flike provides an independent Bayesian calculation using importance sampling. The Hunter River at Singleton record contains 31 annual maximum discharges for 1938-1968; the Wimmera River at Glynwylln record contains 56 annual maxima for 1960-2015. All discharges are in m^3/s . BestFit uses its Bayesian univariate analysis with the Jeffreys scale rule enabled and the posterior-mean point-estimate option. A 90% interval extends from the 5th to the 95th posterior percentile.

Flike case	BestFit setup	Relative limit for every compared mean and interval endpoint
3	Hunter River, LP3, systematic observations only	7.5%
4	Hunter River, LP3, plus one exceedance of 12,525 m^3/s during 1820-1937	5%

Flike case	BestFit setup	Relative limit for every compared mean and interval endpoint
5	Hunter River, LP3, plus a Normal prior on log skewness with mean 0 and standard deviation 0.3	5%
6a	Wimmera River, GEV, without low-outlier censoring	10%
6b	Wimmera River, GEV, with MGBT low-outlier censoring	5%

Procedure. For each case, the point-estimate curve and both 90% interval endpoints are compared at four AEPs. The complete Flike target curves are shown below; these are external benchmark values.

Case	AEP (return period)	Reference mean, m ³ /s	Reference 90% interval, m ³ /s
3	10% (10 years)	3,929	2,229-8,408
3	2% (50 years)	12,786	5,502-51,010
3	1% (100 years)	19,572	7,188-107,122
3	0.2% (500 years)	47,034	11,507-570,635
4	10% (10 years)	3,294	2,181-4,947
4	2% (50 years)	9,350	5,778-16,511
4	1% (100 years)	13,511	7,785-27,687
4	0.2% (500 years)	28,542	12,966-85,583
5	10% (10 years)	3,598	2,172-6,702
5	2% (50 years)	10,535	5,310-26,633
5	1% (100 years)	15,413	7,093-45,087
5	0.2% (500 years)	33,365	12,244-134,107
6a	10% (10 years)	286	172-578
6a	2% (50 years)	1,315	493-4,975
6a	1% (100 years)	2,481	737-12,398
6a	0.2% (500 years)	10,696	1,802-101,034
6b	10% (10 years)	227	177-311
6b	2% (50 years)	423	304-785
6b	1% (100 years)	521	354-1,145
6b	0.2% (500 years)	789	448-2,813

Results. All five methods are verified under the case-specific limits above. The references illustrate the effect the comparison is intended to reproduce: historical information reduces Hunter River's 100-year upper interval limit from 107,122 to 27,687 m³/s; regional-skew information gives 45,087 m³/s. Wimmera's low-outlier treatment changes the 100-year reference mean from 2,481 to 521 m³/s. These substantial differences make censoring and prior specification consequential parts of the verification. The comparisons establish consistency with Flike's specified examples; they do not establish which treatment is appropriate for another catchment or demonstrate 90% empirical coverage.

Kamp at Zwettl information-expansion examples

Setup. The GEV examples of Viglione et al. (2013) and Skahill et al. (2016) examine whether information beyond the systematic record changes the inferred flood-frequency curve in the intended way. The Kamp at Zwettl, Austria, supplies either 51 annual peaks for 1951-2001 or 55 for 1951-2005, in m^3/s . Temporal information adds historical flood intervals: 180-300 m^3/s in 1655, 240-400 in 1803, and 202.5-337.5 in 1829, with representative values 240, 320, and 270. A perception threshold of 300 m^3/s covers 1600-1950. Causal information adds a Normal prior on the 500-year discharge, with mean 480 m^3/s and standard deviation 80 m^3/s . BestFit disables the Jeffreys scale rule for these comparisons.

Procedure. Cases 1-8 compare the posterior-mode GEV parameters and the 100- and 1,000-year discharge estimates and their 90% intervals. Location and scale must agree within 5%; shape within 10% for Case 1 and 5% otherwise; every discharge summary within 5%. The comparison targets are:

Case	Systematic period; added information	Reference location, m^3/s	Reference scale, m^3/s	Reference shape
1	1951-2001; none	42.9	20.2	-0.096
2	1951-2005; none	41.7	20.7	-0.310
3	1951-2001; temporal	43.4	21.7	-0.222
4	1951-2005; temporal	42.6	21.5	-0.281
5	1951-2001; causal	41.9	21.0	-0.313
6	1951-2005; causal	41.6	20.8	-0.333
7	1951-2001; temporal and causal	42.7	21.8	-0.291
8	1951-2005; temporal and causal	42.5	21.5	-0.313

Case	Reference 100-year estimate, m^3/s	Reference 90% interval, m^3/s	Reference 1,000-year estimate, m^3/s	Reference 90% interval, m^3/s
1	160	130-288	241	163-649*
2	253	184-542	543	317-1,853
3	217	176-291	399	278-647
4	244	197-331	497	347-818
5	258	193-307	557	335-702
6	269	217-317	604	418-747
7	253	206-299	527	369-671
8	264	220-308	571	418-708

The Case 1 test compares the 1,000-year lower endpoint with 163 m^3/s ; the published value is 183 m^3/s in Table 2 of Skahill et al. (2016), reference 29. Acceptance against the retained 163 m^3/s target therefore does not establish agreement with that published endpoint.

Case 9 uses the 1951-2001 record with three Normal quantile priors: the 10-year discharge has mean 100 and standard deviation 20 m^3/s ; the 100-year discharge has mean 250 and standard deviation 40 m^3/s ; and the 1,000-year discharge has mean 500 and standard deviation 60 m^3/s . Five posterior summaries for each GEV parameter are compared with EvdBayes, each within 5%. This case uses 95% parameter intervals, unlike the 90% discharge intervals above.

Parameter	EvdBayes mean	EvdBayes standard deviation	EvdBayes 2.5th percentile	EvdBayes median	EvdBayes 97.5th percentile
Location, m ³ /s	41.4307	3.0918	35.5680	41.3467	47.7391
Scale, m ³ /s	20.7700	2.4520	16.3861	20.6195	26.0375
Shape, dimensionless	-0.2694	0.0488	-0.3608	-0.2709	-0.1699

Results. All nine methods are verified against their retained targets, subject to the Case 1 endpoint qualification above. The reference curves show why each design matters. Extending the systematic record alone changes the 100-year estimate from 160 to 253 m³/s. With both temporal and causal information, the corresponding estimates are 253 and 264 m³/s, and the interval widths are much smaller. The ninth comparison examines the shape and spread of the posterior parameter distributions rather than only one fitted curve. The comparisons support the stated combinations of systematic, temporal, and prior information; the target values remain distinct from measured BestFit outputs.

Bootstrap convergence and numerical reliability

The supporting Moose River Example 1 bias-corrected-bootstrap experiment asks a different question: whether repeated fits reach their intended numerical solution. A bootstrap realization is a simulated replacement record that is refitted to assess uncertainty. This experiment fits 1,000 fixed realizations with seed 12345, 68 observations per realization, and regional log-skew information of 0.44 with MSE 0.078. It distinguishes delivery of a fitted result from convergence of the outer GMM iteration, which repeatedly updates moment weights.

Diagnostic	Observed result
Requested and accepted refits	1,000
Outer GMM convergence attained	947
Outer iteration limit reached	53
Fallbacks from the BFGS inner optimizer	77
Thrown fitting exceptions	0

Thus, 1,000 returned fits do not mean 1,000 converged outer iterations. Independent supporting calculations verify the supplied gradients for the roundoff-convergence branch and positive definiteness of the regularized moment matrices. Those checks support the inner numerical work. They do not establish convergence of every outer iteration or coverage of the resulting intervals, and a more accurate inner solution alone does not guarantee a smaller error at a finite outer iteration budget. The supporting experiment retains the configuration and diagnostics.

Numerical Cohn interval values and broad Bulletin 17C interval coverage remain outside the supported claims. The chapter's source and method identifiers are indexed in the report's test-coverage appendix; the setup and interpretation needed to read the comparisons are given above.

CHAPTER 08

Point-Process Analysis

Question and model

Point-process analysis uses all events above a selected threshold rather than only one maximum per year. It combines an event rate with a distribution of event magnitudes. A Poisson process describes the count of exceedances during the observed exposure, and a generalized-Pareto (GPA) distribution describes the excess above the threshold. The corresponding annual-maximum distribution is expressed using generalized extreme value (GEV) parameters.

Verification asks whether BestFit calculates this likelihood correctly, simulates the right rates and tails, and recovers known generating processes. Seasonal tests also ask whether events are assigned to the right part of the year and whether changing the year origin preserves the analysis. All magnitudes below are synthetic and have arbitrary, consistent units; exposure is measured in years and intensity in events per year.

Generating processes and exposure

Every recovery experiment contains exactly 1,000 threshold exceedances, with threshold $u = 80$. The nonseasonal parent has GEV location 100, scale 20, and BestFit shape $\kappa = -0.10$. Its equivalent GPA scale at the threshold is 18, and its annual exceedance rate is 2.867971991. The declared observation exposure is 1,000/2.867971991 years.

Seasonal designs use two GPA populations. The first has scale 12 and BestFit shape $\kappa = -0.10$; the second has scale 40 and shape $\kappa = 0.05$. In the alternative Coles convention, their shapes are $\xi = 0.10$ and $\xi = -0.05$, respectively: $\kappa = -\xi$. This sign conversion matters because it determines whether the tail is bounded.

Recovery design	First/second seasonal intensity	Effective changepoint days	Generation
Nonseasonal	One rate, 2.867971991 per year	Not applicable	BestFit magnitude generator; seed 41001
Seasonal, equal intensity	8 / 8 per year of seasonal exposure	80 and 260	BestFit dated-event generator; seed 42001
Seasonal, unequal intensity	12 / 4 per year of seasonal exposure	80 and 260	BestFit dated-event generator; seed 42101
Calendar-year automatic-prior recovery	8 / 8	170 and 350	Independent counts and GPA magnitudes, uniform within-season dates; seed 47001
October-water-year automatic-prior recovery	8 / 8	170 and 350, in water-year coordinates	Same independent generation and seed, with the date origin shifted

The first season wraps across the year boundary; the second lies between the two changepoints. On the model's 366-day axis their exposure fractions are 186/366 and 180/366 for these designs. Equal-intensity recovery therefore uses 125 exposure years. For unequal intensities, the annual rate is the exposure-weighted sum of the two rates, and exposure is 1,000 divided by that rate. The two production seasonal fits use flat changepoint priors on days 10-100 and 200-360. The calendar/water-year pair instead uses the automatically placed priors.

The fixed-size recovery samples use the declared exposure when estimating magnitude and rate parameters. Separate simulations with random event counts test whether the Poisson count mechanism itself behaves correctly.

Independent likelihood comparisons

The stationary external comparison uses three exact events, with magnitudes 85, 100, and 120, over 1.25 years. It uses threshold 80 and the nonseasonal GEV parent above. SciPy 1.17.1 supplies independent Poisson, GPA, and GEV calculations with the explicit shape-sign conversion.

Quantity	Frozen SciPy reference	BestFit comparison
Threshold exceedance intensity	2.867971990792441 events/year	Passed scaled 10^{-12} agreement
Log likelihood for the three events and exposure	-12.762996830833266	Passed scaled 10^{-12} agreement
Probability an exceedance above 80 also exceeds 120	0.1344306327493119	Passed scaled 10^{-12} agreement
Annual maximum at nonexceedance probability 0.99	216.8195247592646	Passed scaled 10^{-12} agreement

These numbers are stored external references, not saved BestFit fitted parameters. This comparison evaluates the model at specified parameters and does not estimate them from three events.

Two additional calculations test mixed observation types over ten years of exposure:

Comparison	Exact events	Uncertain annual magnitude	Interval observation	Threshold information
Nonseasonal	105 and 125	Normal measurement distribution: mean 115, standard deviation 4	Magnitude between 110 and 126	Threshold 112 over five indexed years, with two above
Seasonal	105 on 20 January and 140 on 30 May	Normal measurement distribution: mean 125, standard deviation 4	Magnitude between 110 and 150	Threshold 120 over three years, with one below and two above

For the seasonal calculation, the first GEV has location 100, scale 20, and BestFit shape -0.10; the second has location 130, scale 30, and shape 0.05. Effective changepoints are 80 and 260. An independent calculation sums the Poisson exposure and exact-event terms, integrates over measurement uncertainty, and calculates the probabilities of the stated intervals and threshold counts. Seasonal non-exact records use the annual maximum of both exposure-adjusted processes. Only the two exact events contribute to the event count. Both mixed-observation comparisons passed the absolute log-likelihood tolerance 2×10^{-7} .

Recovery and deterministic results

Recovery procedure and acceptance

1. Generate each sample and establish agreement among event count, exposure, threshold, season definition, and shape convention. Confirm that the generating parameters are permitted by the priors and explain the sample better than a collapsed alternative.
2. Run Bayesian estimation with the production DEMCzs simulation defaults and seed 12345.
3. For the nonseasonal model, compare all three GEV parameters with their central 95% posterior intervals. Also require the generating intensity and conditional tail probability above 120 inside their corresponding central 95% bands.

4. For seasonal models, compare posterior-mean intensity, GPA scale, and shape with the generating values using their standard errors. Check changepoints, intensity bands, and conditional-tail bands separately, and check the sampling diagnostics.

A season receives information from only its share of the 1,000 events. Its effective event count is

$$N_s = N \frac{w_s \Lambda_s}{\sum_j w_j \Lambda_j}, \quad (\text{P.1})$$

where w_s is the fraction of annual exposure and Λ_s is the seasonal intensity. Equal rates give effective counts 508.197 and 491.803. Rates 12 and 4 give effective counts 756.098 and 243.902; exposure fraction alone would understate the first season's contribution.

Seasonal intensity uses Poisson uncertainty; GPA scale and shape use analytical GPA maximum-likelihood covariance with the effective counts above. Each absolute standardized error must be at most 1.96. Effective changepoints, obtained by flooring their continuous parameters, must lie in their central 95% posterior sets. Each seasonal intensity and conditional tail response must lie in its central 95% band. All monitored GEV and changepoint coordinates require $\hat{R} < 1.10$ and ESS at least 100.

Timing, prior placement, and simulation

The paired calendar/water-year recovery tests hold changepoints at days 170 and 350. Generated magnitudes and block days agree exactly; calendar dates differ by 92 days; the parent log likelihoods agree within 10^{-10} . This isolates the effect of changing the year origin.

Two separate prior-placement experiments each generate 4,000 dated events. They use seasonally clustered PERT timing to create a monthly occurrence pattern: calendar-year targets are days 170 and 350 with seed 45001, and October-water-year targets are days 80 and 260 with seed 46001. An independent calculation rotates and smooths the monthly counts, locates separated peaks and the intervening valleys, and constructs the five-month prior windows. BestFit's initial changepoints and support bounds must match exactly and contain the generating days. These tests assess placement of prior ranges; they do not estimate changepoints from the likelihood.

Three simulation experiments each generate 500 independent 20-year records: nonseasonal, equal-intensity seasonal, and unequal-intensity seasonal. Event totals are compared with their Poisson expectations within five standard errors. Conditional exceedance fractions are compared within five binomial standard errors, using magnitude 120 for the first population and 165 for the second. Seasonal assignment and date-to-block mapping must be exact.

Test group	Number of tests	Outcome
Stationary external likelihood and responses	1	Passed
Mixed-observation likelihood	2	Passed
Automatic seasonal prior placement	2	Passed
Generating-process recovery	5	Passed
Poisson counts, conditional tails, and seasonal dates	3	Passed

All thirteen comparisons passed. SciPy verifies the stationary calculations; independent analytical and recovery designs support the seasonal and year-origin results. Simulation checks alone do not establish parameter recovery. Fitted estimates, posterior endpoints, simulated counts, and prior-window outputs are not retained as result tables. The supporting calculations and stationary reference document the comparisons.

CHAPTER 09

Competing-Risk Analysis

A competing-risk distribution describes the minimum or maximum of several potential outcomes. For a flood maximum, different mechanisms may each produce an event, but only the largest value is observed. For a minimum, the first or weakest outcome controls. Unlike a mixture, this model does not randomly select one population using fixed proportions. Each component's chance of controlling the observed value depends on its distribution and its dependence on the other components.

Verification addresses two questions: does simulation preserve the requested dependence, and can estimation recover the separate component distributions from observations of only their minimum or maximum? The examples are synthetic and use arbitrary, consistent magnitude units.

Dependence theory and independent simulation reference

The four simulation tests use two Normal components with mean 0 and standard deviation 1. Each generates 40,000 maxima with seed 24681357. The fraction of maxima at or below zero is compared with a probability calculated independently. A Gaussian copula supplies the dependence: it links component percentiles through correlated Normal scores, with correlation denoted by ρ .

A second comparison checks rank correlation, which measures whether the two component values tend to rise and fall together. To inspect those ranks through the maximum-generation interface, the test uses two additional runs: one component has mean 1,000 and the other mean 0, both with standard deviation 1, then their means are exchanged. The separation makes each run reveal one component's ordering without reproducing the production sampling algorithm.

For Gaussian dependence, the expected rank correlation ρ_S and zero-threshold probability are

$$\rho_S = \frac{6}{\pi} \arcsin(\rho/2), \quad P[\max(X_1, X_2) \leq 0] = \frac{1}{4} + \frac{\arcsin(\rho)}{2\pi}. \quad (\text{C.1})$$

Dependence setting	Independent target	BestFit result
Independent components, $\rho = 0$	Rank correlation 0; probability of maximum at or below zero 0.25	Passed both comparisons
Perfect positive dependence	Rank correlation 1; probability 0.5	Passed both comparisons
Perfect negative limiting dependence	Equation (C.1), evaluated at $\rho = -1 + \sqrt{\epsilon}$, where ϵ is machine precision	Passed both comparisons
User-supplied correlation, $\rho = 0.6$	Equation (C.1), evaluated at the specified correlation	Passed both comparisons

The rank tolerance is $6/\sqrt{n-1}$. The probability tolerance is six binomial standard errors plus $1/n$, with $n = 40,000$. These bounds allow expected simulation variation. All four dependence settings passed both comparisons.

Identified recovery designs

Each recovery design contains 1,000 observed minima or maxima, generated with seed 12345. A Weibull distribution is specified by its **scale and shape**; a Gumbel distribution by its **location and scale**. These parameters are not interchangeable with means and standard deviations.

Design	First generating component	Second generating component	Dependence
Independent minimum	Weibull scale 50, shape 1	Weibull scale 80, shape 3	Independent
Independent maximum	Weibull scale 100, shape 3	Gumbel location 80, scale 20	Independent
Correlated minimum	Weibull scale 50, shape 1	Weibull scale 80, shape 3	Gaussian correlation fixed at 0.6

The tests deliberately require both components to leave recognizable information in the observed sample. Before fitting, each must control at least 15% of outcomes in theory, win at least 100 of the generated comparisons, and receive at least 100 observations when fractional likelihood memberships are summed. Each must dominate at least 10% of the combined probability range, and the dominant component must change within nonexceedance probabilities 0.10 to 0.90. This change produces the bend sometimes called a *dog leg* in the combined frequency curve.

Design	Theoretical shares, first/second	Observed winning counts, first/second	Probability at the interior change in dominance
Independent minimum	72.7% / 27.3%	720 / 280	0.790
Independent maximum	48.7% / 51.3%	495 / 505	0.474
Correlated minimum	78.7% / 21.3%	776 / 224	0.827

Winning counts are known from the synthetic generation. The fitted model receives only the scalar minimum or maximum, so it must infer component parameters without those labels. The count and dominance checks establish suitability of the experiment; they do not replace parameter recovery.

Estimation and comparison procedure

1. Generate the observed sample, establish the identification checks above, and verify the minimum/maximum convention, dependence, parameter ordering, and prior support.
2. Estimate all component parameters by maximum likelihood (MLE). For two Weibulls, order their parameters by increasing shape. Calculate uncertainty from the curvature of the complete competing-risk likelihood, without adding an artificial term to make the matrix invertible.
3. Require every generating parameter to be within 1.96 estimated standard errors of its MLE. Also check the combined distribution at nonexceedance probabilities 0.10, 0.25, 0.50, 0.75, and 0.90.
4. For the two independent designs, run Bayesian estimation with the standard DEMCzs settings. Require every ordered generating parameter inside its central 95% posterior interval, $\hat{R} < 1.10$, and ESS at least 100. Check the same combined-response probabilities.

Estimation results

Design	MLE	Bayesian analysis
Independent minimum	Passed	Passed
Independent maximum	Passed	Passed with the optional Jeffreys scale multiplier disabled
Correlated minimum	Passed	Not tested

All five recovery comparisons passed. The independent maximum provides a numerical example:

Parameter	Generating value	BestFit MLE, rounded
Weibull scale	100	102.391
Weibull shape	3	3.219
Gumbel location	80	78.491
Gumbel scale	20	19.600

These fitted values satisfied the full-likelihood standard-error rule. Equivalent fitted-parameter and interval tables for the other recovery comparisons, and the realized simulation statistics, are not available for tabulation.

Prior sensitivity and interpretation

The independent maximum is a material boundary case. With the optional Jeffreys scale multiplier enabled, the comparison selected a Weibull scale near 1.11×10^{-16} ; the component effectively disappeared from the maximum. Its 95% interval, approximately $[4.44 \times 10^{-12}, 38.17]$, excluded the generating scale 100. That configuration failed recovery. The passing Bayesian experiment disables the optional multiplier while retaining bounded parameter priors, the generating sample, and the standard sampler settings. It does not establish recovery under the default scale multiplier.

The maximum also has a second dominance change at probability 0.987, where the Gumbel again controls the extreme tail. This does not replace its required interior change at 0.474. Balanced component contributions alone do not guarantee identifiable parameters outside the specified recovery designs.

The results support these specified two-component models and four simulation dependence settings. They do not establish arbitrary-component recovery or correlated Bayesian recovery. Agreement with a separate Numerics estimator is useful corroboration but is not independent scientific evidence. Supporting calculations provide the dependence comparisons, fitted values, and prior-sensitivity results.

CHAPTER 10

Mixture Analysis

Model, identification, and acceptance

A mixture represents observations arising from different populations when the population responsible for each observation is unknown. For example, a sample may combine two flood-generating mechanisms. Each component has its own distribution, and its weight is the proportion of observations expected from that population. An observation comes from one component; the mixture does not add the component values or select their maximum. These tests use synthetic, unitless observations so the generating populations and proportions are known exactly.

The verification asks whether BestFit can recover those proportions, component means, and component standard deviations. It checks two estimation approaches. Expectation-maximization (EM) alternates between estimating each observation's fractional membership and updating the component parameters. Bayesian analysis represents uncertainty by sampling plausible weights and component parameters.

Generating populations

Each design contains exactly 1,000 scalar observations generated with seed 12345. The table gives the **generating values**, not fitted BestFit results. Expected counts describe long-run proportions; the realized counts vary with the generated sample.

Design and component	Generating mean	Generating standard deviation	Share of all observations	Expected count in 1,000
Ordinary two-component: lower-mean Normal	0	1	30%	300
Ordinary two-component: higher-mean Normal	3	0.1	70%	700
Zero-inflated: exact zero	Not applicable	Not applicable	10%	100
Zero-inflated: lower-mean Normal, restricted to positive values	3	0.1	30%	300
Zero-inflated: higher-mean Normal, restricted to positive values	5	2	60%	600
Ordinary three-component: first Normal	0	1	20%	200
Ordinary three-component: second Normal	3	0.1	30%	300
Ordinary three-component: third Normal	5	2	50%	500

The zero-inflated design has a separate probability of an exact zero and a continuous distribution for positive observations. Its listed Normal means and standard deviations describe the component distributions before conditioning on positive values. It therefore tests a positive *hurdle* model, not an ordinary Normal mixture with some zeros appended. BestFit sets the zero probability to the observed fraction of exact zeros. The generating 10% probability is checked against that fraction using binomial sampling uncertainty; it is not sampled as a Bayesian parameter.

Procedure

1. Generate each sample, construct a fresh mixture with the correct component families, and check that the generating parameters lie within the configured priors and explain the sample better than the initial collapsed components.
2. Fit the three samples by EM. Order components by increasing mean so that exchanging component labels cannot create an apparent recovery error. Compare each weight, mean, and standard deviation with its generating value using its estimated standard error.
3. Fit the same designs by Bayesian analysis with the standard DEMCzs simulation settings and the predeclared sampler seeds 22345, 32345, and 42345 for the two-component, hurdle, and three-component designs, respectively.
4. Reconstruct all component weights in every retained draw, order the components by mean, and compare the generating values with the central 95% posterior intervals. Check sampler diagnostics for every sampled parameter.

The weights must sum to one, or to the observed positive fraction in the hurdle model. Consequently, only the first $K - 1$ weights are sampled for K continuous components; the last is the remaining mass. For the hurdle comparison, the final generating weight is expressed on that same observed positive mass. This separates sampling variation in the zero count from estimation of the positive components.

EM acceptance requires the absolute estimation error divided by its standard error to be no greater than 1.96 for every coordinate. Weight uncertainty uses fractional membership counts; component uncertainty uses the curvature of the observed likelihood. Bayesian acceptance requires all ordered generating coordinates inside their central 95% intervals, chain agreement $\hat{R} < 1.10$, and effective sample size (ESS) at least 100 for each sampled coordinate.

Recovery results

Design	EM result	Bayesian result	Meaning of the pass
Ordinary two-component Normal mixture	Passed	Passed	Both component proportions, means, and standard deviations satisfied their respective uncertainty rules
Positive-hurdle two-component Normal mixture	Passed	Passed	The zero frequency satisfied its binomial rule; positive-component parameters satisfied their respective uncertainty rules
Ordinary three-component Normal mixture	Passed	Passed	All three ordered components satisfied their respective uncertainty rules

All six comparisons passed their stated uncertainty rules. Recovery allows fitted values to differ from their generating values by an amount consistent with the estimated uncertainty.

External evidence, limitations, and provenance

A separate comparison uses an independently generated sample of 1,000 observations from the ordinary two-component parent above. NumPy's PCG64 generator uses seed 12345; this is a different sample from the BestFit-generated recovery sample despite the matching seed number. The reference fit is from scikit-learn 1.9.0, with Python 3.12.13, NumPy 2.5.2, and SciPy 1.18.1. Its diagonal variances are converted to standard deviations before comparison.

Quantity	Frozen independent reference	BestFit comparison
Lower-mean component weight	0.304407173	Within the scaled coordinate tolerance
Lower-mean component mean	0.053369416	Within the scaled coordinate tolerance
Lower-mean component standard deviation	1.042725781	Within the scaled coordinate tolerance
Higher-mean component weight	0.695592827	Within the scaled coordinate tolerance
Higher-mean component mean	3.002147278	Within the scaled coordinate tolerance
Higher-mean component standard deviation	0.099250007	Within the scaled coordinate tolerance
Log likelihood at the generating parameters	-429.767425439	Agreement within 10^{-9}
Log likelihood at the scikit-learn fitted parameters	-428.760759412	Agreement within 10^{-9}
Fitted probability of an observation at or below 3	0.645482673	Agreement within 10^{-12}

The procedure first evaluates the likelihood and seven cumulative probabilities at the fixed parent and scikit-learn parameters, then runs BestFit EM and compares the six fitted coordinates. Each coordinate tolerance is $10^{-4} \max(1, |\text{reference}|)$. This comparison passed. The available results do not tabulate BestFit's fitted coordinates, recovery interval endpoints, or observed zero count; the numerical reference column contains scikit-learn/SciPy values.

Together, recovery and the external comparison support the stated two- and three-component designs. They do not establish recovery for arbitrary overlapping populations or all possible prior choices. scikit-learn does not implement the positive-conditioned hurdle law, so no external-package agreement is claimed for that design. Supporting calculations and the external numerical reference provide the detailed comparison.

CHAPTER 11

Composite Analysis

Question and combination rules

Composite analysis combines distributions that have already been estimated in separate child analyses. It does not fit a new set of component parameters to pooled observations. Verification therefore asks whether the chosen combination rule produces the correct probabilities and whether uncertainty from the child analyses is carried into the combined frequency curve.

Three rules have different physical meanings. A **mixture** selects a child population with a specified probability. A **maximum** describes the largest of the child outcomes, and a **minimum** describes the smallest. **Model averaging** weights alternative models; an equal-weight average has the same mathematical distribution as an equal-weight mixture, although the interpretation of the weights differs. The tests check these distinctions explicitly.

Let $F_i(x)$ be the probability that child i is at or below magnitude x .

Combination	Independent reference probability at or below x
Mixture, with nonnegative weights summing to one	$\sum_i w_i F_i(x)$
Maximum of independent children	$\prod_i F_i(x)$
Minimum of independent children	$1 - \prod_i [1 - F_i(x)]$
Maximum under perfect positive dependence	$\min_i F_i(x)$
Minimum under perfect positive dependence	$\max_i F_i(x)$

Perfect positive dependence means all children occupy the same percentile together. The minimum and maximum rules cannot be substituted for a mixture without changing the result.

Deterministic and recovery results

Fixed-distribution setup and procedure

The fixed calculations use three synthetic Normal children from Tables 44-45 of the 2024 *Verification of the RMC-TotalRisk Software* report. Magnitudes have arbitrary, consistent units.

Child	Mean	Standard deviation	Mixture weight
First	10	2	0.30
Second	20	1	0.20
Third	30	5	0.50

The tests supply these known child distributions directly, without fitting data. They calculate the reference probabilities with individual Normal formulas, then compare BestFit's combined distribution over magnitudes 0 to 55. Quantiles are values associated with specified nonexceedance probabilities; their verification uses both independent numerical inversion and a published R *mistr* table.

The table below reproduces all 25 rounded reference quantiles. Annual exceedance probability (AEP) is the probability of exceeding the listed magnitude. BestFit passed the published table's 1% relative comparison allowance at every row. The allowance is for this rounded published comparison, not an estimation or uncertainty-coverage rule.

AEP	Published R <i>mistr</i> mixture quantile	AEP	Published R <i>mistr</i> mixture quantile
0.000001	53.10	0.01	40.27
0.000002	52.34	0.02	38.75
0.000005	51.33	0.05	36.40
0.00001	50.54	0.10	34.21
0.00002	49.72	0.20	31.26
0.00005	48.60	0.30	28.72
0.0001	47.70	0.50	21.38
0.0002	46.76	0.70	15.58
0.0005	45.45	0.80	10.88
0.001	44.39	0.90	9.15
0.002	43.26	0.95	8.07
0.005	41.63	0.98	7.00
-	-	0.99	6.34

A separate inversion check substitutes each BestFit quantile back into the independent weighted Normal probability. Its probability-error bound is $\max[10^{-8}, 0.005 \min(\text{AEP}, 1 - \text{AEP})]$. This checks inversion more directly than the rounded magnitude table.

Controlled propagation of uncertainty

Three additional experiments give each child 20 equally spaced possible mean values: 9.8-10.2, 19.8-20.2, and 29.8-30.2. Standard deviations remain 2, 1, and 5. Repeating those supports supplies 5,000 retained parameter draws per child; these are controlled uncertainty inputs, not 5,000 observed events or newly fitted posterior samples.

For mixture, maximum, and minimum separately, the independent calculation enumerates every one of the $20 \times 20 \times 20 = 8,000$ combinations. BestFit's resampled result, using seed 20260803, is compared with that complete calculation at nonexceedance probabilities 0.05, 0.25, 0.50, 0.75, and 0.95. Reported mean-curve magnitudes must agree within 0.02; endpoints of the central 90% intervals must agree within 0.05. Each fixed-parent quantile must also lie within its interval.

A separate correlated calculation uses two Normal children, each with mean 10 and standard deviation 1, linked by Gaussian correlation 0.6. At their common median 10, the exact maximum probability is $1/4 + \arcsin(0.6)/(2\pi)$; the minimum probability is its complement. This checks the correlation setting against a known joint probability.

Fixed-distribution or controlled-uncertainty comparison	Number of tests	Acceptance	Result
Weighted three-Normal probability	1	Absolute error at most 10^{-12}	Passed
Published R <i>mistr</i> quantiles	1	1% relative magnitude difference at all 25 probabilities	Passed
Quantile inversion	1	Probability-dependent bound stated above	Passed
Independent and perfectly positively dependent maxima	1	Probability error at most 10^{-10}	Passed
Independent and perfectly positively dependent minima	1	Probability error at most 10^{-10}	Passed
Distinction and theoretical ordering of combination rules	1	Bounds with 10^{-12} slack; at least 0.10 probability separation somewhere on the comparison grid	Passed
Mixture, maximum, and minimum uncertainty propagation	3	Mean curve 0.02; central-90% endpoints 0.05; parent quantile inside interval	Passed
Correlated minimum and maximum at the median	1	Probability error at most 10^{-8}	Passed

End-to-end experiments with fitted children

Four experiments then check the full path from data to propagated uncertainty. They use two independent samples, each of 1,000 observations: the first Normal parent has mean 10 and standard deviation 2, and the second has mean 22 and standard deviation 3. Generation seeds are 51001 and 51002. These parents differ from the three-child published example above.

1. Fit each child by Bayesian analysis with the production simulation settings and seed 12345. Require its true mean and standard deviation inside central 95% intervals, $\hat{R} < 1.10$, and ESS at least 100.
2. Construct the selected composite from the fitted child analyses and propagate their uncertainty.
3. Independently calculate the generating composite's quantiles from the Normal formulas at nonexceedance probabilities 0.10, 0.25, 0.50, 0.75, and 0.90.
4. Require each generating quantile inside the corresponding central 95% propagated interval.

Combination tested	Weights or dependence	Result
Mixture	First child 0.35; second child 0.65	Passed child recovery and all five response comparisons
Maximum	Independent children	Passed child recovery and all five response comparisons
Minimum	Independent children	Passed child recovery and all five response comparisons
Equal-weight model average	Computed weights exactly 0.50 and 0.50	Passed weight, child-recovery, and all five response comparisons

The ten fixed/controlled comparisons and four fitted-child experiments passed. The BestFit child estimates, combined quantiles, and interval endpoints are not available for a numerical results table.

Independence of child uncertainty

An additional experiment tests the maximum of two Normal children, both with standard deviation 0.20. Each receives 100 equally spaced possible means: 10-14 for the first child and 11-15 for the second, repeated to form 5,000 supplied draws per child. The independent reference considers all 10,000 combinations of means at nonexceedance probability 0.50.

With resampling seed 20260803, BestFit's mean-curve magnitude must agree within 0.02 and its central 50% interval endpoints within 0.05. The comparison is repeated after reversing the second child's draw order and after exchanging the children. All three variants passed. As a negative control, pairing draws only by their stored positions misses the independent reference by at least 0.10. This demonstrates that the check detects artificial dependence caused by matching unrelated draws. The companion coincident-frequency experiment is described in the next chapter.

Interpretation and limitations

These results support the specified combination rules, dependence settings, and propagation at the declared response probabilities. The controlled 8,000 combinations are not independent repetitions of data collection, and the four fitted-child examples do not establish repeated-sample simultaneous coverage of an entire curve. Supporting calculations provide the published reference, analytical comparisons, and resampling details.

CHAPTER 12

Bivariate and Coincident-Frequency Analyses

A bivariate analysis describes two measurements from the same event. Each variable has its own *marginal distribution*, describing its values considered separately. A **copula** describes how their percentiles occur together. This permits the dependence to be changed while retaining the same marginal distributions. Coincident-frequency analysis then combines those paired inputs through a response relationship, such as the sum of two contributing flows. The numerical examples use arbitrary, consistent input units.

Bivariate copula estimation

Data and comparison procedure

Six one-parameter copula families are tested with both maximum pseudo-likelihood (MPL) and inference functions for margins (IFM), using twelve fixed samples of 100 matched observations. MPL replaces each measurement by its ascending rank divided by $n + 1$ and estimates dependence from those probabilities. IFM first fits a Normal distribution to each marginal and then estimates dependence using the fitted marginal probabilities. The independent Normal fits use the sample mean and the maximum-likelihood standard deviation, whose variance denominator is n .

The Student-t copula adds a second parameter, degrees of freedom ν , which controls joint tail behavior. Its external comparison uses an independently generated sample of 1,000 pairs: X has mean 100 and standard deviation 15, Y has mean 80 and standard deviation 25, and the parent copula has correlation $\rho = 0.8$ and $\nu = 4$. Generation uses NumPy PCG64 seed 20260830. This sample is also analyzed by both MPL and IFM.

1. Independently calculate each copula density and likelihood using Python 3.12.13, NumPy 2.3.5, and SciPy 1.18.1. Check the one-parameter density formulas against cumulative-probability derivatives or, for the Gaussian family, a Normal-density ratio. Check the Student-t formula against SciPy's bivariate-t to univariate-t density ratio.
2. Find the independent likelihood maximum and preserve the observations, fitted parameters, and maximum log likelihood in a frozen reference file.
3. Evaluate BestFit's likelihood at those same reference parameters to check that both calculations describe the same probability model.
4. Run BestFit's production Differential Evolution estimator with its default tolerances, and compare the likelihood attained with the independent maximum.

Reference values and results

The parameter values and log likelihoods below are **independent reference results**. For the first five families the parameter is θ ; for Gaussian it is correlation ρ . The Student-t rows name both parameters explicitly.

Family	Method	Independent fitted parameter	Independent maximum log likelihood	BestFit comparison
Ali-Mikhail-Haq	MPL	$\theta = 0.8321521$	7.245322	Passed
Ali-Mikhail-Haq	IFM	$\theta = 0.8392378$	7.500811	Passed
Clayton	MPL	$\theta = 1.5340162$	30.370897	Passed

Family	Method	Independent fitted parameter	Independent maximum log likelihood	BestFit comparison
Clayton	IFM	$\theta = 1.4852343$	30.315461	Passed
Frank	MPL	$\theta = 7.7187608$	46.461072	Passed
Frank	IFM	$\theta = 8.1303213$	49.134008	Passed
Gumbel	MPL	$\theta = 2.0979531$	39.077868	Passed
Gumbel	IFM	$\theta = 2.0311985$	40.512847	Passed
Joe	MPL	$\theta = 2.6643256$	38.271996	Passed
Joe	IFM	$\theta = 2.9656127$	44.762409	Passed
Gaussian	MPL	$\rho = 0.8000853$	48.086892	Passed
Gaussian	IFM	$\rho = 0.7871334$	48.323828	Passed
Student t	MPL	$\rho = 0.8157372; \nu = 5.0704726$	557.465595	Passed
Student t	IFM	$\rho = 0.8142171; \nu = 5.0055361$	559.839653	Passed

All fourteen comparisons passed. BestFit's likelihood at the reference parameters must agree within 10^{-8} for a one-parameter family and 10^{-5} for Student t. Optimization is judged separately: twice the absolute difference between reference and BestFit maximum log likelihood must not exceed the joint 95% likelihood-ratio cutoff, 3.841458820694124 for one parameter or 5.991464547107979 for two. Passing therefore establishes agreement under the stated likelihood rule, not equality of every printed parameter digit.

R *copula* estimates also support the first six families, but their unrecorded package version prevents a fully reproducible package comparison. The independent likelihood calculations above define the present acceptance rule. No R-package comparison is claimed for Student t.

Bivariate recovery and coincident frequency

Recovery of known dependence

Each recovery experiment generates exactly 1,000 matched pairs with Normal marginals: X has mean 100 and standard deviation 15; Y has mean 80 and standard deviation 25. The same observation index pairs X with Y . The copula families and generating parameters are:

Copula family	Generating dependence parameter	Generation seed	Recovery result
Ali-Mikhail-Haq	$\theta = 0.8$	13050	Bayesian passed
Clayton	$\theta = 1.5$	13049	Bayesian passed
Frank	$\theta = 8$	13048	Bayesian passed
Gumbel	$\theta = 2$	13047	Bayesian passed
Joe	$\theta = 3$	13046	Bayesian passed
Gaussian	$\rho = 0.8$	13045	Bayesian passed
Student t	$\rho = 0.8; \nu = 4$	13051	Conditional MLE passed for correlation and symmetric tail dependence

The procedure first fits each Normal marginal by maximum likelihood and requires its mean and standard deviation within 1.96 estimated standard errors of the generating values. It then holds those marginal fits fixed while estimating the copula. Each generating copula must be inside prior support and explain the sample better than the corresponding independence limit.

The six one-parameter families use the standard DEMCzs simulation settings and seed 12345. Their generating dependence parameter must lie inside its central 95% posterior interval, with $\hat{R} < 1.10$ and ESS at least 100. Student t uses conditional MLE. Its correlation must satisfy the 1.96-standard-error rule; its combined tail dependence must satisfy that rule using uncertainty from both fitted coordinates and their covariance. Tail dependence describes the tendency for both variables to be extreme together. The weakly identified raw degrees-of-freedom parameter is not claimed recovered. All seven recovery tests passed.

Linear coincident-frequency response

The three linear experiments use $Z = X + Y$, the same Normal marginal parents, and 1,000 pairs generated with seed 12345. Correlation is set to 0, 0.5, or -0.5. Positive dependence makes the sum more variable, while negative dependence reduces its variation. Its exact Normal law is

$$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2 + 2\rho\sigma_X\sigma_Y). \quad (\text{B.1})$$

Here the second argument denotes variance. The generating sum has mean 180 and variance 850, 1,225, or 475 for correlations 0, 0.5, or -0.5, respectively.

For each experiment, fit the Normal marginals by MLE and the Gaussian copula by Bayesian analysis. Build a five-by-five response table from each fitted marginal's 5th, 25th, 50th, 75th, and 95th percentiles, setting every response to the sum of its two inputs. BestFit converts that table into an exceedance curve over 50 output bins. Compare it at every output bin with Equation (B.1) evaluated using the **fitted** means, standard deviations, and posterior-mean correlation.

All three tests passed: maximum absolute AEP error was within 0.05, mean absolute AEP error within 0.01, and exceedance probability decreased as response increased. Separately, the generating correlation lay inside its central 95% posterior interval and met the R-hat/ESS rules. The response comparison measures table-integration error conditional on the fitted inputs. It does not establish recovery of the entire generating response curve or propagation of marginal estimation uncertainty; marginal uncertainty samples are not supplied in these three experiments.

Nonlinear coincident-frequency response

A fourth experiment uses $Z = \exp(0.01X + 0.01Y)$, correlation 0.5, the same Normal marginal parents, and 1,000 pairs generated with seed 13055. Its independent reference is Lognormal because $\log Z$ is exactly Normal with mean 1.8 and standard deviation 0.35.

The procedure fits and checks both marginal distributions, then fits and checks the Gaussian copula. It builds a 17-by-17 response table spanning marginal probabilities 0.001 to 0.999 and uses 61 output bins. At the fitted parameters, the numerical exceedance curve must agree with the exact Lognormal law within 0.015 AEP. This comparison isolates response-table error.

Uncertainty propagation is checked separately. The experiment supplies 2,000 independent asymptotic MLE uncertainty draws for each Normal marginal, with seeds 24680 and 24681, and combines them with the retained copula posterior draws. The marginal draws are frequentist approximations, not Bayesian posterior samples. At generating nonexceedance probabilities 0.10, 0.25, 0.50, 0.75, and 0.90, the known response AEP must lie inside the central 95% propagated band. The nonlinear experiment passed the fitting, numerical-error, and response-band checks.

Independent combination of marginal uncertainty

A separate sum-response experiment supplies 100 equally spaced possible means for each Normal marginal: 0-5 for X and 1-6 for Y , with standard deviation 0.70 for both and fixed correlation 0. Each support is repeated to form 5,000 parameter draws. A 35-by-35 response table over input values -5 to 12 is evaluated in 33 output bins using resampling seed 20260803.

At each output bin, the independent calculation evaluates the closed-form Normal-sum AEP for all 10,000 combinations of means. BestFit's mean AEP must agree within 0.02 and its central 50% interval endpoints within 0.05, both before and after reversing the second marginal's draw order. Both variants passed. Pairing draws only by their stored positions produces a mean-AEP error of at least 0.10 somewhere on the response curve, confirming that the comparison detects artificial dependence between separately estimated marginals.

Conclusion

The results support fourteen optimization comparisons, seven copula recoveries, three linear and one nonlinear response experiment, and independent uncertainty resampling. Actual BestFit fit coordinates, attained likelihoods, maximum response errors, and interval endpoints are not available for numerical results tables.

Bivariate copula intervals remain conditional on fixed marginal estimates. The results apply to paired observations, the stated marginal distributions and parameter regions, and the specified response tables. They do not establish raw Student-t degrees-of-freedom recovery or general nonlinear-response accuracy. Supporting calculations and the independent fitted values provide the detailed comparisons.

CHAPTER 13

Rating-Curve Analysis

A rating curve converts a measured river stage into discharge. The practical verification question is whether BestFit can reproduce a specified stage-discharge relation, including the additional flow that begins when water reaches a second or third hydraulic control. These examples use synthetic gaugings with known generating curves, so both the discharge calculation and the fitted uncertainty can be checked.

The 19 methods comprise three likelihood comparisons, three analytical continuity checks, three worked-example maximum-likelihood comparisons, and five generating-model designs fitted by both maximum likelihood and Bayesian estimation. The numerical examples retain their supplied stage and discharge units. Activation stages have stage units, exponents are dimensionless, and a control coefficient has discharge units divided by stage units raised to its exponent.

Discharge-space likelihood

Example setup and control definitions

The worked example starts with 300 daily pairs dated from 1 January 2000. Stored workbook draws place stage uniformly between 1 and 20. At each stage, discharge is calculated from the controls below and multiplied by a random base-10 lognormal error. The standard deviation of log10 error is 0.05. These are simulated measurements, not observations from a gauging station.

Hydraulic control	Activation stage h_k	Discharge coefficient α_k	Exponent β_k	Included in
Triangular main channel	1	1.753463	2.666667	All three models
First rectangular overbank	10	883.689894	1.67	Two- and three-control models
Second rectangular overbank	15	4069.426757	1.67	Three-control model

The workbook describes rectangular overbank widths of 20 and 50 ft. The numerical comparison uses its stored coefficients and stage-discharge values unchanged; it does not independently validate the hydraulic derivation of those coefficients. Previously active controls continue carrying flow when a new control activates. With K controls, the median discharge and observation law are

$$q(h) = \sum_{k=1}^K \alpha_k (h - h_k)^{\beta_k} \mathbf{1}_{h > h_k}, \quad \log_{10} Q_i = \log_{10} q(h_i) + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2). \quad (\text{R.1})$$

Here $q(h)$ is the underlying median rating relation; Q_i is a simulated observed discharge, and σ describes measurement scatter on the log10 scale. The indicator contributes zero until that control's activation stage is exceeded. BestFit estimates $a_k = \log_{10} \alpha_k$ internally.

Comparison and results

The independent Python/SciPy calculation evaluates the probability density of every positive discharge. Transforming the observations to log10 discharge requires a change-of-variables factor $1/(Q_i \ln 10)$ when expressing the density in discharge units. Omitting that factor would change reported likelihood values and model-comparison criteria, even though it would not move the optimum.

Model, 300 observations	Independent discharge-space log likelihood at generating truth	Absolute tolerance	BestFit result
One control	-1577.7594156516989	10^{-8}	Passed
Two controls	-1850.3134085800252	10^{-8}	Passed
Three controls	-1888.5687896304166	10^{-8}	Passed

BestFit's total, per-observation, and component likelihood calculations reproduce the independent discharge-space density. A separate base-10 LogNormal density calculation also reproduces the terms. This verifies the observation-density calculation at the declared parameters; it is separate from whether an estimator can find suitable parameters from data.

Activation continuity

A practitioner expects discharge to change smoothly as water first reaches an overbank. The continuity checks isolate the added control's contribution just below and above its activation stage. For positive exponent β_k , that contribution is $\alpha_k \epsilon^{\beta_k}$ at a stage offset ϵ above activation and approaches zero as the offset shrinks.

Three analytical checks cover the following cases.

Setup	Independent target and result
Second control; exponents 0.1, 1, 1.67, and 2.5; offsets 10^{-3} , 10^{-6} , and 10^{-9}	The two-sided discharge increment equals the main-channel increment plus the added power-law term, within 10^{-10} relative to the discharge scale. Passed.
Deliberate zero exponent	The new control jumps by its full coefficient at activation. The analytical jump is reproduced within 10^{-9} relative. Passed. This demonstrates why zero must be excluded.
Three controls at the default exponent lower bound, 0.1	Added contributions decrease as the stage offset shrinks from 10^{-6} to 10^{-12} . Passed.

For the last case, the analytical ratio is

$$(10^{-12}/10^{-6})^{0.1} = 0.2511886432. \quad (\text{R.2})$$

The small exponent approaches zero slowly, but it remains continuous. Fast tests separately check that the default bounds exclude zero. Continuity alone does not establish that the chosen controls adequately describe a real channel.

Recovery

Fitting the three worked-example models

The three maximum-likelihood tests use a **separate 1,000-pair replication** of the workbook recipe, with generator seed 20260822, stages between 1 and 20, and log10 error standard deviation 0.05. The independent calculation minimizes squared log10 residuals from multiple starting points in SciPy and estimates the error scale from those residuals. BestFit fits the same data with Differential Evolution, unchanged optimizer tolerances, and the example's parameter bounds and prior settings.

The following are generating values and **independent SciPy comparison targets**, not a table of BestFit fitted coordinates.

Quantity	Generating value	One-control reference fit	Two-control reference fit	Three-control reference fit
Main-channel activation h_1	1	1.000633	1.000989	1.001032
Main-channel log coefficient a_1	0.243897	0.244804	0.246160	0.246300
Main-channel exponent β_1	2.666667	2.664527	2.661335	2.660905
First overbank activation h_2	10	-	9.981416	9.936705
First overbank log coefficient a_2	2.946300	-	2.936743	2.899111
First overbank exponent β_2	1.67	-	1.678032	1.730227
Second overbank activation h_3	15	-	-	15.332091
Second overbank log coefficient a_3	3.609533	-	-	3.828382
Second overbank exponent β_3	1.67	-	-	1.341175
Log10 error standard deviation σ	0.05	0.052251	0.052240	0.051935

First, BestFit's likelihood at the independent optimum must agree within 10^{-8} . Second, the BestFit optimum must lie within the joint 95% likelihood-ratio region around that independent optimum, and the generating parent must lie within the corresponding region around the BestFit optimum. The likelihood-ratio comparison measures the loss of fit for the entire parameter combination; its cutoffs are 9.487729, 14.067140, and 18.307038 for 4, 7, and 10 fitted quantities, respectively.

All three comparisons pass. The available results do not include BestFit's fitted coordinates, so the reference values above must not be presented as observed BestFit output. The three-control reference fit also illustrates why individual control parameters can move appreciably while combined flow remains constrained by the gaugings.

Generating-model recovery

Five calibration designs

These ten tests ask whether estimation recovers the known curve under distinct combinations of calibration range, observation noise, and overbank activation. Each design contains 1,000 aligned stage-discharge pairs with independent Normal error in log10 discharge. Stage is treated as exact. Both estimators use the same data within each design. The following coefficients are on the physical discharge scale, rather than their internally stored logarithms.

Design and generator seed	Calibration-stage range	Known controls: activation; coefficient; exponent	Log10 error SD
Standard single control; 12345	1-10	Main channel: 0.5; 10; 2	0.05
Low-noise single control; 54321	1-12	Main channel: 0.3; 15; 1.8	0.02
Wide calibration range; 99999	1-25	Main channel: 0; 5; 2	0.05
Bankfull transition; 44444	0.5-12	Main channel: 0.3; 12; 2.2. Overbank: 6; 190.4; 1.6667	0.04
Three-control activation; 66666	0.5-10	Low flow: 0.2; 5; 2.5. Main channel: 3; 50; 1.6667. Overbank: 7; 150; 1.6667	0.04

The response is checked at the following stages, specified before examining fitted results.

Design	Stages used for the complete discharge comparison
Standard single control	1.25, 3, 6, 9.5
Low-noise single control	1.25, 3, 7, 11.5
Wide calibration range	1.25, 5, 15, 24.5
Bankfull transition	0.75, 2, 5.5, 6.5, 9, 11.5
Three-control activation	0.75, 2, 2.75, 3.25, 5.5, 6.75, 7.25, 8.5, 9.75

The bankfull sample has 495 observations below stage 6 and 505 at or above it. The three-control sample has 270 below stage 3, 406 between stages 3 and 7, and 324 at or above 7. Thus the highest control is informed by fewer gaugings than the entire curve. These counts help explain identification; they are not substitutes for the covariance or posterior used to calculate uncertainty.

Estimation and acceptance

For a single control, every MLE parameter must be within 1.96 estimated standard errors of its parent. The standard errors come from the unregularized curvature of the fitted likelihood. In multi-control models, several hydraulic quantities can compensate for each other; the coordinate rule therefore applies to the identifiable error scale, while the combined discharge is checked on the complete grid.

MLE response uncertainty is formed from 20,000 bounded multivariate-Normal parameter draws using the full fitted covariance, using seed 20260831. Independent log10 observation errors, using seed 20260901, are then added to each simulated curve. The largest standardized departure across all stages in each draw determines a single simultaneous 95% predictive band. This makes the acceptance statement apply to the whole declared grid.

Bayesian estimation uses retained posterior draws for parameter uncertainty and independent log10 errors with seed 20260902 for observation uncertainty. Its simultaneous predictive band is centred on the posterior-mode (MAP) curve. Every sampled parameter must have $\hat{R} < 1.10$ and effective sample size at least 100; these assess chain agreement and the amount of independent information in the draws. All single-control parent parameters, and the multi-control error scale, must lie in their central 95% posterior intervals. Every parent must be inside its prior support. These recovery designs keep their prior settings with automatic flat priors and the additional Jeffreys scale rule disabled.

Results and interpretation

All five MLE and all five Bayesian tests pass. The MLE outputs for the three single-control designs provide direct fitted-versus-parent comparisons:

Design	Zero-flow stage: parent / fitted	Log10 coefficient: parent / fitted	Exponent: parent / fitted	Error SD: parent / fitted
Standard	0.5 / 0.490708	1 / 0.991367	2 / 2.009298	0.05 / 0.049202
Low noise	0.3 / 0.309034	1.176091 / 1.180771	1.8 / 1.796011	0.02 / 0.019849
Wide range	0 / -0.026289	0.698970 / 0.680915	2 / 2.013448	0.05 / 0.049292

Selected Bayesian discharge outputs and multi-control intervals are shown below. These are examples from the full-grid checks, not replacements for checking every stage.

Design and stage	Parent discharge	BestFit Bayesian MAP discharge	BestFit simultaneous 95% Bayesian predictive band
Standard, stage 6	302.500	302.276	[227.740, 401.206]
Low noise, stage 7	460.284	460.611	[410.797, 516.466]
Wide range, stage 15	1125.000	1123.643	[845.862, 1492.647]
Bankfull transition, stage 6.5	724.391	741.266	[576.299, 953.454]
Three-control activation, stage 8.5	2144.093	2197.933	[1701.739, 2838.807]

The MLE predictive bands at the last two comparison points are [578.013, 950.262] and [1695.479, 2851.016], respectively, and both contain the generating discharge. For the low-noise design, the narrower relative predictive band is consistent with its smaller observation-error scale. In the multi-control designs, passing response checks does not establish separate recovery of every hydraulic coefficient or activation stage.

These are inclusion checks for fixed generated datasets. They do not estimate how often a nominal 95% band covers truth over repeated new datasets, and they do not establish calibration beyond the observed stage range. Equivalence with R `bdrc` is not claimed: matching control law, error density, segmentation, and parameterization has not been established.

CHAPTER 14

Time-Series Analyses

Time-series models describe how an observation depends on earlier observations, earlier unexplained shocks, and external information such as a dated covariate. Verification must therefore check more than a fitted coefficient: a correct fit can still produce a wrong forecast if dates, differences, transformations, or the last observed value are handled incorrectly.

This chapter describes 25 verification methods: eight estimator-recovery tests, thirteen shared calculation checks, and four additional model-interaction checks. The recovery data are synthetic numerical series, with no assigned discharge or other physical unit. An innovation is the new, unpredictable Normal error at a time step; its standard deviation is measured on the model's working scale.

AR, MA, ARIMA, and ARIMAX oracle design

An autoregressive model (AR) carries forward departures of past observations from their mean. A moving-average model (MA) carries forward past innovations. ARIMA combines those mechanisms after differencing, which means subtracting the preceding value to model change rather than level. ARIMAX adds external, dated covariates. In $\text{ARIMA}(p, d, q)$, the three orders are the number of autoregressive lags, the number of differences, and the number of moving-average lags.

The independent R generator discards 110 initialization steps before retaining 1,000 observations for each recovery design. AR and MA use monthly observations. ARIMA and ARIMAX use daily observations; one difference leaves 999 model-scale changes before further conditioning on lags. The starting state and innovation history are specified in the generator, rather than obtained from BestFit.

Shared calculation checks

The thirteen checks below isolate operations used across the four model families. An independent reference calculation is called an oracle. These calculations are specified directly in R or in the test mathematics, rather than obtained by calling another BestFit analysis.

Calculation and setup	Independent comparison and outcome
Scale prior, all four families	Four scalar evaluations verify the Jeffreys scale contribution, whose log density includes $-\ln \sigma$, and its metadata to 10^{-12} . Passed.
Invalid innovation scale, all four families	Positive-scale Gaussian density and prior values agree to 10^{-12} ; zero and negative scales are rejected as impossible in every declared scalar and per-observation path. Passed.
Automatically fitted Box-Cox and Yeo-Johnson transforms	R fits the transformation using six training observations only. Changing the three later observations cannot change the fitted transformation. Saved/restored settings are also checked. Passed.
Manually specified transformation	Eight training values, Yeo-Johnson power 0.6, AR coefficient 0.35, MA coefficient -0.25, and innovation SD 0.8 are transformed independently before reconstructing the likelihood. Passed.
Dated ARIMAX likelihood	An R calculation independently matches response differences with same-date covariates, accounts for lags and transformation density, and checks every likelihood contribution to 10^{-10} . Changing holdout responses leaves the training likelihood unchanged. Passed.

Calculation and setup	Independent comparison and outcome
Lagged-covariate ARIMAX conditioning	An R calculation independently conditions on $\max(q, p + b)$ steps, so every evaluated mean and every autoregressive-lag mean carries all b covariate lags. Four cases with $p > 0$ and $b > 0$ (untransformed, Box-Cox with one difference, and Yeo-Johnson with two covariates and negative responses) check the number of evaluated steps, residuals, per-observation transformation density terms, every likelihood contribution, and conditional predictions to 10^{-10} . Passed.
Reconstructing levels after differencing	Short ARIMA and ARIMAX examples deliberately depart from the recurrence within training. Hand calculations require forecasts to begin at the last observed training level, without using the holdout value. Agreement is within 10^{-10} . Passed.
Forecast uncertainty at the training boundary	An independently calculated innovation-variance recurrence checks when forecast uncertainty starts accumulating. Variances and standard deviations agree to 10^{-10} . Passed.
Transformed stochastic forecasts	Forecasts evolve on the transformed model scale and undergo one inverse transformation. Deterministic values agree to 10^{-10} ; 1,000-seed moment comparisons meet their analytical sampling-error bounds. Passed.
Transformed AR and MA generation	Independent algebraic recurrences and Gaussian mean/variance calculations check generated series, including 1,000-seed moment experiments. Passed.
Transformed and differenced ARIMA generation	An independent recurrence checks changes, reconstructed levels, and final-horizon Gaussian moments over 1,000 seeds. Passed.
Transformed and differenced ARIMAX generation	Dates, covariate contributions, model-scale changes, and reconstruction are independently calculated; deterministic and 1,000-seed moment checks pass.
Information criteria, all four families	AIC and BIC are recalculated from observation likelihood at the stored MAP, excluding the prior. Agreement is within 10^{-10} , and the prior-inclusive alternative is discriminated. Associated flat-prior Normal fits also satisfy declared uncertainty and joint-likelihood checks. Passed.

For the automatically fitted transformations, the Box-Cox training values are 1.1, 1.3, 1.8, 2.7, 5, and 12; the independent fitted power is -0.5411999130. The Yeo-Johnson training values are -4, -2, -0.5, 0.5, 2, and 4; the fitted power is approximately 1. Both automatic and manual comparisons use the stored cross-language rule of 10^{-8} absolute or 10^{-7} relative agreement. The holdout experiment establishes that future observations do not leak into transformation fitting.

A small second-difference example makes the forecast-boundary question concrete. The observed training levels are 1, 4, and 10. With a next second difference of 2, the first forecast is 18 and the next is 28. A later stored value of 999 is a deliberate holdout sentinel and must not influence either forecast. In the uncertainty check, a random walk with innovation variance 1 has forecast variances 1, 2, and 3 at horizons one, two, and three. Its in-training conditional variances do not accumulate as if the historical observations were unobserved forecasts.

Recovery

Each family is fitted by maximum likelihood (MLE) and Bayesian estimation. For Bayesian recovery, every generating parameter must lie in its central 95% posterior interval, every $\hat{R}$ must be below 1.10, and every effective sample size must be at least 100. These diagnostics check agreement between chains and the independent information in the retained draws. The production sampler settings and declared priors are retained. MLE acceptance is stated separately for each model below.

Autoregressive model

Setup. The AR(1) series contains 1,000 monthly observations beginning in January 2002, generated with seed 12345. Its long-run mean is $\mu = 10$, lag-one coefficient is $\phi = 0.6$, and innovation standard deviation is $\sigma = 5$. There is no response transform, differencing, or covariate.

$$Y_t = \mu + \phi(Y_{t-1} - \mu) + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma^2). \quad (\text{TS.1})$$

Procedure. BestFit estimates the mean, lag coefficient, and innovation scale from the retained series. The MLE uses the required Differential Evolution configuration and unregularized observed information: each fitted quantity must be within 1.96 estimated standard errors of its parent, and the fitted likelihood must be at least as high as the parent likelihood. Bayesian estimation uses the interval and diagnostic rules above. A separate Python least-squares calculation supplies a closed-form conditional optimum.

Results and meaning. Both recovery tests pass. The independent optimum is mean 10.127594, lag coefficient 0.610755, and innovation SD 4.939809. These are external reference values. At the generating parameters, the independent next observation with no new innovation is 5.421625410, and BestFit's recurrence agrees within 10^{-12} . The result checks persistence about a stable mean, not a trending or seasonal AR model.

Moving-average model

Setup. The MA(1) series contains 1,000 monthly observations beginning in January 2003, with seed 12345, mean $\mu = 10$, lag-one innovation coefficient $\theta = 0.6$, and innovation SD $\sigma = 5$. There is no transform, differencing, or covariate. Its specified positive-sign convention is

$$Y_t = \mu + \epsilon_t + \theta\epsilon_{t-1}. \quad (\text{TS.2})$$

Procedure. Innovations are reconstructed recursively from the observations. The conditional sum-of-squares objective is the sum of their squared values, with the Normal density supplying the likelihood and scale terms. BestFit fits the three quantities by MLE and Bayesian estimation. MLE requires each parent within 1.96 unregularized observed-information standard errors and a likelihood no worse than at the parent. Independent Python optimization evaluates the same recurrence.

Results and meaning. Both recovery tests pass. The independent reference optimum is mean 10.073259, MA coefficient 0.610861, and innovation SD 4.941178. The next zero-innovation value at the generating parameters is 5.547955597 and is reproduced within 10^{-12} . Unlike AR persistence, this model's direct dependence on a shock ends after the declared MA lag; the recurrence test checks that distinction.

ARIMA model

Setup. The ARIMA(1,1,1) example contains 1,000 positive daily values beginning on 1 January 2004, with initial level 100 and seed 51037. It takes the natural logarithm of the response, then one difference. The resulting changes have AR coefficient 0.45, MA coefficient 0.25, and innovation SD 0.04. There is no intercept or drift and no covariate. Thus the error scale is in log-response-change units, not the units of the original positive observations.

Procedure. R independently generates the changes and optimizes their conditional likelihood, including the transformation density. BestFit fits the same three parameters. Its maximum must be within the joint 95% likelihood-ratio region of the independent R maximum, with cutoff 7.814728. Each generating parameter must also fall within its independent one-parameter 95% likelihood profile, which allows the other parameters to be refitted while that parameter is varied. Bayesian recovery uses the common central-95% and diagnostic rules, plus an independent data/prior likelihood comparison.

Parameter	Generating value	Independent R optimum	Independent 95% profile interval
AR coefficient	0.45	0.400116	[0.311504, 0.484293]
MA coefficient	0.25	0.323088	[0.232573, 0.408449]
Innovation SD	0.04	0.039205	[0.037546, 0.040990]

Results and meaning. Both recovery tests pass. At the generating parameters, the next zero-innovation log difference is -0.0290494944510721. Adding it to the last observed log level and exponentiating gives the independent next-response target 332.259097907995; BestFit agrees within 10^{-10} . This target verifies reconstruction at known parameters, not an observed forecast from a fitted parameter set. The tests support the stated log/difference combination and its boundary handling.

ARIMAX model

Setup. The ARIMAX(1,1,0) series contains 1,000 daily levels beginning on 1 January 2005, with initial level 10 and seed 51038. It uses one difference, no response transform, and no MA term. A dimensionless dated covariate is the sum of a 17-day sine wave and half a 31-day cosine wave. The expected daily change is $m_t = 0.25 + 1.5X_t$; departures from that expectation have AR coefficient 0.4 and innovation SD 0.5. The covariate stays on its original level scale and is matched by date.

Procedure. Both estimators fit the change intercept, covariate effect, AR coefficient, and error scale. The production and independent R maxima must occupy the four-parameter joint 95% likelihood-ratio region, and the generating point must be in that region; the cutoff is 9.487729. Independent calculations also check same-point likelihood, prior contribution, and the date-aligned forecast. Bayesian recovery applies the same interval and diagnostic rules used above.

Parameter	Generating value	Independent R optimum
Mean-change intercept	0.25	0.236897
Covariate coefficient	1.5	1.540371
AR coefficient	0.4	0.364772
Innovation SD	0.5	0.505265

Results and meaning. Both recovery tests pass. At the generating parameters, the next covariate value is -0.9204878758, the next zero-innovation change is -1.2555476231495, and the reconstructed response is 251.211255651498. BestFit agrees with the forecast recurrence within 10^{-10} . The values in the table are independent R estimates.

Current time-series coverage

Four additional Python comparisons extend beyond the first-order recovery cases. They evaluate known parameter settings and independent objectives; they do not add unperformed higher-order estimator-recovery claims.

Check	Setup	Numerical target and result
First-order conditional objectives	The AR(1) and MA(1) recovery series above; independent residuals, optimum, likelihood, and observed-information uncertainty	Parent and optimum likelihoods and residual prefixes agree within 10^{-9} . The independent parent standardized errors are all below 1.96. Passed.
Second-order AR and MA	1,000 values after 110 discarded steps. AR(2): mean 12, lag coefficients 0.55 and -0.22, error SD 1.75. MA(2): mean 12, lag coefficients 0.5 and -0.28, error SD 1.75	One-step targets are 12.104644614 and 12.027055310. Residuals, likelihoods, and responses agree within 10^{-9} ; independently calculated roots confirm stable/invertible responses. Passed.
Pure AR and MA through ARIMA	The same second-order series evaluated as ARIMA(2,0,0) and ARIMA(0,0,2), without differencing	Both have 998 scored conditional terms. The ARIMA MA initialization differs from the standalone MA's 1,000-term convention and is checked explicitly. Likelihoods and forecasts agree within 10^{-9} . Passed.
Trend, seasonality, and two covariates together	ARIMAX(1,0,1), 1,000 monthly observations from January 2000. Intercept 20; trend 0.015 per step; annual sine/cosine coefficients 3 and -1.5; covariate coefficients 1.25 and -0.8; AR 0.45; MA 0.3; error SD 1.2	Correct date alignment gives log likelihood -1597.134695090. Deliberately matching by array position gives -939039.172443445. The correct 999-term result agrees within 10^{-9} . Passed.

For the last case, the covariates begin one and two months before the response series. This creates a test that can detect matching values by position rather than calendar date. The second-order checks also examine how a single shock propagates through future values, rather than accepting a likelihood comparison alone. The first-order objective check includes an inferior AR optimizer coordinate as a negative control: independently recalculating its likelihood prevents a reported optimizer success from being mistaken for scientific recovery.

Conclusion

The eight recovery tests pass their declared parameter-uncertainty and forecast-recurrence rules. The seventeen independent calculation checks support the specified transformations, date alignment, likelihoods, forecast boundaries, generators, and criteria. The available recovery results do not include a full table of BestFit estimates and credible intervals; external optima are therefore labelled as references throughout this chapter.

A known-parent inclusion result for one dataset does not measure repeated-sample 95% interval coverage. The separate simulation-moment checks assess generated means and variances, not fitted interval coverage. Practical forecast conclusions remain conditional on the tested orders, chronology, transforms, covariates, and Normal innovation law.

CHAPTER 15

Spatial-Extremes Analysis

Regional flood analysis uses information from several sites. The practical questions are whether BestFit represents dependence between sites, handles a missing observation without inventing a flood, and carries uncertainty into predictions at an ungauged location or into a regional frequency curve.

The examples below use synthetic networks with specified generalized extreme-value (GEV) distributions. Location controls the distribution's level, scale its spread, and shape its tail. Flow magnitudes are in consistent numerical units, without assignment to a named river or a discharge unit. Projected recovery-network distances are in kilometres. A Gaussian copula joins the site distributions through dependence between their Normal-transformed probabilities; a latent Gaussian process instead describes spatial departures in a parameter such as location. Those two uses of spatial correlation are checked separately.

The chapter covers 31 verification methods, supported by independent R or Python calculations, analytical targets, or known generating parents.

Likelihood oracle

Network and missing-data setup

The observation-likelihood example contains twelve row/year blocks at five sites. A row is the collection of floods for one synthetic year. All sites have GEV location 100, scale 30, and shape -0.1 in the Hosking/Numerics sign convention. Location and scale are represented by natural-log parameters internally, while shape is used directly. Exponential copula correlation is $\rho(d) = \exp(-d/15)$, where d is distance in the projected coordinate units.

Site	Projected X coordinate	Projected Y coordinate
1	0	0
2	12	3
3	25	0
4	8	18
5	30	22

Some years omit one or more sites. R evaluates the GEV densities of the observations that exist and the Gaussian-copula density using only those sites' correlation submatrix. This is marginalization: the absent site is integrated out, not replaced by zero or another placeholder. A wholly missing year contributes zero to the log likelihood.

Row/year	Sites with observations	Independent row log-likelihood target
5	1, 3, 4, 5	-23.7536861453931
6	2, 3, 4	-14.894701
7	1, 2, 4, 5	-21.970089
8	1, 5	-9.952734

Row/year	Sites with observations	Independent row log-likelihood target
9	None	0

Comparisons and results

Eight methods assess the observation density and the separation between data and prior information. A companion seven-complete-row example removes missingness. A further example gives each site's log-location a fixed spatial departure: 0.05, -0.03, 0.02, -0.04, and 0.01 at sites 1-5. The Gaussian process for those departures has standard deviation 0.06 and range 20. Its probability density is a process/prior contribution, not an extra observed flood.

Comparison	Independent target or rule	BestFit result
Missing-site total likelihood	-238.89272936973; absolute agreement within 10^{-8}	Passed
Missing-site contribution of each year	Each R row within 10^{-10} ; row sum equals total within 10^{-8}	Passed
Seven complete rows with copula	Total -168.321519223767; total tolerance 10^{-8} and row tolerance 10^{-10}	Passed
GEV margins without copula	Missing-data total -238.363666518069 within 10^{-8}	Passed
Observation likelihood with location departures	Observation-only total -169.071665098156 within 10^{-8} , excluding process log density 8.46428373793472	Passed
Equivalent posterior decompositions	Data plus prior gives the same posterior kernel within 10^{-8} under both decompositions	Passed
Total and component decomposition	Year contributions sum to data likelihood; prior components sum to prior; data plus prior equals posterior, within 10^{-8}	Passed
Likelihood gradients	Independently calculated small-perturbation derivatives of the total and summed rows agree within 10^{-4}	Passed

These results check probability calculations for the stated missingness patterns and parameter settings. They do not demonstrate that ignoring an unobserved site's value is appropriate for every missing-data mechanism in an actual flood record.

Information criteria

A separate Bayesian fit of the missing-site example checks that model-comparison criteria use annual vectors as their observation units. There are twelve rows, of which eleven contain data. AIC and BIC are independently reconstructed from the observation likelihood at the sampled posterior mode (MAP); BIC uses eleven nonempty years. WAIC is independently reconstructed from the retained per-year likelihoods. One Pareto diagnostic for leave-one-out assessment is retained per row, including the empty row.

BestFit reports data log likelihood -237.5295, AIC 483.0590, BIC 484.6506, and WAIC 482.7811. AIC/BIC agree with independent calculations within 10^{-8} , and WAIC and its effective-parameter term agree within 10^{-9} relative. Maximum $\hat{R}$ is 1.0004 and minimum effective sample size is 9,233. These are BestFit estimation outputs, not the fixed likelihood targets in the previous table. The criteria check passes.

Prediction, distance, uncertainty, and recovery

Distance and conditional spatial prediction

Before fitting or prediction can be trusted, distances and spatial interpolation must be correct. Two methods check distance conventions. Cartesian distances on the five-site network equal direct Euclidean distances exactly. The geodesic example uses latitude/longitude pairs (38.90, -77.04), (39.29, -76.61), (40.44, -79.99), (37.54, -77.44), and (41.88, -87.63). R applies the haversine formula with Earth radius 6,371.0088 km. Distances agree within 10^{-9} km; exponential correlations with 150-km range and five conditional prediction moments agree within 10^{-10} . Both methods pass.

A third method checks simple kriging: the conditional mean and remaining variance of a Gaussian spatial departure at a target, given the departures at observed sites. R independently solves the covariance equations on the five-site projected network. Five targets, including an observed site, are evaluated under three process settings: SD/range 0.06/20, 0.15/6, and 0.30/80, giving fifteen comparisons. All means and variances agree within 10^{-10} ; uncertainty at the observed site is zero within 10^{-9} .

For example, at target (10, 10) under the first setting, the independent conditional departure is -0.02367145054 with variance 0.001297213485. Adding the departure to log-location $\ln 100$ and exponentiating gives physical location 97.66065206. These reference quantities show why prediction must distinguish a log-scale departure from a physical flow parameter.

Dependent simulation

The simulation check configures a five-site GEV model with exponential copula range 35 and generates 20,000 joint site vectors using seed 20260822. No estimator is run in this check. Each simulated value is mapped through its configured GEV distribution to a Normal score, so empirical intersite correlations can be compared with the specified copula matrix independently of marginal skewness.

Every pairwise correlation is within 0.02 of its configured target. At every site, the empirical 10th, 50th, and 90th percentiles are within 3% of the analytical GEV quantiles. The check passes. It supports the stated simulation law and tolerances, not a fitted model's prediction accuracy for an external river network.

N=1000 recovery

The recovery experiments contain ten sites with 100 complete observations per site. Thus there are 1,000 scalar flood values, but only 100 ten-site row/year vectors. With a copula, each year is one multivariate likelihood contribution; neither the scalar count nor the MCMC draw count is an interchangeable count of independent years.

The independent homogeneous networks use ten sites on a partial Cartesian grid with 25-km spacing: four sites in each of the first two rows and two in the third. Copula and regression networks use the corresponding layout with 100/3-km spacing. The following table specifies all eight estimator-recovery cases. Location and scale are fitted on their natural-log scales.

Estimator and design	Generating location, scale, shape, and additional quantities	Generator seed	BestFit result
MLE, independent homogeneous sites	Location 10,000; scale 3,000; shape -0.1	54321	Passed
MLE, homogeneous sites with copula	Location 10,000; scale 3,000; shape -0.1; exponential range 40 km	66666	Passed
Bayesian, independent homogeneous sites	Location 10,000; scale 3,000; shape -0.1	12345	Passed
Bayesian, homogeneous sites with copula	Location 10,000; scale 3,000; shape -0.1; exponential range 40 km	33333	Passed
Bayesian, location regression	Log-location intercept 8.987; X effect 0.005/km; Y effect 0.008/km; scale 3,000; shape -0.1	66666	Passed
Bayesian, positive shape	Location 10,000; scale 3,000; shape 0.1	11111	Passed
Bayesian, zero shape	Location 10,000; scale 3,000; shape 0, the Gumbel limit	22222	Passed
Bayesian, negative shape	Location 10,000; scale 3,000; shape -0.2	33333	Passed

The regression case gives different site locations according to

$$\ln \mu_j = 8.987 + 0.005X_j + 0.008Y_j, \quad \sigma_j = 3000, \quad \kappa_j = -0.1. \quad (\text{S.1})$$

Here X_j and Y_j are a site's coordinates in kilometres, μ_j is its GEV location, σ_j its GEV scale, and κ_j its shape. The homogeneous models replace that regression with a single common location. The copula models additionally estimate range in $\rho(d) = \exp(-d/\text{range})$. The signed-shape cases exercise positive, zero, and negative values under the declared GEV convention without changing the estimation method.

For MLE, every generating parameter must be within 1.96 standard errors of its fitted value. The standard errors come from unregularized observed-information covariance; singular or regularized covariance is a failure. Bayesian tests require every parent inside its central 95% posterior interval, $\hat{R} < 1.10$, and effective sample size at least 100 for each fitted quantity. If an already valid interval is narrower than 5% of a nonzero parent's magnitude, the secondary point estimate must also be within 5%; that rule never replaces the interval requirement. Every parent must lie inside its unchanged parameter bounds and prior support.

MLE uses Differential Evolution with estimator seed 12345. Bayesian estimation uses production DEMCzs settings: 3,500 iterations, 1,750 warmup iterations, retained output length 10,000, and seed 12345. Three-, four-, and five-parameter cases have 6, 8, and 10 chains, respectively, with thinning 30, 40, and 50 and initial populations 300, 400, and 500. All eight recovery tests pass. The available results do not include fitted coordinates or interval endpoints, so the table reports the generating parents and the outcomes of the declared comparisons.

These recovery models have no latent spatial-error field. Their intervals concern uncertainty in estimated parameters, not residual spatial-process variation or a future year's flood. Inclusion of truth in one generated dataset does not establish repeated-sample 95% coverage.

Independent correlation and held-out prediction checks

Three analytical comparisons ask whether the selected correlation model gives the intended dependence at distances 0, 5, 10, 20, and 30, using range 20. All five values per family agree with the independent Python calculation within 10^{-9} .

Correlation family	Definition or characteristic	Reference at distance 10	Reference at distance 30
Exponential	$\rho(d) = \exp(-d/20)$	0.606530660	0.223130160
Powered exponential	$\rho(d) = \exp[-(d/20)^{1.6}]$	0.719012183	0.147616623
Spherical	Cubic decrease to exactly zero at range 20, and zero beyond	0.312500000	0

Two further checks assess prediction at a withheld site. The copula example fits 120 complete three-site years at projected coordinates (0, 0), (10, 0), and (0, 12), then predicts at (9, 9). An independent SciPy fit supplies both the optimum and an unregularized uncertainty calculation. BestFit's likelihood at that optimum agrees within 10^{-7} ; its fitted optimum lies within the four-parameter joint 95% likelihood-ratio region, whose cutoff is 9.487729.

Held-out annual exceedance probability	Independent quantile target	Independent standard error	BestFit comparison
0.50	20.635440	0.232776	BestFit within 1.96 standard errors; passed
0.10	25.306526	0.433969	BestFit within 1.96 standard errors; passed
0.02	29.205737	0.823428	BestFit within 1.96 standard errors; passed

The covariate example uses four fixed site log-locations 2.1, 2.5, 2.7, and 3.15, with two covariates per site: (-1, 0), (0, 1), (1, -1), and (2, 0.5). Independent ordinary least squares predicts the held-out covariate row (0.5, -0.25). The target log-location is 2.586206897, or physical location 13.27930617; BestFit agrees within 10^{-9} . Parameter prediction variance is 0.00001808363, while adding residual variation gives 0.00007555489. This check passes and keeps those uncertainty sources distinct. It does not fit a GEV sample at each of the four supplied site summaries.

Posterior prediction and uncertainty

Five independent checks distinguish uncertainty about parameters, unobserved spatial departures, and future observations. Fixed-draw examples verify the calculation applied to supplied draws; they assess prediction formulas separately from posterior estimation.

Calculation	Setup, independent procedure, and result
Ungauged prediction for each draw	Three latitude/longitude sites near Denver, Boulder, and Colorado Springs predict at (39.5501, -105.7821). Four fixed process-parameter draws each get an independent haversine/covariance solve and log-link transformation. All conditional means, variances, and physical locations agree within 10^{-9} . Passed.
Regional posterior curve	Nine fixed parameter draws, three site covariates (-1, 0.5, 1.5), and exceedance probabilities 0.5, 0.1, and 0.02. For each draw, calculate each site's GEV quantile and then average across sites; summarize those regional averages with linearly interpolated 2.5th/97.5th percentiles. Means and endpoints agree within 10^{-9} . Passed.
Godambe covariance	Three sites and 24 complete years. Independent finite differences give likelihood curvature H and the sum of annual score outer products J . The unregularized covariance $H^{-1} J H^{-1}$ agrees with BestFit within 2×10^{-5} relative; reconstruction from stored matrices agrees within 10^{-9} . Passed.

Calculation	Setup, independent procedure, and result
Temporal block bootstrap	Twelve three-site annual vectors are resampled in wrapping blocks of four complete years, preserving the sites within each row. Five fixed replicates are fitted independently in SciPy and by BestFit. All five BestFit fits succeed; physical-parameter, site-quantile, and regional-quantile interval endpoints meet 2% relative tolerance, with 0.02 absolute tolerance near zero. Passed.
Variance inflation	Ten three-site years give independent average absolute intersite correlation 0.9997195942. The resulting variance inflation factor is 2.9994391884. Site and regional interval transformations agree with the independent calculation within 10^{-9} . Passed.

The regional fixed-draw example yields the following independent targets. Because each draw's regional average is formed before taking percentiles, the interval is for the regional average curve, not a collection of site intervals and not a region-wide flood maximum.

Exceedance probability	Regional mean quantile	Central 95% interval from the supplied draws
0.50	21.951292	[20.841282, 23.119005]
0.10	27.064969	[25.834934, 28.310139]
0.02	31.247433	[29.648213, 32.894444]

Variance inflation keeps an interval's midpoint and multiplies its half-width by $\sqrt{2.9994391884} = 1.731888908$; scale lower limits are truncated at zero. The five-replicate bootstrap is a reproducible calculation check, not a sufficiently large experiment to establish the long-run coverage of bootstrap intervals.

Conclusion

The 31 methods comprise eight likelihood checks, one criteria check, three distance/kriging checks, one simulation check, eight estimator recoveries, three correlation checks, two held-out prediction checks, and five posterior-prediction/uncertainty checks. The calculations and recovery outcomes support the stated network sizes, missingness patterns, covariates, dependence laws, and uncertainty sources. Large-network performance, recovery of a latent spatial-error field, and simultaneous predictive coverage remain outside the demonstrated recovery evidence.

CHAPTER 16

Evidence Boundaries and Conclusions

Supported conclusions

The 329-method verification library provides analytical, independently implemented, external-package, published-source, and recovery evidence across the model families described in this report. The strongest conclusions concern a specified calculation under a defined parameterization: distribution functions and likelihoods, estimator objectives and covariance, diagnostic statistics, recurrence relations, and uncertainty transformations.

Recovery experiments add evidence that the tested estimators recover their generating parameters or responses under the declared sample sizes, seeds, priors, and acceptance rules. Published examples connect these calculations to established flood-frequency applications. Appendix A gives the complete correspondence between report scope and executable tests.

Limits of the supported claims

Area	Boundary
Bulletin 17C intervals	Numerical Cohn interval values and broad repeated-sample interval coverage are not established by the current evidence.
Bulletin 17C bootstrap fitting	Accepted output is not synonymous with outer GMM convergence; 53 of 1,000 fits in the inspected experiment reached the iteration cap.
Published GEV comparison	The systematic-only Kamp example tests a 1,000-year lower endpoint of 163 m ³ /s, while Skahill et al. (2016), Table 2, gives 183 m ³ /s. That endpoint does not establish agreement with the published value.
Joint-prior sampling	Independent parameter-prior draws do not sample additional coupled quantile, Jeffreys, or spatial prior factors.
GMM deletion influence	The one-step magnitude is not calibrated to exact deletion scale, although the tested outlier-ranking result is supported.
Competing risks	Five recovery methods cover three specified generating designs. General correlated Bayesian recovery is not established.
Bivariate estimation	Copula fitting conditions on fitted marginals; joint marginal-copula posterior estimation is not established.
Rating curves	Independent likelihood and recovery evidence does not establish equivalence to R ² , repeated-realization simultaneous coverage, or extrapolation calibration.
Spatial extremes	Large-network performance, latent-error recovery, and simultaneous predictive coverage are not established by the ten-site recovery designs.
Spatial weighting	Correlation weights control relative site influence; they are not pairwise composite likelihood, effective sample size, or a Godambe correction.
PSIS-LOO	Pareto diagnostics are supplied, but exact leave-one-out refits, moment matching, and chain-relative-efficiency adjustment are not performed.
Engineering application	Numerical verification does not establish the physical adequacy of a model for a particular watershed or risk decision.

CHAPTER 17

Appendix A. Complete Verification Test Index

The source inventory audited on 27 September 2026 contains **329 methods in 71 active test classes**. Every executable verification method appears exactly once below; helper methods are not counted as tests.

The catalog records 327 methods as verified and 2 as accepted limitations. **Verified** identifies a check with a declared oracle, acceptance rule, and verified catalog disposition. Acceptance rules, sample sizes, seeds, oracle descriptions, artifact paths, and original evidence-ledger anchors remain in the machine-readable catalog (source baseline; current snapshot attached). The PDF embeds the current catalog as an attachment; its repository links identify the source-review baseline. Each method links to the source file containing its assertions, including partial classes.

Report area	Active classes	Methods
Estimation and diagnostics	20	56
Distribution fitting	8	52
Univariate recovery	3	23
Published univariate comparisons	2	14
Bulletin 17C	5	41
Point process	4	13
Competing risks	2	9
Mixtures	2	7
Composite analyses	2	14
Bivariate and coincident frequency	3	25
Rating curves	5	19
Time-series models	7	25
Spatial extremes	8	31
Total	71	329

Estimation and diagnostics

Interpretation and results: Estimation and diagnostics.

BayesianAnalysisRecoveryTests

Source: BayesianAnalysisRecoveryTests.cs. Namespace: `RMC.BestFit.Verification.ModelEstimation`.

Scope: UnivariateDistribution (Normal).

Method	Evidence	Disposition
<code>Test_Estimate_PosteriorMean_NearTrueValue</code>	Recovery	Verified
<code>Test_N1000_DefaultPosteriorIntervals_ContainGeneratingParameters</code>	Recovery	Verified

BayesianConjugateOracleVerificationTests

Source: BayesianConjugateOracleVerificationTests.cs. Namespace: `RMC.BestFit.Verification.ModelEstimation`.

Scope: Known-scale Normal mean with Normal(120,5) location prior.

Method	Evidence	Disposition
<code>InformativePrior_NormalNormalConjugatePosterior_MatchesClosedForm</code>	Analytical	Verified

GmmGeneratedParentRecoveryTests

Source: GmmGeneratedParentRecoveryTests.cs. Namespace: `RMC.BestFit.Verification.ModelEstimation`.

Scope: One-parameter two-step GMM with two response/instrument moments.

Method	Evidence	Disposition
<code>TwoStepGmm_RecoversGeneratedParent_WithinStandardizedError</code>	Recovery	Verified

GmmInfluenceDiagnosticsVerificationTests

Source: GmmInfluenceDiagnosticsVerificationTests.cs. Namespace: `RMC.BestFit.Verification.ModelEstimation`.

Scope: Bulletin17CDistribution (Log10-Normal moments with quadratic penalty); Bulletin17CDistribution (Log10-Normal moments).

Method	Evidence	Disposition
<code>Log10NormalObservationInfluence_MatchesRGmmOracle</code>	External package	Verified
<code>VanishingCenteredPenalty_PreservesObservationInfluenceScale</code>	Analytical	Verified

GmmObjectiveGradientVerificationTests

Source: GmmObjectiveGradientVerificationTests.cs. Namespace: `RMC.BestFit.Verification.ModelEstimation`.

Scope: Inline one-parameter nonlinear moment model.

Method	Evidence	Disposition
<code>PenalizedObjectiveGradient_MatchesIndependentCentralDifference</code>	Analytical	Verified
<code>UnpenalizedEstimatingGradient_IsHalfConventionalObjectiveDerivative</code>	Analytical	Verified

GmmSpecificationVerificationTests

Source: GmmSpecificationVerificationTests.cs. Namespace: `RMC.BestFit.Verification.ModelEstimation`.

Scope: One-parameter, two-moment overidentified GMM fixture.

Method	Evidence	Disposition
<code>HansenJ_MatchesRGmmSelectedWeightStatistic</code>	External package	Verified
<code>OveridentifiedFixedWeightSandwichCovariance_MatchesRGmmOracle</code>	External package	Verified
<code>OveridentifiedOneStep_MatchesRGmmFixedWeightOracle</code>	External package	Verified
<code>OveridentifiedTwoStepSandwichCovariance_MatchesRGmmOracle</code>	External package	Verified

InformationCriterionOracleTests

Source: InformationCriterionOracleTests.cs. Namespace: RMC.BestFit.Verification.ModelEstimation.

Scope: UnivariateDistribution (Normal) with fixed posterior draws.

Method	Evidence	Disposition
DIC_MatchesRBayesianToolsOracle	External package	Verified
WAIC_MatchesRLooOracle	External package	Verified

JointPriorSamplingVerificationTests

Source: JointPriorSamplingVerificationTests.cs. Namespace: RMC.BestFit.Verification.ModelEstimation.

Scope: Soft-coupled two-parameter prior model.

Method	Evidence	Disposition
SampleFromPriors_SoftJointPrior_CurrentlyDrawsIndependentMarginals	Analytical	Accepted limitation

Limitation: The generic sampler draws independent marginal priors and does not sample the additional soft joint-prior coupling.

Log10NormalEstimationEquivalenceTests

Source: Log10NormalEstimationEquivalenceTests.cs. Namespace: RMC.BestFit.Verification.ModelEstimation.

Scope: Log10-Normal UnivariateDistribution / Bulletin17CDistribution.

Method	Evidence	Disposition
FlatPriorMap_MatchesClosedFormLog10NormalMle	Analytical	Verified
GmmMuPenaltyRegimes_MatchAnalyticalFitAndVarianceInfluence	Analytical	Verified
MapAndGmmMuPosterior_InverseVarianceReferenceInsideCentral95Intervals	Analytical	Verified
MapMuPriorRegimes_MatchAnalyticalFitAndVarianceInfluence	Analytical	Verified
UnpenalizedB17CGmm_MatchesExactSampleMoments	Analytical	Verified

Log10NormalInfluenceVerificationTests

Source: Log10NormalInfluenceVerificationTests.cs. Namespace: RMC.BestFit.Verification.ModelEstimation.

Scope: Bulletin17CDistribution (Log10-Normal moments).

Method	Evidence	Disposition
GmmObservationInfluence_CurrentMagnitudeDiffersFromExactDeletionScale	Analytical	Accepted limitation

Limitation: The GMM one-step influence magnitude is not calibrated to exact deletion scale; the tested outlier-ranking result remains valid.

MLEIntegrationTests

Source: MLEIntegrationTests.cs. Namespace: RMC.BestFit.Verification.ModelEstimation.

Scope: UnivariateDistribution (Exponential); UnivariateDistribution (GammaDistribution); UnivariateDistribution (GeneralizedExtremeValue); UnivariateDistribution (GeneralizedLogistic); UnivariateDistribution (GeneralizedNormal); UnivariateDistribution (GeneralizedPareto); UnivariateDistribution (Gumbel); UnivariateDistribution (LnNormal); UnivariateDistribution (LogPearsonTypeIII); UnivariateDistribution (Logistic); UnivariateDistribution (Normal); UnivariateDistribution (PearsonTypeIII); UnivariateDistribution (Weibull).

Method	Evidence	Disposition
Test_MLE_Exponential_RecoversTrueParameters	Recovery	Verified
Test_MLE_GEV_RecoversTrueParameters	Recovery	Verified
Test_MLE_Gamma_RecoversTrueParameters	Recovery	Verified
Test_MLE_GeneralizedLogistic_RecoversTrueParameters	Recovery	Verified
Test_MLE_GeneralizedNormal_RecoversTrueParameters	Recovery	Verified
Test_MLE_GeneralizedPareto_RecoversTrueParameters	Recovery	Verified
Test_MLE_Gumbel_RecoversTrueParameters	Recovery	Verified
Test_MLE_LnNormal_MatchesClosedFormLogSampleMLE	Analytical	Verified
Test_MLE_LnNormal_RecoversGeneratingParameters	Recovery	Verified
Test_MLE_LogPearsonTypeIII_RecoversTrueParameters	Recovery	Verified
Test_MLE_Logistic_RecoversTrueParameters	Recovery	Verified
Test_MLE_Normal_RecoversTrueParameters	Recovery	Verified
Test_MLE_PearsonTypeIII_RecoversTrueParameters	Recovery	Verified
Test_MLE_Weibull_RecoversTrueParameters	Recovery	Verified

MaximumAPosteriorInformationCriteriaOracleTests

Source: MaximumAPosteriorInformationCriteriaOracleTests.cs. Namespace: RMC.BestFit.Verification.ModelEstimation.

Scope: Inline Normal likelihood with bounded Uniform priors.

Method	Evidence	Disposition
AIC_UsesDataLikelihoodAtMap_ExcludesPriorDensity	Analytical	Verified
BIC_UsesDataLikelihoodAtMap_ExcludesPriorDensity	Analytical	Verified

MaximumAPosterioriRecoveryTests

Source: MaximumAPosterioriRecoveryTests.cs. Namespace: `RMC.BestFit.Verification.ModelEstimation`.

Scope: Inline Normal likelihood with bounded Uniform priors.

Method	Evidence	Disposition
<code>Test_Estimate_NormalData_FindsMode</code>	Recovery	Verified

MaximumLikelihoodCovarianceVerificationTests

Source: MaximumLikelihoodCovarianceVerificationTests.cs. Namespace: `RMC.BestFit.Verification.ModelEstimation`.

Scope: Known-scale Normal mean model; Two-parameter bounded quadratic likelihood.

Method	Evidence	Disposition
<code>MLE_AndFlatPriorMAP_ReportTheSameCovariance</code>	Analytical	Verified
<code>MLE_OneParameterModel_CovarianceMatchesClosedFormInformation</code>	Analytical	Verified

McmcNumericalVerificationTests

Source: McmcNumericalVerificationTests.cs. Namespace: `RMC.BestFit.Verification.ModelEstimation`.

Scope: Nine deterministic multi-chain diagnostic fixtures; Two-parameter smooth Gaussian-plus-coupled-prior target; Two-parameter spike target.

Method	Evidence	Disposition
<code>Arwmh_BestFitWiringRecordsEveryRealizedStateInAdaptiveCovariance</code>	Published source	Verified
<code>ConservativeEss_MatchesRPosteriorBulkAndTailOracle</code>	External package	Verified
<code>Nuts_BestFitNumericalGradientMatchesPosteriorGradient</code>	Analytical	Verified
<code>RankNormalizedRhat_MatchesRPosteriorOracle</code>	External package	Verified

PosteriorResamplingVerificationTests

Source: PosteriorResamplingVerificationTests.cs. Namespace: `RMC.BestFit.Verification.ModelEstimation`.

Scope: Composite competing-risk maximum of two Normal child analyses; Independent Normal copula marginals with additive response surface.

Method	Evidence	Disposition
<code>CoincidentFrequencyPosteriorResampling_MatchesIndependentClosedFormOracle</code>	Independent calculation	Verified
<code>CompositePosteriorResampling_MatchesIndependentCartesianOracle</code>	Independent calculation	Verified

ProfileLikelihoodGridPointFailureTests

Source: ProfileLikelihoodGridPointFailureTests.cs. Namespace: RMC.BestFit.Verification.ModelEstimation.

Scope: Support-restricted correlated quadratic likelihood.

Method	Evidence	Disposition
MAP_ProfileLikelihood_SupportedGridMatchesClosedFormProfile	Analytical	Verified
MLE_ProfileLikelihood_SupportedGridMatchesClosedFormProfile	Analytical	Verified

ProfileLikelihoodVerificationTests

Source: ProfileLikelihoodVerificationTests.cs. Namespace: RMC.BestFit.Verification.ModelEstimation.

Scope: Correlated two-parameter quadratic likelihood.

Method	Evidence	Disposition
MAP_ProfileLikelihood_WithFlatPriors_MatchesRTrueProfile	External package	Verified
MAP_ProfileLikelihood_WithInformativePrior_ProfilesFullPosteriorKernel	Analytical	Verified
MLE_ProfileLikelihood_MatchesRTrueProfile	External package	Verified

ProfileQRecoveryTests

Source: ProfileQRecoveryTests.cs. Namespace: RMC.BestFit.Verification.ModelEstimation.

Scope: Bulletin17CDistribution (LogPearsonTypeIII).

Method	Evidence	Disposition
TrueProfile_LP3Q99_RecoversGeneratingResponse	Recovery	Verified

PsisLooOracleVerificationTests

Source: PsisLooOracleVerificationTests.cs. Namespace: RMC.BestFit.Verification.ModelEstimation.

Scope: Fixed Normal pointwise log-likelihood matrix.

Method	Evidence	Disposition
PSISL00_MatchesRLooOracle	External package	Verified
ParetoInfluence_UsesRLoo210DiagnosticThreshold	External package	Verified
PsisTailRegimes_MatchRLooOracle	External package	Verified
RLooOracle_InternalIdentitiesAreConsistent	Independent calculation	Verified

Distribution fitting

Interpretation and results: Distribution fitting.

FittingAnalysisCriteriaVerificationTests

Source: FittingAnalysisCriteriaVerificationTests.cs. Namespace: `RMC.BestFit.Verification.DistributionFitting`.

Scope: DataFrame exact-series Weibull plotting positions; Gumbel, Normal, and Logistic candidates on common data; Gumbel, Normal, and Logistic criteria/ranking/weights.

Method	Evidence	Disposition
<code>CriteriaRankingWeightsAndConfiguredOrder_MatchIndependentOracle</code>	External package	Verified
<code>ExactSeriesReplacement_ProducesAnalyticalWeibullPositions</code>	Analytical	Verified
<code>FittedParameters_MatchIndependentOracle</code>	External package	Verified

FittingAnalysisRecoveryTests

Source: FittingAnalysisRecoveryTests.cs. Namespace: `RMC.BestFit.Verification.DistributionFitting`.

Scope: Exponential(xi=10, alpha=25); Gamma(theta=5, kappa=3); GeneralizedExtremeValue(xi=50, alpha=15, kappa=0.1); GeneralizedLogistic(xi=75, alpha=10, kappa=0.15); GeneralizedNormal(xi=50, alpha=10, kappa=-0.3); GeneralizedPareto(xi=0, alpha=20, kappa=0.15); Gumbel(xi=50, alpha=15); KappaFour(xi=100, alpha=20, kappa=-0.1, hondo=0.1); LnNormal(real-space mean=3.5, real-space standard deviation=0.4; covariance coordinates Mu, Sigma^2); LogNormal base-10(mu=3, sigma=0.5); LogPearsonTypeIII(moment coordinates mu=2, sigma=0.3, gamma=0.5; MLE covariance coordinates Mu, 1/Beta, Alpha); Logistic(xi=75, alpha=10); Normal(mu=100, sigma=15); PearsonTypeIII(moment coordinates mu=100, sigma=20, gamma=0.8; MLE covariance coordinates Mu, 1/Beta, Alpha); Weibull(lambda=100, kappa=2.5).

Method	Evidence	Disposition
<code>Exponential_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>Gamma_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>GeneralizedExtremeValue_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>GeneralizedLogistic_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>GeneralizedNormal_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>GeneralizedPareto_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>Gumbel_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>KappaFour_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>LnNormal_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>LogNormal_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>LogPearsonTypeIII_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>Logistic_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>Normal_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>PearsonTypeIII_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified
<code>Weibull_N1000_FittingAnalysisRecoversGeneratingFamily</code>	Recovery	Verified

GoodnessOfFitRmseVerificationTests

Source: GoodnessOfFitRmseVerificationTests.cs. Namespace: `RMC.BestFit.Verification.DistributionFitting`.

Scope: Parameter-adjusted RMSE.

Method	Evidence	Disposition
<code>ParameterAdjustedRmse_UsesAllResidualsAndIsPermutationInvariant</code>	Analytical	Verified

KappaFourZeroShapeVerificationTests

Source: KappaFourZeroShapeVerificationTests.cs. Namespace: `RMC.BestFit.Verification.DistributionFitting`.

Scope: KappaFour ($\xi=0$, $\alpha=1$, $\kappa=0$, $\text{hondo}=0.2$).

Method	Evidence	Disposition
<code>ZeroShapeNonzeroHondo_PdfMatchesAnalyticalDerivative</code>	Analytical	Verified
<code>ZeroShapeNonzeroHondo_QuantileInvertsCdf</code>	Analytical	Verified

LmomcoDistributionFittingVerificationTests

Source: LmomcoDistributionFittingVerificationTests.cs. Namespace: `RMC.BestFit.Verification.DistributionFitting`.

Scope: UnivariateDistribution (GeneralizedLogistic); UnivariateDistribution (GeneralizedNormal).

Method	Evidence	Disposition
<code>GeneralizedLogistic_MleAndDistributionFunctionsMatchLmomco</code>	External package	Verified
<code>GeneralizedNormal_MleAndDistributionFunctionsMatchLmomco</code>	External package	Verified

Log10NormalFittingVerificationTests

Source: Log10NormalFittingVerificationTests.cs. Namespace: RMC.BestFit.Verification.DistributionFitting.

Scope: UnivariateDistribution (LogNormal, base-10).

Method	Evidence	Disposition
ClosedFormMle_LikelihoodCdfAndQuantileMatchAnalyticalOracle	Analytical	Verified

ScipyDistributionFittingVerificationTests

Source: ScipyDistributionFittingVerificationTests.cs. Namespace: RMC.BestFit.Verification.DistributionFitting.

Scope: UnivariateDistribution (Exponential); UnivariateDistribution (GammaDistribution); UnivariateDistribution (GeneralizedExtremeValue); UnivariateDistribution (GeneralizedPareto); UnivariateDistribution (Gumbel); UnivariateDistribution (KappaFour); UnivariateDistribution (LnNormal); UnivariateDistribution (LogNormal); UnivariateDistribution (LogPearsonTypeIII); UnivariateDistribution (Logistic); UnivariateDistribution (Normal); UnivariateDistribution (PearsonTypeIII); UnivariateDistribution (Weibull).

Method	Evidence	Disposition
Exponential_MleAndDistributionFunctionsMatchScipy	External package	Verified
Gamma_MleAndDistributionFunctionsMatchScipy	External package	Verified
GeneralizedExtremeValue_MleAndDistributionFunctionsMatchScipy	External package	Verified
GeneralizedPareto_MleAndDistributionFunctionsMatchScipy	External package	Verified
Gumbel_MleAndDistributionFunctionsMatchScipy	External package	Verified
KappaFour_MleAndDistributionFunctionsMatchScipy	External package	Verified
LnNormal_MleAndDistributionFunctionsMatchScipy	External package	Verified
LogNormal_MleAndDistributionFunctionsMatchScipy	External package	Verified
LogPearsonTypeIII_MleAndDistributionFunctionsMatchScipy	External package	Verified
Logistic_MleAndDistributionFunctionsMatchScipy	External package	Verified
Normal_MleAndDistributionFunctionsMatchScipy	External package	Verified
PearsonTypeIII_MleAndDistributionFunctionsMatchScipy	External package	Verified
Weibull_MleAndDistributionFunctionsMatchScipy	External package	Verified

UnivariateDistributionMLETests

Source: UnivariateDistributionMLETests.cs. Namespace: `RMC.BestFit.Verification.DistributionFitting`.

Scope: Exponential (xi, alpha); GammaDistribution (theta, kappa); GeneralizedExtremeValue (xi, alpha, kappa); GeneralizedLogistic (xi, alpha, kappa); GeneralizedNormal (xi, alpha, kappa); GeneralizedPareto (xi, alpha, kappa); Gumbel (xi, alpha); KappaFour (xi, alpha, kappa, hondo); LnNormal (natural-log mu, sigma); LogNormal (base-10-log mu, sigma); LogPearsonTypeIII (base-10-log mu, sigma, gamma); Logistic (xi, alpha); Normal (mu, sigma); PearsonTypeIII (mu, sigma, gamma); Weibull (lambda, kappa).

Method	Evidence	Disposition
Test_Exponential_MLE	Published source	Verified
Test_GammaDist_MLE	Published source	Verified
Test_GeneralizedExtremeValue_MLE	Published source	Verified
Test_GeneralizedLogistic_MLE	Published source	Verified
Test_GeneralizedNormal_MLE	Published source	Verified
Test_GeneralizedPareto_MLE	Published source	Verified
Test_Gumbel_MLE	Published source	Verified
Test_Kappa4_MLE	Published source	Verified
Test_LnNormal_MLE	Published source	Verified
Test_LogNormal_MLE	Published source	Verified
Test_LogPearsonTypeIII_MLE	Published source	Verified
Test_Logistic_MLE	Published source	Verified
Test_Normal_MLE	Published source	Verified
Test_PearsonTypeIII_MLE	Published source	Verified
Test_Weibull_MLE	Published source	Verified

Univariate recovery

Interpretation and results: Univariate recovery.

NonstationaryParentTrendCoverageTests

Source: NonstationaryParentTrendCoverageTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.ValidationTests`.

Scope: GeneralizedExtremeValue with constant location/scale and linear shape; GeneralizedPareto with linear location, exponential scale, and constant shape; LogPearsonTypeIII with linear log-mean, exponential log-scale, and constant skew; Normal with constant mean and reciprocal scale; Normal with constant mean and sinusoidal scale.

Method	Evidence	Disposition
GeneralizedExtremeValue_LinearShape_RecoversParentResponses	Recovery	Verified
GeneralizedPareto_LinearLocationExponentialScale_RecoversParentResponses	Recovery	Verified
LogPearsonTypeIII_LinearLogMeanExponentialLogScale_RecoversParentResponses	Recovery	Verified
Normal_ReciprocalSigma_RecoversParentResponses	Recovery	Verified
Normal_SinusoidalSigma_RecoversParentResponses	Recovery	Verified

NonstationaryValidationTests

Source: NonstationaryValidationTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.ValidationTests`.

Scope: Normal with reciprocal mean and constant scale; Normal; constant mean and constant sigma trends; Reciprocal trend $f(t)=1/(a+b(t-\text{StartIndex}))$.

Method	Evidence	Disposition
Nonstationary_ConstantTrend_RecoversTrueParameters	Recovery	Verified
Nonstationary_ReciprocalMeanTrend_RecoversParentResponses	Recovery	Verified
ReciprocalTrend_ValuesAndParameterDerivativesMatchAnalyticalFormula	Analytical	Verified

UnivariateValidationTests

Source: UnivariateValidationTests.cs. Namespace: RMC.BestFit.Verification.Univariate.ValidationTests.

Scope: UnivariateDistribution (Exponential), stationary; UnivariateDistribution (Gamma), stationary; UnivariateDistribution (GeneralizedExtremeValue), stationary; UnivariateDistribution (GeneralizedLogistic), stationary; UnivariateDistribution (GeneralizedNormal), stationary; UnivariateDistribution (GeneralizedPareto), stationary; UnivariateDistribution (Gumbel), stationary; UnivariateDistribution (KappaFour), stationary; UnivariateDistribution (LnNormal), stationary; UnivariateDistribution (LogNormal), stationary; UnivariateDistribution (LogPearsonTypeIII), stationary; UnivariateDistribution (Logistic), stationary; UnivariateDistribution (Normal), stationary; UnivariateDistribution (PearsonTypeIII), stationary; UnivariateDistribution (Weibull), stationary.

Method	Evidence	Disposition
Exponential_RecoversTrueParameters	Recovery	Verified
Gamma_RecoversTrueParameters	Recovery	Verified
GeneralizedExtremeValue_RecoversTrueParameters	Recovery	Verified
GeneralizedLogistic_RecoversTrueParameters	Recovery	Verified
GeneralizedNormal_RecoversTrueParameters	Recovery	Verified
GeneralizedPareto_RecoversTrueParameters	Recovery	Verified
Gumbel_RecoversTrueParameters	Recovery	Verified
KappaFour_RecoversTrueParameters	Recovery	Verified
LnNormal_RecoversTrueParameters	Recovery	Verified
LogNormal_RecoversTrueParameters	Recovery	Verified
LogPearsonTypeIII_RecoversTrueParameters	Recovery	Verified
Logistic_RecoversTrueParameters	Recovery	Verified
Normal_RecoversTrueParameters	Recovery	Verified
PearsonTypeIII_RecoversTrueParameters	Recovery	Verified
Weibull_RecoversTrueParameters	Recovery	Verified

Published univariate comparisons

Interpretation and results: Published univariate comparisons.

ArrFlikeTests

Source: ArrFlikeTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.VerificationReportTests`.

Scope: GeneralizedExtremeValue with low-outlier censoring; ARR Flike example 6b; GeneralizedExtremeValue; ARR Flike example 6a; LogPearsonTypeIII with historical threshold; ARR Flike example 4; LogPearsonTypeIII with regional-skew prior; ARR Flike example 5; LogPearsonTypeIII; ARR Flike example 3.

Method	Evidence	Disposition
TestExample3	External package	Verified
TestExample4	External package	Verified
TestExample5	External package	Verified
TestExample6a	External package	Verified
TestExample6b	External package	Verified

ViglioneEtAlTests

Source: ViglioneEtAlTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.VerificationReportTests`.

Scope: GeneralizedExtremeValue; Viglione information-expansion case 1; GeneralizedExtremeValue; Viglione information-expansion case 2; GeneralizedExtremeValue; Viglione information-expansion case 3; GeneralizedExtremeValue; Viglione information-expansion case 4; GeneralizedExtremeValue; Viglione information-expansion case 5; GeneralizedExtremeValue; Viglione information-expansion case 6; GeneralizedExtremeValue; Viglione information-expansion case 7; GeneralizedExtremeValue; Viglione information-expansion case 8; GeneralizedExtremeValue; Viglione information-expansion case 9.

Method	Evidence	Disposition
Test1_ExactData_1951_2001	Published source	Verified
Test2_ExactData_1951_2005	Published source	Verified
Test3_ExactData_1951_2001_TemporalExpansion	Published source	Verified
Test4_ExactData_1951_2005_TemporalExpansion	Published source	Verified
Test5_ExactData_1951_2001_CausalExpansion	Published source	Verified
Test6_ExactData_1951_2005_CausalExpansion	Published source	Verified
Test7_ExactData_1951_2001_TemporalAndCausalExpansion	Published source	Verified
Test8_ExactData_1951_2005_TemporalAndCausalExpansion	Published source	Verified
Test9_ExactData_1951_2001_CausalExpansion_TheePriors	External package	Verified

Bulletin 17C

Interpretation and results: Bulletin 17C.

B17CCovarianceTests

Source: B17CCovarianceTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.Bulletin17CTests`.

Scope: Exponential GMM covariance; Gamma GMM covariance; LogNormal GMM covariance; LogPearsonTypeIII GMM covariance; Normal GMM covariance; PearsonTypeIII GMM covariance.

Method	Evidence	Disposition
Exponential_Covariance_N100	Analytical	Verified
Exponential_Covariance_N25	Analytical	Verified
Gamma_Covariance_N100	Analytical	Verified
Gamma_Covariance_N25	Analytical	Verified
LogNormal_Covariance_N100	Analytical	Verified
LogNormal_Covariance_N25	Analytical	Verified
LogPearsonTypeIII_Covariance_Example1	Analytical	Verified
LogPearsonTypeIII_Covariance_N100	Analytical	Verified
LogPearsonTypeIII_Covariance_N25	Analytical	Verified
Normal_Covariance_N100	Analytical	Verified
Normal_Covariance_N25	Analytical	Verified
PearsonTypeIII_Covariance_N100	Analytical	Verified
PearsonTypeIII_Covariance_N25	Analytical	Verified

B17CExampleTests

Source: B17CExampleTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.Bulletin17CTests`.

Scope: LogPearsonTypeIII; Bulletin 17C worked Example 1; LogPearsonTypeIII; Bulletin 17C worked Example 2; LogPearsonTypeIII; Bulletin 17C worked Example 3; LogPearsonTypeIII; Bulletin 17C worked Example 4; LogPearsonTypeIII; Bulletin 17C worked Example 5; LogPearsonTypeIII; Bulletin 17C worked Example 6; LogPearsonTypeIII; Bulletin 17C worked Example 7.

Method	Evidence	Disposition
Test_Example1	Published source	Verified
Test_Example2	Published source	Verified
Test_Example3	Published source	Verified
Test_Example4	Published source	Verified
Test_Example5	Published source	Verified
Test_Example6	Published source	Verified
Test_Example7	Published source	Verified

B17CPenaltyTests

Source: B17CPenaltyTests.cs. Namespace: RMC.BestFit.Verification.Univariate.Bulletin17CTests.

Scope: LogNormal regional parameter/quantile penalties; LogPearsonTypeIII regional parameter/quantile penalties.

Method	Evidence	Disposition
LogNormal_PenaltyOnMuAndSigma_N100	Analytical	Verified
LogNormal_PenaltyOnMuAndSigma_N25	Analytical	Verified
LogNormal_PenaltyOnMuSigmaAndQ99_N25	Analytical	Verified
LogNormal_PenaltyOnMu_N100	Analytical	Verified
LogNormal_PenaltyOnMu_N25	Analytical	Verified
LogNormal_PenaltyOnQ99_N100	Analytical	Verified
LogNormal_PenaltyOnQ99_N25	Analytical	Verified
LogNormal_PenaltyOnSigma_N100	Analytical	Verified
LogNormal_PenaltyOnSigma_N25	Analytical	Verified
LogPearsonTypeIII_PenaltyOnAllParamsAndQ99_N25	Analytical	Verified
LogPearsonTypeIII_PenaltyOnAllParams_N25	Analytical	Verified
LogPearsonTypeIII_PenaltyOnMuAndGamma_N25	Analytical	Verified

B17CSyntheticDataTests

Source: B17CSyntheticDataTests.cs. Namespace: RMC.BestFit.Verification.Univariate.Bulletin17CTests.

Scope: Bulletin17CDistribution (Exponential); Bulletin17CDistribution (Gamma); Bulletin17CDistribution (LogNormal); Bulletin17CDistribution (LogPearsonTypeIII); Bulletin17CDistribution (Normal); Bulletin17CDistribution (PearsonTypeIII).

Method	Evidence	Disposition
Exponential_GmmRecoversGeneratingParent	Analytical	Verified
Gamma_GmmRecoversGeneratingParent	Analytical	Verified
LogNormal_GmmRecoversGeneratingLog10Parent	Analytical	Verified
LogPearsonTypeIII_GmmRecoversGeneratingLog10ParentAndQ99	Analytical	Verified
Normal_GmmRecoversGeneratingParent	Analytical	Verified
PearsonTypeIII_GmmRecoversGeneratingParentAndQ99	Analytical	Verified

HirschStedingerPlottingPositionVerificationTests

Source: B17CExampleTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.Bulletin17CTests`.

Scope: Hirsch-Stedinger plotting positions; worked Example 4; Hirsch-Stedinger plotting positions; worked Example 5; Hirsch-Stedinger plotting positions; worked Example 7.

Method	Evidence	Disposition
<code>Example4_MatchesPeakFQ</code>	Published source	Verified
<code>Example5_MatchesPeakFQ</code>	Published source	Verified
<code>Example7_MatchesPeakFQ</code>	Published source	Verified

Point process

Interpretation and results: Point process.

PointProcessExternalPackageOracleTests

Source: PointProcessExternalPackageOracleTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.PointProcessTests`.

Scope: stationary Poisson-GPA threshold process.

Method	Evidence	Disposition
<code>StationaryPoissonGpa_MatchesSciPyArtifact</code>	External package	Verified

PointProcessLikelihoodOracleTests

Source: PointProcessLikelihoodOracleTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.PointProcessTests`.

Scope: nonseasonal mixed-observation POT likelihood; seasonal mixed-observation annual-maximum likelihood.

Method	Evidence	Disposition
<code>NonseasonalMixedObservations_MatchIndependentLikelihood</code>	Analytical	Verified
<code>SeasonalMixedObservations_MatchIndependentAnnualMaximumLikelihood</code>	Analytical	Verified

PointProcessPriorTests

Source: PointProcessPriorTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.PointProcessTests`.

Scope: seasonal PointProcessModel; October water year automatic changepoint priors; seasonal PointProcessModel; calendar year automatic changepoint priors.

Method	Evidence	Disposition
<code>Test_CalendarYearPertHistogram_MatchesIndependentPriorPlacementOracle</code>	Analytical	Verified
<code>Test_WaterYearPertHistogram_MatchesIndependentPriorPlacementOracle</code>	Analytical	Verified

PointProcessRecoveryTests

Source: PointProcessRecoveryTests.Uniform.cs, PointProcessRecoveryTests.cs. Namespace: RMC.BestFit.Verification.Univariate.PointProcessTests.

Scope: nonseasonal PointProcessModel; nonseasonal Poisson/POT simulation; seasonal PointProcessModel with two changepoints; seasonal Poisson/POT simulation.

Method	Evidence	Disposition
Test_CalendarYearUniformSeasonality_AutomaticPriorsRecoverParentAndBothChangePoints	Recovery	Verified
Test_NonSeasonalProductionGenerator_RecoversParent	Recovery	Verified
Test_NonSeasonalSimulation_MatchesPoissonRateAndConditionalTail	Analytical	Verified
Test_SeasonalProductionGenerator_RecoversParentAndBothChangePoints	Recovery	Verified
Test_SeasonalProductionGenerator_WithUnequalIntensities_RecoversParentAndBothChangePoints	Recovery	Verified
Test_SeasonalSimulation_MatchesSeasonRatesAssignmentsAndConditionalTails	Analytical	Verified
Test_SeasonalSimulation_WithUnequalIntensities_MatchesSeasonRatesAssignmentsAndConditionalTails	Analytical	Verified
Test_WaterYearUniformSeasonality_AutomaticPriorsRecoverParentAndBothChangePoints	Recovery	Verified

Competing risks

Interpretation and results: Competing risks.

CompetingRiskDependencyVerificationTests

Source: CompetingRiskDependencyVerificationTests.cs. Namespace: RMC.BestFit.Verification.Univariate.CompetingRiskTests.

Scope: two-Normal competing-risk maximum under the named Gaussian-copula dependence mode.

Method	Evidence	Disposition
Test_CorrelationMatrixSimulation_MatchesRankDependenceAndCompositeCdf	Analytical	Verified
Test_IndependentSimulation_MatchesRankDependenceAndCompositeCdf	Analytical	Verified
Test_PerfectlyNegativeSimulation_MatchesRankDependenceAndCompositeCdf	Analytical	Verified
Test_PerfectlyPositiveSimulation_MatchesRankDependenceAndCompositeCdf	Analytical	Verified

CompetingRiskRecoveryTests

Source: CompetingRiskRecoveryTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.CompetingRiskTests`.

Scope: Maximum WeibullGumbelDogLeg; Minimum CorrelatedTwoWeibullDogLeg; Minimum TwoWeibullDogLeg.

Method	Evidence	Disposition
<code>Bayesian_Maximum_WeibullGumbelDogLeg_RecoversParent</code>	Recovery	Verified
<code>Bayesian_Minimum_TwoWeibullDogLeg_RecoversParent</code>	Recovery	Verified
<code>MLE_Maximum_WeibullGumbelDogLeg_RecoversParent</code>	Recovery	Verified
<code>MLE_Minimum_CorrelatedTwoWeibullDogLeg_RecoversParent</code>	Recovery	Verified
<code>MLE_Minimum_TwoWeibullDogLeg_RecoversParent</code>	Recovery	Verified

Mixtures

Interpretation and results: Mixtures.

MixtureExternalPackageOracleTests

Source: MixtureExternalPackageOracleTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.MixtureTests`.

Scope: ordinary two-component Normal mixture.

Method	Evidence	Disposition
<code>OrdinaryNormalMixture2D_MatchesScikitLearnArtifact</code>	External package	Verified

MixtureRecoveryTests

Source: MixtureRecoveryTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.MixtureTests`.

Scope: three-component Normal mixture; two-component Normal mixture; zero-inflated two-component Normal mixture.

Method	Evidence	Disposition
<code>NormalMixture2D_BayesianRecovery</code>	Recovery	Verified
<code>NormalMixture2D_Recovery_Parity</code>	Recovery	Verified
<code>NormalMixture3D_BayesianRecovery</code>	Recovery	Verified
<code>NormalMixture3D_Recovery_Parity</code>	Recovery	Verified
<code>ZeroInflatedNormalMixture2D_BayesianRecovery</code>	Recovery	Verified
<code>ZeroInflatedNormalMixture2D_Recovery_Parity</code>	Recovery	Verified

Composite analyses

Interpretation and results: Composite analyses.

CompositeOracleVerificationTests

Source: CompositeOracleVerificationTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.CompositeTests`.

Scope: CombinationRules SatisfyTheoreticalBracketingAndRemainDistinct; CorrelationMatrix MinimumAndMaximumMatchBivariateNormalOrthants; MaximumComposite IndependentAndComonotonicClosedForms; MaximumPosterior CompleteCartesianOracle; MinimumComposite IndependentAndComonotonicClosedForms; MinimumPosterior CompleteCartesianOracle; MixtureCdf ExactWeightedNormalSum; MixturePosterior CompleteCartesianOracle; MixtureQuantiles InvertAnalyticWeightedNormalCdf; MixtureQuantiles MatchPublishedRMistrTable45.

Method	Evidence	Disposition
CombinationRules_SatisfyTheoreticalBracketingAndRemainDistinct	Analytical	Verified
CorrelationMatrix_MinimumAndMaximumMatchBivariateNormalOrthants	Analytical	Verified
MaximumComposite_MatchesIndependentAndComonotonicClosedForms	Analytical	Verified
MaximumPosterior_MatchesCompleteCartesianOracle	Analytical	Verified
MinimumComposite_MatchesIndependentAndComonotonicClosedForms	Analytical	Verified
MinimumPosterior_MatchesCompleteCartesianOracle	Analytical	Verified
MixtureCdf_MatchesExactWeightedNormalSum	Analytical	Verified
MixturePosterior_MatchesCompleteCartesianOracle	Analytical	Verified
MixtureQuantiles_InvertAnalyticWeightedNormalCdf	Analytical	Verified
MixtureQuantiles_MatchPublishedRMistrTable45	External package	Verified

CompositePredictiveRecoveryTests

Source: CompositePredictiveRecoveryTests.cs. Namespace: `RMC.BestFit.Verification.Univariate.CompositeTests`.

Scope: equal-weight model average of two Normal child analyses; independent maximum of two Normal child analyses; independent minimum of two Normal child analyses; unequal-weight mixture of two Normal child analyses.

Method	Evidence	Disposition
EqualWeightModelAverage_EndToEndPredictiveRecovery	Recovery	Verified
MaximumComposite_EndToEndPredictiveRecovery	Recovery	Verified
MinimumComposite_EndToEndPredictiveRecovery	Recovery	Verified
MixtureComposite_EndToEndPredictiveRecovery	Recovery	Verified

Bivariate and coincident frequency

Interpretation and results: Bivariate and coincident frequency.

BivariateAnalysisParameterRecoveryTests

Source: BivariateAnalysisParameterRecoveryTests.cs. Namespace: `RMC.BestFit.Verification.Bivariate`.

Scope: AMH copula with two Normal marginals; Clayton copula with two Normal marginals; Frank copula with two Normal marginals; Gumbel copula with two Normal marginals; Joe copula with two Normal marginals; Normal copula with two Normal marginals; StudentT copula with two Normal marginals.

Method	Evidence	Disposition
RecoverAMHCopulaParameters	Recovery	Verified
RecoverClaytonCopulaParameters	Recovery	Verified
RecoverFrankCopulaParameters	Recovery	Verified
RecoverGumbelCopulaParameters	Recovery	Verified
RecoverJoeCopulaParameters	Recovery	Verified
RecoverNormalCopulaParameters	Recovery	Verified
RecoverStudentTCopulaParametersWithMaximumLikelihood	Recovery	Verified

CoincidentFrequencyAnalysisTests

Source: CoincidentFrequencyAnalysisTests.cs. Namespace: `RMC.BestFit.Verification.Bivariate`.

Scope: monotone nonlinear $\exp(0.01 X + 0.01 Y)$ response under Normal marginals and positive Gaussian-copula dependence; sum of correlated Normal marginals; negative rho; sum of correlated Normal marginals; positive rho; sum of correlated Normal marginals; rho=0.

Method	Evidence	Disposition
ExponentialLinearCombination_ParentResponseInsidePredictiveBands	Recovery	Verified
SumOfNormals_RhoNegative_MatchesClosedForm	Analytical	Verified
SumOfNormals_RhoPositive_MatchesClosedForm	Analytical	Verified
SumOfNormals_RhoZero_MatchesClosedForm	Analytical	Verified

CopulaEstimationOracleTests

Source: CopulaEstimationOracleTests.cs. Namespace: `RMC.BestFit.Verification.Bivariate`.

Scope: AliMikhailHaq copula; inference from margins; AliMikhailHaq copula; maximum pseudo-likelihood; Clayton copula; inference from margins; Clayton copula; maximum pseudo-likelihood; Frank copula; inference from margins; Frank copula; maximum pseudo-likelihood; Gumbel copula; inference from margins; Gumbel copula; maximum pseudo-likelihood; Joe copula; inference from margins; Joe copula; maximum pseudo-likelihood; Normal copula; inference from margins; Normal copula; maximum pseudo-likelihood; StudentT copula; inference from margins; StudentT copula; maximum pseudo-likelihood.

Method	Evidence	Disposition
AliMikhailHaq_InferenceFromMargins_MatchesIndependentOptimum	External package	Verified
AliMikhailHaq_PseudoLikelihood_MatchesIndependentOptimum	External package	Verified
Clayton_InferenceFromMargins_MatchesIndependentOptimum	External package	Verified
Clayton_PseudoLikelihood_MatchesIndependentOptimum	External package	Verified
Frank_InferenceFromMargins_MatchesIndependentOptimum	External package	Verified
Frank_PseudoLikelihood_MatchesIndependentOptimum	External package	Verified
Gumbel_InferenceFromMargins_MatchesIndependentOptimum	External package	Verified
Gumbel_PseudoLikelihood_MatchesIndependentOptimum	External package	Verified
Joe_InferenceFromMargins_MatchesIndependentOptimum	External package	Verified
Joe_PseudoLikelihood_MatchesIndependentOptimum	External package	Verified
Normal_InferenceFromMargins_MatchesIndependentOptimum	External package	Verified
Normal_PseudoLikelihood_MatchesIndependentOptimum	External package	Verified
StudentT_InferenceFromMargins_MatchesIndependentOptimum	External package	Verified
StudentT_PseudoLikelihood_MatchesIndependentOptimum	External package	Verified

Rating curves

Interpretation and results: Rating curves.

RatingCurveBayesianRecoveryTests

Source: RatingCurveBayesianRecoveryTests.cs. Namespace: `RMC.BestFit.Verification.RatingCurve`.

Scope: one-control synthetic rating curve; SingleSegment_Default; one-control synthetic rating curve; SingleSegment_LowNoise; one-control synthetic rating curve; SingleSegment_WideRange; three-control synthetic rating curve; ThreeSegment_MultipleControl; two-control synthetic rating curve; TwoSegment_BankfullTransition.

Method	Evidence	Disposition
Test_EstimateParameters_SingleSegment_Default	Recovery	Verified
Test_EstimateParameters_SingleSegment_LowNoise	Recovery	Verified
Test_EstimateParameters_SingleSegment_WideRange	Recovery	Verified
Test_EstimateParameters_ThreeSegment_MultipleControl	Recovery	Verified
Test_EstimateParameters_TwoSegment_BankfullTransition	Recovery	Verified

RatingCurveContinuityVerificationTests

Source: RatingCurveContinuityVerificationTests.cs. Namespace: RMC.BestFit.Verification.RatingCurve.

Scope: piecewise added-control power law at activation.

Method	Evidence	Disposition
AddedControl_TwoSidedIncrementAtActivation_MatchesAnalyticalPowerLaw	Analytical	Verified
ExponentsAtDefaultLowerBound_AddedControlIncrementsVanishAtActivation	Analytical	Verified
ZeroExponent_AddedControlJumpsByItsCoefficientAtActivation	Analytical	Verified

RatingCurveExampleRecoveryTests

Source: RatingCurveExampleRecoveryTests.cs. Namespace: RMC.BestFit.Verification.RatingCurve.

Scope: one-control piecewise power-law rating curve; three-control piecewise power-law rating curve; two-control piecewise power-law rating curve.

Method	Evidence	Disposition
Mle_OneSegment_RecoversExampleCurve	External package	Verified
Mle_ThreeSegment_RecoversExampleCurve	External package	Verified
Mle_TwoSegment_RecoversExampleCurve	External package	Verified

RatingCurveLikelihoodOracleTests

Source: RatingCurveLikelihoodOracleTests.cs. Namespace: RMC.BestFit.Verification.RatingCurve.

Scope: one-control piecewise power-law rating curve; three-control piecewise power-law rating curve; two-control piecewise power-law rating curve.

Method	Evidence	Disposition
OneSegment_DataLogLikelihood_IsDischargeSpaceDensity	External package	Verified
ThreeSegment_DataLogLikelihood_IsDischargeSpaceDensity	External package	Verified
TwoSegment_DataLogLikelihood_IsDischargeSpaceDensity	External package	Verified

RatingCurveMLERecoveryTests

Source: RatingCurveMLERecoveryTests.cs. Namespace: RMC.BestFit.Verification.RatingCurve.

Scope: one-control synthetic rating curve; SingleSegment_Default; one-control synthetic rating curve; SingleSegment_LowNoise; one-control synthetic rating curve; SingleSegment_WideRange; three-control synthetic rating curve; ThreeSegment_MultipleControl; two-control synthetic rating curve; TwoSegment_BankfullTransition.

Method	Evidence	Disposition
Test_EstimateParameters_SingleSegment_Default	Recovery	Verified
Test_EstimateParameters_SingleSegment_LowNoise	Recovery	Verified
Test_EstimateParameters_SingleSegment_WideRange	Recovery	Verified
Test_EstimateParameters_ThreeSegment_MultipleControl	Recovery	Verified
Test_EstimateParameters_TwoSegment_BankfullTransition	Recovery	Verified

Time-series models

Interpretation and results: Time-series models.

ARAnalysisTests

Source: ARAnalysisTests.cs. Namespace: `RMC.BestFit.Verification.TimeSeriesAnalysis`.

Scope: AR(1).

Method	Evidence	Disposition
Test_EstimateParameters_AR1	External package	Verified

AutoRegressiveMLERecoveryTests

Source: AutoRegressiveMLERecoveryTests.cs. Namespace: `RMC.BestFit.Verification.TimeSeriesAnalysis`.

Scope: AR(1).

Method	Evidence	Disposition
Test_EstimateParameters_AR1	External package	Verified

MAAnalysisTests

Source: MAAnalysisTests.cs. Namespace: `RMC.BestFit.Verification.TimeSeriesAnalysis`.

Scope: MA(1).

Method	Evidence	Disposition
Test_EstimateParameters_MA1	External package	Verified

MovingAverageMLERecoveryTests

Source: MovingAverageMLERecoveryTests.cs. Namespace: `RMC.BestFit.Verification.TimeSeriesAnalysis`.

Scope: MA(1).

Method	Evidence	Disposition
Test_EstimateParameters_MA1	External package	Verified

TimeSeriesChunk13OracleTests

Source: TimeSeriesChunk13OracleTests.cs. Namespace: `RMC.BestFit.Verification.TimeSeriesAnalysis`.

Scope: ARIMA(2,0,0) and ARIMA(0,0,2); ARIMAX(1,0,1), intercept + linear trend + monthly Fourier seasonality + two level covariates; AutoRegressive(1) and MovingAverage(1); AutoRegressive(2) and MovingAverage(2).

Method	Evidence	Disposition
ArimaxTrendSeasonalityAndCovariatesMatchIndependentPythonOracle	Independent calculation	Verified
FirstOrderConditionalObjectivesMatchIndependentPythonOracle	Independent calculation	Verified
HigherOrderArAndMaResponsesMatchIndependentPythonOracle	Independent calculation	Verified

Method	Evidence	Disposition
PureArAndMaThroughArimaMatchIndependentPythonOracle	Independent calculation	Verified

TimeSeriesIndependentOracleTests

Source: TimeSeriesIndependentOracleTests.cs. Namespace: `RMC.BestFit.Verification.TimeSeriesAnalysis`.

Scope: AR, MA, ARIMA, and ARIMAX automatic Box-Cox/Yeo-Johnson fitting; AR, MA, ARIMA, and ARIMAX information criteria; AR, MA, ARIMA, and ARIMAX invalid innovation scales; AR, MA, ARIMA, and ARIMAX manual Box-Cox/Yeo-Johnson transforms; AR, MA, ARIMA, and ARIMAX scale priors; ARIMA and ARIMAX forecast variance recurrence; ARIMA(1,1,1) log transform and ARIMAX(1,1,0) level covariate; ARIMA(1,1,1) transformed/differenced generator; ARIMAX transformed/differenced date and covariate generator; ARIMAX with $p > 0$ and date-indexed covariates lagged $b > 0$ steps; ARIMAX(1,0,0) $b=2$, ARIMAX(1,0,3) $b=1$, Box-Cox ARIMAX(2,1,1) $b=1$, Yeo-Johnson ARIMAX(1,0,1) $b=1$ with two covariates; ARIMAX(1,1,0), date-indexed contemporaneous covariate; Transformed AR(1) and MA(1) generators; Transformed ARIMA and ARIMAX stochastic forecasts.

Method	Evidence	Disposition
ArAndMaTransformedGeneratorsMatchIndependentOracle	External package	Verified
ArimaAndArimaxPredictionReintegrationMatchesHandRecurrenceOracle	External package	Verified
ArimaAndArimaxPredictionUncertaintyBeginsAtForecastBoundary	Analytical	Verified
ArimaDifferencedTransformedGeneratorMatchesIndependentOracle	External package	Verified
ArimaxDifferencedLikelihoodMatchesDateIndexedIndependentOracle	External package	Verified
ArimaxDistributedLagConditioningMatchesIndependentOracle	External package	Verified
ArimaxTransformedDifferencedGeneratorMatchesIndependentOracle	External package	Verified
InformationCriteriaUseDataLikelihoodAtMapAndExcludePrior	Analytical	Verified
InvalidScaleBehaviorMatchesScalarAndPointwiseOracle	External package	Verified
JeffreysScaleMetadataMatchesIndependentPriorOracle	External package	Verified
ManualTransformLambdaRebuildMatchesIndependentLikelihoodOracle	External package	Verified
TransformLambdaMatchesIndependentTrainingOnlyOracle	External package	Verified
TransformedArimaAndArimaxForecastsMatchModelScaleOracle	Analytical	Verified

TimeSeriesIndependentRecoveryTests

Source: TimeSeriesIndependentRecoveryTests.cs. Namespace: `RMC.BestFit.Verification.TimeSeriesAnalysis`.

Scope: ARIMA(1,1,1), logarithmic response transform, no intercept; ARIMAX(1,1,0), intercept and one level covariate, no response transform.

Method	Evidence	Disposition
BayesianArima111LogD1RecoversGeneratingParameters	External package	Verified
BayesianArimax10D1LevelCovariateRecoversGeneratingParameters	External package	Verified
MleArima111LogD1RecoversGeneratingParameters	External package	Verified
MleArimax10D1LevelCovariateRecoversGeneratingParameters	External package	Verified

Spatial extremes

Interpretation and results: Spatial extremes.

SpatialGEVBayesianRecoveryTests

Source: SpatialGEVBayesianRecoveryTests.cs. Namespace: `RMC.BestFit.Verification.SpatialExtremes`.

Scope: Homogeneous SpatialGEV with exponential Gaussian copula; Homogeneous SpatialGEV without copula; Homogeneous SpatialGEV without copula, shape -0.2; Homogeneous SpatialGEV without copula, shape 0; Homogeneous SpatialGEV without copula, shape 0.1; SpatialGEV with X/Y location regression, no copula.

Method	Evidence	Disposition
Bayesian_BasicHomogeneous_RecoversParameters	Recovery	Verified
Bayesian_WithCopula_RecoversParameters	Recovery	Verified
Bayesian_WithLocationRegression_RecoversParameters	Recovery	Verified
Bayesian_WithNegativeShape_RecoversParameters	Recovery	Verified
Bayesian_WithPositiveShape_RecoversParameters	Recovery	Verified
Bayesian_WithZeroShape_RecoversParameters	Recovery	Verified

SpatialGEVChunk14OracleTests

Source: SpatialGEVChunk14OracleTests.cs. Namespace: `RMC.BestFit.Verification.SpatialExtremes`.

Scope: Basic exponential correlation with Cartesian distance; Four-site fixed log-location summaries with two location covariates and one held-out target row; Powered exponential correlation with range 20 and smoothness 1.6; Spherical correlation with compact support at range 20; Ten complete row/year vectors across three sites with fixed posterior draws; Three-site Cartesian exponential Gaussian-copula training network with one held-out target; Three-site geodesic exponential latent-error network with four fixed parameter draws; Three-site intercept-only SpatialGEV on 24 complete row/year blocks; Three-site location-regression SpatialGEV with nine fixed independently supplied draws; Twelve row/year vectors across three sites, wrapping blocks of four, five fixed replicates.

Method	Evidence	Disposition
BasicExponentialCorrelation_MatchesAnalyticalGrid	Analytical	Verified
GodambeSensitivityVariabilityAndSandwich_MatchIndependentOracle	Independent calculation	Verified
HeldOutCopulaFold_MatchesIndependentFittedOracle	Independent calculation	Verified
HeldOutCovariateFold_MatchesIndependentRegressionOracle	Independent calculation	Verified
PoweredExponentialCorrelation_MatchesAnalyticalGrid	Analytical	Verified
RegionalFixedDrawAggregation_MatchesIndependentPosteriorOracle	Independent calculation	Verified
SphericalCorrelation_MatchesAnalyticalGridAndCompactSupport	Analytical	Verified
TemporalBlockBootstrap_MatchesIndependentWholeRowOracle	Independent calculation	Verified
UngaugedDrawSpecificPrediction_MatchesIndependentGeodesicGaussianOracle	Independent calculation	Verified
VarianceInflation_UsesExactIndependentAnalyticalTransformation	Independent calculation	Verified

SpatialGEVDistanceOracleTests

Source: SpatialGEVDistanceOracleTests.cs. Namespace: `RMC.BestFit.Verification.SpatialExtremes`.

Scope: Five-site Cartesian distance with exponential correlation components; Five-site exponential correlation with geodesic distance.

Method	Evidence	Disposition
<code>CartesianMetric_IsPlanarEuclidean</code>	Analytical	Verified
<code>GeodesicMetric_MatchesHaversineOracle</code>	External package	Verified

SpatialGEVInformationCriteriaTests

Source: SpatialGEVInformationCriteriaTests.cs. Namespace: `RMC.BestFit.Verification.SpatialExtremes`.

Scope: Five-site SpatialGEV with exponential Gaussian copula and structured missing sites.

Method	Evidence	Disposition
<code>MissingSiteModel_InformationCriteria_UseRowYearBlocks</code>	Analytical	Verified

SpatialGEVKrigingOracleTests

Source: SpatialGEVKrigingOracleTests.cs. Namespace: `RMC.BestFit.Verification.SpatialExtremes`.

Scope: Five-site exponential latent Gaussian process.

Method	Evidence	Disposition
<code>KrigingPrediction_MatchesConditionalGaussianProcessOracle</code>	External package	Verified

SpatialGEVLikelihoodOracleTests

Source: SpatialGEVLikelihoodOracleTests.cs. Namespace: `RMC.BestFit.Verification.SpatialExtremes`.

Scope: Five-site SpatialGEV with exponential latent location-error Gaussian process; Five-site homogeneous SpatialGEV margins without copula, structured missing sites; Five-site homogeneous SpatialGEV, exponential Gaussian copula, complete rows; Five-site homogeneous SpatialGEV, exponential Gaussian copula, structured missing sites.

Method	Evidence	Disposition
<code>CompleteRows_CopulaLikelihood_MatchesIndependentOracle</code>	External package	Verified
<code>LocationErrorModel_DataLogLikelihood_ExcludesProcessDensity</code>	External package	Verified
<code>LocationErrorModel_PosteriorKernel_IsInvariantToTheDecomposition</code>	External package	Verified
<code>LocationErrorModel_ScalarAndPointwiseDecompositionsAgree</code>	External package	Verified
<code>LocationErrorModel_ScalarAndPointwiseGradientsAgree</code>	External package	Verified
<code>MarginalOnly_WithoutCopula_MatchesIndependentOracle</code>	External package	Verified
<code>MissingSites_DataLogLikelihood_UsesObservedSiteCopulaSubmatrix</code>	External package	Verified
<code>MissingSites_PointwiseRows_MatchObservedSubsetOracle</code>	External package	Verified

SpatialGEVMLERecoveryTests

Source: SpatialGEVMLERecoveryTests.cs. Namespace: RMC.BestFit.Verification.SpatialExtremes.

Scope: Homogeneous SpatialGEV with exponential Gaussian copula; Homogeneous SpatialGEV without copula.

Method	Evidence	Disposition
MLE_BasicHomogeneous_RecoversParameters	Recovery	Verified
MLE_WithCopula_RecoversParameters	Recovery	Verified

SpatialGEVSimulationVerificationTests

Source: SpatialGEVSimulationVerificationTests.cs. Namespace: RMC.BestFit.Verification.SpatialExtremes.

Scope: Five-site homogeneous SpatialGEV with exponential Gaussian copula.

Method	Evidence	Disposition
GenerateRandomValues_WithCopula_ReproducesTheFittedIntersiteDependence	Analytical	Verified

CHAPTER 18

References

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